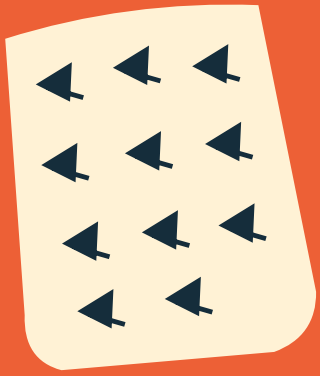
THE BEER DETECTIVE

Uncovering the Myths, Mysteries and Untold Stories of Beer



A WORLD BEER WIKI BOOK

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A WORLD BEER WIKI BOOK PROJECT

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About this edition

The Beer Detective is a free, original narrative nonfiction book developed from the research routes in World Beer Wiki. Its research baseline is edition 1.19, repository commit `3b985c19a9dcccef8d334be54b5405e42093e812`.

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Research, drafting, internal editing and production were assisted by AI. This review edition awaits independent human editorial, specialist and community review. It draws on published work rather than new interviews or field visits. The research note and chapter notes describe its method, source access and remaining limits.

No ISBN has been assigned. This book is an investigation of history, science and culture, rather than a brewing or water-treatment manual.

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The Case for Beer

A glass can smell of bananas even when no banana went near the brewery. The resemblance is convincing. The conclusion is wrong.

That small discrepancy is a good reason to begin an investigation. A drink arrives with information already attached to it: a name, a color, an ingredient list, perhaps a story about a monk or a king. Then it supplies information of its own. Something rises from the glass and suggests a fruit. Something on the tongue seems to confirm a number printed on the label. The evidence appears to agree, until someone changes the light, hides the label or asks the yeast what it has been making.[1]

Beer rewards this sort of interference. Its apparent simplicity gives us plenty to get wrong. Grain, water and fermentation can be described in a sentence; accounting for what happens between the field and the glass takes rather longer. There are organisms doing chemical work, people deciding which work is desirable, and institutions deciding who may do it. The result is a remarkably convenient meeting place for different kinds of knowledge. A chemist and a historian can disagree about entirely different things while studying the same bottle.

They also inherit different kinds of evidence. The chemist may be able to make another batch. The historian cannot ask an eighteenth-century brewer to repeat the day on which a famous beer was supposedly invented. An archaeologist may have the container without the contents. A researcher studying a household business may find that a published price records the finished drink while leaving out hours spent carrying its ingredients. The problem is often to recognize which part of an event has survived as a fact.

This book follows twenty-four such problems. They range from a hollow in a prehistoric stone to a modern experiment in expectation. The route passes through poems, law courts, cold rooms and kitchens. Along the way, a number on a label becomes less informative, a dry account becomes more interesting, and a famous date acquires a longer history on either side of it. Some cases have clear chemical resolutions. Others end with an interpretation that remains open to argument. Both can make satisfying reading, provided the question is worth asking.

There is a temptation, when collecting these stories, to choose the most spectacular claim in each source and hurry on. The oldest beer. The first brewer. The ingredient that changed everything. Such claims supply their own headlines and spare the reader an untidy middle. They also have a way of becoming less interesting under examination. The question of whether beer caused agriculture is narrower than the question of what people knew how to do with wild plants. A single inventor can obscure the more absorbing business of how a new product became possible, profitable and recognizable.

The people come into better focus when the headline loosens its grip. An official's record of an offense can preserve a working woman's refusal. A communal brewhouse can turn out to be only one address in a drink's production. A starter analyzed in a laboratory can contain more kinds of life than a familiar strain name suggests. These discoveries redirect the inquiry toward a sign taken down, liquid carried to another building, and organisms whose work overlaps.[2]

Nor does every fermented drink in these pages need to be squeezed into the same category. Names carry local distinctions. Pito, chicha, makgeolli, mbege and masato enter the book through particular studies and accounts, with different ingredients, histories and social settings. Similar questions can be asked of them without proposing a shared origin or treating one as an imperfect version of another. A useful comparison keeps the differences visible.

The itinerary moves through places where ancient evidence and present-day choices coexist. A household enterprise is a current arrangement of work and money, with decisions still to be made. Its maker may have reasons to change a method, keep it, or use different methods for different purposes. The laboratory can help explain a transformation; the people doing the work can explain what they want it to achieve. A living tradition has a future under discussion as well as a past.

The reader will encounter some technical terms. They earn their place by solving a problem. An enzyme becomes useful when we need to understand how grain can yield fermentable sugar. A hybrid matters when a yeast's family history refuses to fit a single branch. A statistical comparison matters when a result that failed to appear is as revealing as one that did. There is no examination at the end, and very little reward for memorizing the vocabulary before meeting the evidence.

The book's four parts change the vantage point. First come things that survived after the drink was gone. Then we look inside the glass, following processes too small or too rapid to see unaided. The third part follows the people

and arrangements that make a drink available. The last returns to stories that remain powerful even after their foundations have been questioned, before looking toward the energy and crops on which future brewing will depend.

World Beer Wiki supplied the initial map. The route then followed papers, translations, museum records and local accounts, all identified in the notes. Readers who want to examine a disputed interpretation or follow a promising detour can begin there. The research note explains how those sources were used.

The detective in the title has no superior sense of smell and no secret archive. The useful equipment is ordinary: attention to dates, a willingness to compare explanations, and enough patience to notice when a source has changed the subject. It helps to be curious about a failed prediction. It helps even more to enjoy a story after discovering that its most familiar version was mistaken.

The banana will return. First, there is a stone.

PART I

What Survives

A stone, a poem, a tablet, a pot. Each preserves something different. The first six investigations follow the distance between an object and the story it can support.

01

Before the Field

The object that might overturn your idea of a brewery is a hollow in a stone. There is no copper kettle, no chimney, no building with a sign above the door. At Raqefet Cave on Mount Carmel, in present-day Israel, researchers examined three mortars associated with a Natufian burial site. In 2018, Li Liu and her colleagues proposed that some of the traces preserved there came from brewing roughly 13,000 years ago. If their interpretation is right, people were making a cereal-based alcoholic drink before the region's domesticated cereals appeared.[1]

It is a finding with an irresistible sequel. First there was beer; then someone decided to plant enough grain to keep the beer coming; then came farms, villages, cities and, eventually, the person who asks you to fill in a form. Civilization as the unintended consequence of a drinks order.

The stone does not tell that story. It poses a better question: what would count as evidence that someone brewed before there were farmers?

We need to begin with the grain. A wild cereal already contains starch. Domestication did not invent this useful material, and a person did not need to own a field to collect it. The difficulty for a brewer is that starch is a store of linked sugar units, whereas ordinary brewing yeast needs sugars it can take up and use. A pile of grain and some water are therefore the beginning of a problem, not a complete solution. The potential alcohol is locked away.

Germination offers one way in. A sprouting seed must mobilize its food reserve to support its own growth. Enzymes help break the reserve into smaller, usable molecules. A brewer who moistens grain, lets it germinate and then manages what happens next is borrowing a plant's machinery. Heating a mixture of crushed grain and water can make starch more accessible; under suitable conditions, enzymes continue the conversion. Yeast can then turn available sugars into alcohol and carbon dioxide. These are distinguishable operations, although no early brewer needed to name an enzyme to discover a useful sequence.[2]

This distinction matters archaeologically. Evidence that people possessed grain is not evidence that they brewed it. Evidence that they ground grain is not

enough either. Grinding can lead toward bread, porridge or other foods. Even a vessel suited to holding liquid gives us a range of possibilities. To identify brewing, researchers must find something more discriminating than a plausible container and an appetite.

The Raqefet investigation combined the wear on stone surfaces, microscopic plant residues and experiments in processing cereals. The team identified a range of plant materials, including wheat or barley, and interpreted two of the mortars as storage facilities and another as a place for processing food, including beer. Experimental comparisons helped them ask whether the ancient starch had suffered changes consistent with malting, heating and fermentation. This was an argument built from several kinds of trace, rather than the discovery of ancient liquid.[1]

A starch granule is small, but it has a structure. Processing can pit it, distort it or destroy part of it. The principle resembles the reading of a damaged tool: the damage may preserve information about the forces that acted on it. Yet a mark is useful only if we understand how distinctive it is. A scratch on a door establishes contact with something hard. It does not, by itself, identify a burglar. The scientific work lies in comparing the mark with alternatives, including uneventful ones.

That is where this case became a dispute. In a published response in 2019, Liu and colleagues addressed objections raised by David Eitam about mortar functions, starch damage, vessel size and the proposed heating method. They defended their interpretation as a convergence of evidence and emphasized that they were not assigning a brewing function to every Natufian mortar. The exchange matters because an ancient beer claim can sound conclusive once detached from the discussion that followed it. Here the discussion remained technical and consequential: how do we distinguish deliberate brewing from other changes to grain?[3]

The disagreement does not oblige us to dismiss the find. Nor can we resolve it by voting for the more appealing account. It tells us where the weight rests. If several different processes can produce a particular feature, that feature must be read with its companions. If an experimental procedure produces a similar residue, the experiment demonstrates a possible pathway. Additional evidence is needed to show that this was the pathway taken at the site. A successful reconstruction is a valuable result; it is not a recording.

Then there is the setting. Raqefet was a place where people buried their dead. A separate study, published in 2013 by Dani Nadel and colleagues, reported plant impressions interpreted as linings in four Natufian graves, including

impressions identified as flowering stems. The research concerns the preparation of graves, not beer. Its relevance is that the cave preserves deliberate activities around burial. The brewing researchers' proposal of drink served at ritual feasts belongs within this context, but a burial ground cannot tell us who drank, what words were spoken or how grief felt.[4]

Suppose, for the moment, that the brewing interpretation proves correct. Have we established that beer came before agriculture? We have made progress, but the question still needs better wording. Collecting wild plants, deliberately cultivating them, and depending on domesticated crops are different activities. They need not begin together. The history of agriculture is not a switch with a hunter-gatherer position on the left and a farmer position on the right.

In 2006, Ken-ichi Tanno and George Willcox examined archaeological wheat remains and argued that domestication was slower than some previous estimates suggested. Ears that retained their seeds, a useful property for harvesting, appeared gradually; wild forms persisted in cultivated fields. Their study concerned wheat domestication, not an alcoholic motive for it. It helps explain why a phrase such as "before farming" can conceal several different chronological claims.[5]

A person can work hard to obtain wild grain. Another can sow grain whose inherited characteristics still resemble those of wild populations. A community can plant some food while continuing to gather and hunt for much of the rest. A brewing discovery must be placed within that changing landscape of practices. It cannot be made to stand on one side of a sharp boundary simply because the boundary is useful for a headline.

There is also a rival in the supposed contest for the first use of grain. In 2018, Amaia Arranz-Otaegui and colleagues reported charred bread-like remains from Shubayqa 1 in northeastern Jordan, dating to about 14,400 years ago. Their analyses identified evidence of prepared foods using wild cereals and other plant materials, well before an agricultural way of life. The oldest surviving evidence for a food is not its birth certificate, but these remains make a cereal world with several possibilities more credible than a single-file procession from beer to bread, or bread to beer.[6]

The usual argument asks which product deserves credit for agriculture. Another way to approach it is to notice what both require: knowledge of plants, repeated handling and an interest in turning something difficult to eat into something worth making. Several foods might encourage attention to the same grain. A favored drink could matter without being the sole cause of a

transformed food system. Causation is not a prize that only one ingredient can win.

To prove that brewing drove cultivation, we would need evidence connecting choices over time: grain acquired or grown for that purpose, repeated investment, consequences that distinguish the brewing explanation from other explanations. A single early brewing site, however persuasive, cannot supply the whole sequence. Earlier in time does not mean responsible for what followed. The first person in a queue has not necessarily summoned everyone else.

The most interesting result at Raqefet survives this subtraction. A society without domesticated cereal fields may have possessed intricate knowledge of how grain changes. The people responsible would not have been waiting for a later civilization to teach them technology. If the mortars record brewing, they record practical accomplishment in its own setting, with purposes we can only partly recover. If the interpretation changes, the work of comparing residues, tools and experiments will still have made the question more precise.[7]

For now, we can leave the grand procession of civilization outside the cave. Inside, there are three studied mortars, microscopic traces, an argument and a much larger collection of unexamined possibilities. The gap between those things and a confident creation story is not empty. It is where the next investigation belongs.

02

A Goddess in the Brewhouse

Ninkasi has a large shovel. This is one of the pleasures of meeting a goddess in a brewing poem: among the divine relationships and ceremonial language, there is equipment. The hymn places her among dough, grain, an oven, reed mats and vessels. It ends with beer pouring like the great rivers of Mesopotamia. A modern reader can recognize enough of a working process to become dangerously confident.[1]

Here, apparently, is the ancient recipe everyone has been looking for. Follow the goddess through her day and recover the drink.

Before lighting a fire, however, it is worth looking at the page. The Oxford translation contains gaps, a question mark and a fragmentary passage. Lines recur, with Ninkasi's name inserted into their repetition. The form keeps telling us what kind of thing we are reading. Someone has made brewing into praise. We are the ones trying to turn the praise into instructions.

That difference should not be exaggerated. A poem can preserve practical knowledge; a recipe need not have a table of weights. Many excellent cooks work from terse reminders that would defeat an inexperienced stranger. Calling a text poetic does not empty it of information. The question is how much the hymn tells a reader who lacks the working knowledge its original audience may have brought to it. Can we identify a process, or can we reproduce one particular process without filling in decisive steps ourselves?

Miguel Civil's 1964 study, a foundational scholarly edition of the hymn and its companion drinking song, makes the problem visible before a single brewing term is resolved. He worked with several tablets preserving different portions of the compositions. One copy included alternative readings added by an ancient scribe. The text was already being compared and reconciled long before modern beer enthusiasts encountered it. A document that looks like one smooth poem in English rests on a history of surviving pieces, variants and editorial decisions.[2]

The surviving copies are associated with the Old Babylonian period, around the early second millennium BCE. A date for a surviving copy is not necessarily the date when a composition began, much less when a brewing practice began. Its

value lies in what people chose to say about Ninkasi and the operations they associated with her.[3]

Take *bappir*. In the Oxford translation it appears as beerbread. That is an inviting word: two familiar foods joined together, offering a convenient bridge from loaf to drink. But an English compound is not an excavated ingredient. Civil discussed it as a prepared cereal material whose composition and use required interpretation. He explicitly noted unresolved questions about how the materials in the hymn were combined. The tidy phrase can make the problem look smaller than it is.[2]

For an experimenter, the choices quickly become physical. A dry cake and a wet fermented dough behave differently. A material added before heating need not have the same effect as one added after cooling. A substance can contribute starch, sugar, microorganisms, flavor or several of these at once. The name of an ingredient does not by itself choose among those roles. Even if two interpretations produce drinkable beer, their success does not make them historically equivalent.

The same difficulty appears in the order of operations. We can see a movement from handling ingredients toward pouring. It is tempting to treat each stanza as the next line of a procedure. Yet activities can happen alongside one another, and a hymn can arrange them for rhythm or emphasis. A poem celebrating a baker might praise mixing, kneading and the oven while leaving the waiting entirely unmentioned. The omission would be harmless to the audience that already understood dough, disastrous to a novice trying to follow it literally.

Peter Damerow made this difficulty central to his 2012 analysis of Sumerian brewing. He questioned whether familiar translations of technical terms had allowed modern brewing knowledge to determine the sequence we then claimed to discover in the text. In particular, the hymn does not unambiguously settle how separate ingredients and intermediate products relate. His objection is methodological: we risk supplying the process in order to translate the words, then citing the translated words as proof of the process.[4]

This circle is easy to enter because useful background knowledge and preconception are close neighbors. An experienced brewer may notice an operation that a linguist would miss. The brewer may also assume that a successful drink must have passed through the stages used in the brewer's own practice. The solution is to put the assumptions on the table, where they can be changed. It helps to write down what a proposed reconstruction needs from the poem, and what it needs from the person carrying the shovel.

The poem also insists on features a recipe might discard. It gives Ninkasi parents and places her labor in a divine world. Its repeated address directs attention to an agent, not simply to a transformation of matter. The final comparison with the Tigris and Euphrates gives the pouring grandeur that a measured quantity could not supply. Reading these passages as ornamental packaging around a technical core misses part of the document's purpose. The brewer's work is worthy of praise.[1]

The poem makes some of that labor unusually visible. Materials must be handled; vessels arranged. A goddess works among equipment, and the repeated praise falls on what she does with it. Even where a brewing term remains unresolved, the text gives us this evidence of how the work was valued. Precision about a mash temperature would answer a different question.

Repetition helps create that effect. Each return to Ninkasi's name asks us to dwell on an action rather than hurry through a checklist. On a practical instruction sheet, a repeated line might look like an editing error. Here it gives the work emphasis and the composition shape. We cannot recover a particular performance from the printed translation, but we can read its recurring addresses. The form carries information about what deserves attention, even when it leaves the handling of an ingredient uncertain.

Civil himself cautioned that a short lyrical composition was a difficult starting point for a technological investigation. His edition is sometimes encountered today only through the smooth translation that circulated beyond it. The commentary is less portable. There is an old problem in this very modern pattern: the sentence that survives quotation can sound more certain than the scholar who wrote the page.[2]

Fortunately, uncertainty does not prevent making an experiment. Archaeologist Tate Paulette describes a collaboration involving the University of Chicago's Oriental Institute and Great Lakes Brewing Company, joined by a ceramicist and a baker. Over several years the team brewed with replica vessels and explored possible roles for ingredients, including *bappir*. Paulette describes the undertaking as a way to generate questions, rather than a tightly controlled program proving a single reconstruction. Two of the resulting beers were named Enkibru and Gilgamash. Ancient history has room for a pun.[5]

The names are amusing; the unresolved questions are more useful. A vessel with a hole in its base forces a practical decision about what the hole does. A bread-like material demands a choice of texture and timing. The experiment cannot proceed on a cloud of approximate nouns. Somebody must decide where

to put the liquid. In that sense, practical work can expose uncertainty that polished prose has concealed.

It also changes what a failed trial means. Suppose a proposed interpretation produces no useful conversion of starch, or a vessel arrangement cannot perform the assigned task. That would be information about the proposal and its conditions. Conversely, a drinkable result would establish that the chosen pathway works. Neither outcome grants access to the exact beer consumed by people who knew a different grain supply, a different microbial environment and an inherited craft. Reconstruction is most informative when it records those choices rather than hiding them behind the word "authentic."

What should we call the hymn, then? A brewing song that preserves technical clues is a good beginning. Someone using "recipe" broadly may reasonably see a recipe-like account in it. The stronger claim, that it provides an unambiguous formula for recreating a particular ancient beer, asks the text to do work it cannot do alone. That distinction leaves room for both the literary composition and the experiments it has inspired.[6]

The best way to end this encounter may be to let the poem keep its final gesture. Ninkasi pours. The rivers supply the comparison; the reader supplies no invented measurement. The craft has been honored, the beer has reached its vessel, and the goddess has never promised to explain every step to us.

03

The Accountant's Beer

The twentieth day of the second month, in the first year of King Ibbi-Sin: someone had provisions to account for. A tablet now in the Metropolitan Museum of Art records beer, bread, oil and onions intended for messengers. The museum dates it to about 2028 BCE and associates it with Umma, in southern Mesopotamia. It measures roughly three centimeters across. A day's administrative business fits in a space smaller than many modern drinks receipts.[1]

We can know the date more precisely than we know the drink. That asymmetry is the beginning of this case.

A beer-ration tablet seems wonderfully cooperative. Here are commodities, people and quantities, marked in clay. Unlike a residue that must be coaxed into revealing what it once contained, a written account was designed to communicate. Someone meant another person to understand it. Surely this is where a historian can finally put down the magnifying glass and pour a reliable reconstruction.

But communication depends on its audience. The person who checked the allocation needed to know whether the provisions balanced, not how a brewer had converted grain into beer. A useful record leaves out things its users can safely assume. The feature that made the tablet efficient in its own time makes it frustrating in ours. It tells us what somebody needed to remember, which is not necessarily what we most want to ask.

Another Ur III tablet, held at Valdosta State University's Odum Library and described by the Cuneiform Digital Library Initiative, supplies the names of three messengers: Agu'a, Ikalla and Dugamu. The account distinguishes two beer forms among its provisions. The modern catalog explains that beer compensation in the Ur III evidence was irregular, often connected with messengers on special service. That qualification changes the familiar picture of an entire ancient workforce routinely taking its wages in beer.[2]

The three names make the arrangement less abstract. An empire communicating with itself required someone to carry the communication. The provisions in this record are one trace of the material support for that work. We

cannot recover the travelers' conversation, the destinations of every journey or their opinion of the onions. We can recognize that official movement had costs, and that those costs were recorded through things people could eat and drink. Administration was dependent on bodies.

Calling this "payment in beer" is useful up to a point. Calling it a modern salary translated into liquid can obscure the arrangement. Compensation, travel provision, subsistence and hospitality overlap without being identical. A supply for a mission does not automatically establish a daily wage. Before turning an ancient allocation into an annual income, we would have to know its frequency, the obligations it met and what else the recipient received. A multiplication can be flawless while its historical premise is wrong.

The distinction between kinds of beer is equally tempting. "Good-quality beer" sounds like an invitation to construct a tasting hierarchy. One imagines a superior flavor, perhaps a stronger drink, perhaps a finer filtration. Yet the adjective enters our account through translation, and the administrative category has its own purpose. Quality might be related to materials, value, concentration or recognized product distinctions. It need not coincide with what a present-day judge would reward in a glass.

A messenger tablet in the Maxwell Museum in New Mexico makes the differentiation visible. Its published transliteration and translation list individual allocations, distinguishing good-quality beer from beer without that description, and recording different amounts of bread and beer. The text belongs to the reign of Shu-Sin. Here, at least, the archive is not presenting one undifferentiated beverage handed in one identical portion to every traveler.[3]

There is enough to investigate an institution. There is still not enough to write a tasting note.

This is where numbers can be especially deceptive. Numbers look like an escape from linguistic ambiguity. A quantity seems solid even when a name is unfamiliar. If we can establish the amount of grain associated with a quantity of beer, it feels as though we are very close to a brew sheet. The remaining step would be to scale down the ancient operation into a kitchen pot.

Peter Damerow's analysis of Sumerian accounts shows why the calculation needs another question: what is the number doing? Some records express a product's value as an equivalent quantity of barley. An equivalent is an accounting relationship. It is not necessarily a statement of what physically went into the product. His discussion of these conversions is a reminder that the material named beside a number can be a unit of value rather than an ingredient.[4]

A modern analogy helps. Suppose a receipt values a meal at the price of three loaves. It would be a mistake to conclude that the cook used three loaves in making it. Even if the cost of bread helped determine the comparison, the receipt and the recipe remain different documents. Likewise, converting one grain into the value of another does not require anyone to transform one plant species into another. A system can bring unlike things into a common account without making them physically alike.

Different rates may still help distinguish products, and repeated records can reveal regularities that a single entry cannot. Identifying what each equivalence measures makes those comparisons possible. The arithmetic deserves as much attention as the clay.

In a published Metropolitan Museum catalog, another messenger text from Umma is laid out in a modern table. It distinguishes provisions for named travelers going toward a frontier from those of a returning traveler, then gives totals for the commodities. The presentation makes an ancient account unusually legible to an eye trained on rows and columns. But the helpful modern grid is itself an act of scholarly interpretation. It organizes a transliteration and translation; it does not replace them.[5]

Beer appears in administrative writing considerably earlier than the Ur III messenger records. A British Museum tablet dated to around 3100–3000 BCE records beer rations using an earlier writing system. The museum identifies beer through a sign resembling a pointed vessel, and an eating sign through a bowl directed toward a head. This object is separated from the messenger tablets by roughly a millennium. The distance matters: early Mesopotamian records should not be compressed into a single office with an unchanging procedure.[6]

Still, the repeated appearance of beer in accounts changes its place in our mental picture. A drink can be enjoyable and also be something that must be counted, authorized, allocated and transported. The archive records that second life unusually well. It is less interested in the sensation of drinking than in the relation between a supply and a recipient. A person can be present as a name while absent as a voice.

The brewers can be absent too. In the messenger records we have examined, a finished allocation has come to the fore. The making has moved out of the frame. Grain had to be obtained and prepared; vessels and labor had to be available; someone had to organize the result. Those are necessary conditions of a supply, but the record does not automatically identify who fulfilled them. We should resist assigning the work to a familiar stock character simply because the recipient is easier to see than the maker.

There is another journey to consider: the tablets' movement into modern collections. The Metropolitan's small Ibbi-Sin record was a gift in 1985. The Maxwell Museum identifies a 1967 donation and placed its messenger tablet in an exhibition concerned with cultural heritage and displaced artifacts. These are parts of the objects' modern histories. They do not supply an excavation context where one is missing, and they do not erase the difference between an ancient place of use and a present place of custody.[13]

That distance need not make the encounter cold. There is something strikingly intimate about a record that never intended intimacy. A person entitled to provisions was distinguished from another person. Someone checked the amount. A date mattered. The task was ordinary enough to be written down economically, and consequential enough to preserve in a durable material. Its survival has made an administrative moment available long after the institution that needed it disappeared.

The mystery is therefore not that the tablet refuses to reveal the taste of beer. It is that we keep expecting an account to become a recipe because the account contains a drink. Read on its own terms, it offers another kind of history: how beer participated in the movement of people, the expression of value and the practical running of a state.[7]

The three-centimeter tablet has done its job rather well. More than four thousand years later, we can still see that the messengers were to receive provisions. The fact that we cannot also brew their beer is not a defect in the paperwork.

04

What the Pottery Remembered

In 2004, a team studying pottery from Jiahu, a Neolithic site in China's Henan province, proposed a drink made with rice, honey and fruit. In 2026, another team returned to the site's brewing problem and gave a prominent role to molds. The first account names things you might put in a pot. The second asks what could help transform them. Both belong to the same investigation.[12]

The apparent change is less about replacing one ingredient list with another than about asking what makes an ingredient list work. Rice, honey and fruit can name materials without explaining the transformations that put them into a drink. A molecule surviving in pottery and the remains of a microscopic organism answer different questions. Jiahu is an unusually good place to watch those answers meet.

The 2004 research, led by Patrick McGovern with Chinese and other international collaborators, examined organic residues absorbed into ancient pottery. Its argument combined chemical results with archaeological and botanical context. It proposed fruit that could include hawthorn or grape, rather than identifying a single fruit beyond doubt. Crucially, the researchers had not detected surviving ethanol. They inferred fermentation from the combination of ingredients and their properties.[1]

That last distinction deserves to be kept in view. Ancient drink is commonly reported as though someone has uncorked a bottle and sent its contents to a laboratory. Jiahu's pottery offered traces of what had contacted or penetrated its surfaces. The drink itself had disappeared. An absence of ethanol in material this old is hardly surprising, but it means the scientific argument must be reconstructed from other evidence. We are examining what a container retained, not tasting what a container held.

The chemical approach is a kind of comparison. A sample produces signals; researchers ask which substances can account for them. Different methods can separate components or reveal features of their molecular structure. Candidate

ingredients must fit the signals and the archaeological setting. Agreement across methods strengthens an interpretation, while a signal shared by several possible sources demands more work. Chemistry can narrow the field remarkably without giving every molecule a unique postal address.

The early report illustrates the problem. Tartaric acid helped raise the possibility of fruit, but the authors discussed more than one possible source. Wax-like compounds supported an argument involving honey while leaving room for chemically similar plant materials. These qualifications were present in the research, even though "rice, honey and fruit" was the version easiest to remember. A list is portable. The reasoning behind a list is heavier luggage.[1]

The public impact was immediate enough to be dated. The University of Pennsylvania announced the findings on 8 December 2004. Its account also described much later liquids preserved in tightly lidded bronze vessels. Those liquids were a separate set of evidence, separated from Neolithic Jiahu by thousands of years. Put both discoveries into one short story and it becomes easy to transfer the wonderful survival of liquid in bronze to the much older pottery. The actual materials require us to keep the investigations apart.[3]

Now consider what the three proposed ingredients imply. Honey and fruit can supply fermentable sugars directly. Rice is rich in starch. To turn much of that starch into alcohol, a brewing process needs an effective way to break it down. Naming the grain leaves a biochemical question unanswered. This is where microscopic organisms become more than accidental guests.

Some molds produce enzymes that convert starch into smaller sugars. Yeasts can ferment suitable sugars as they become available, so conversion and fermentation can overlap. The tasks need not be performed by one organism. Nor does starch conversion have to depend exclusively on sprouting the grain, the route familiar from malt. A *qu* is a grain-based fermentation starter whose molds, yeasts and bacteria can perform complementary work. Mixed into other prepared grain, it brings organisms and enzymes to the material being transformed. In such a system, mold is part of the proposed technology. Treating every mold as evidence that food has gone wrong would prevent us from understanding how the food was made.[4]

Li Liu and colleagues' 2026 Jiahu study examined microfossils in sixteen pottery sherds. They reported starch, plant silica bodies called phytoliths, and fungal remains, including structures assigned to *Monascus*. The combined evidence led them to propose predominantly mold-assisted starch conversion, with signs compatible with malt-based processing as well. They did not require

honey and fruit to disappear from the picture; those materials could still have supplied additional sugars.[2]

At first glance, a microscopic shape seems more tangible than a chemical signal. Here is something that once had a structure. Yet identification has its own demands. A damaged granule can lose the features that distinguish one plant from another. A cell's outline may resemble that of a familiar yeast without establishing its species. Recognizing a possible organism is also different from reconstructing every member of an ancient fermentation community. A few surviving forms cannot give us a complete roster.

The newer paper makes a further boundary explicit: it did not perform a parallel modern chemical reanalysis of the original pottery samples. It offers a complementary line of inquiry. We should therefore avoid describing the research as two laboratories testing one identical ancient portion and producing contradictory results. The shared point is the site and the historical problem. The samples, methods and questions must still be specified.[2]

Two other Chinese investigations help explain the value of the additional approach. In work published in 2016, Jiajing Wang and colleagues studied residues from pottery associated with pits at Mijiaya, in Shaanxi. Their evidence included starch, phytoliths and chemistry, together with equipment they interpreted as a brewing assemblage. They proposed a combination of cereals and other starchy plants. This was a later site, around five thousand years old, with its own materials and context. Its relevance here is methodological: an interpretation can draw on the arrangement of equipment as well as on what survived inside it.[5]

At Qiaotou in southern China, research published in 2021 examined pottery from a platform mound associated with burials. The team used microfossil evidence to propose fermented drinks involving rice and other plant foods, with mold-assisted processing. Again, this was neither Jiahu nor a duplicate of it. It showed why a researcher returning to Jiahu might ask questions that a chemical ingredient list had not fully answered. Neighboring problems can suggest a new examination without dictating its result.[6]

There is a risk in lining up these finds on a map. The line connecting points can start to look like a route, and the route can become a story of invention and transmission. Similar evidence might indeed support historical connections, but shared materials or processes do not demonstrate who learned from whom. Establishing movement needs chronology, geography and further evidence. An analytical resemblance is a clue to investigate, not a family tree already drawn.

Even within one site, the contents of a pot can be harder to separate than they first appear. An object may have been reused. Materials could have reached it at

different times. A vessel involved in brewing might later have held a food containing fermentation residue. These are possibilities to evaluate against the context, not reasons to assume that every result is contamination. The container preserves an accumulation; the researcher is trying to recover events.

One useful test is to ask how the account would change if an ingredient vanished from it. Suppose future work weakened the evidence for honey. That would matter to the proposed mixture, but it would not automatically erase every indication of cereal processing. Suppose a particular fungal identification were revised. That would affect a proposed mechanism, while leaving other plant traces available for study. Claims should be connected closely enough to support an argument but not fused so tightly that the correction of one requires burning down the whole account.

The research at Jiahu has moved from identifying a plausible combination of materials toward investigating how some of those materials became available for fermentation. The questions have become more specific, and the surviving traces have been asked to do different work.[7]

What emerges is a less familiar image of a brewery. Its decisive machinery need not be a large, visible object. Grain can be transformed by organisms working alongside one another, with human practice arranging the conditions. The most consequential part of the operation may once have been too small for its makers to see. Thousands of years later, it has become part of what makes their skill visible to us.

05

The Last Service

The drinking vessels were broken. The building had burned. Yet the excavators of the brewery at Cerro Baúl, in southern Peru, did not interpret this combination as an ordinary accident or a sudden attack. They proposed an organized ending: brewing, feasting and drinking, followed by deliberate destruction as the inhabitants withdrew. A place made to bring people together had been taken apart in a final act of gathering.[1]

It is a striking way to close a brewery. It is also a difficult one to investigate. Ash can establish a fire without identifying the person who lit it. A broken cup can record a fall, a blow or an act of deliberate sacrifice. The interpretation has to come from the arrangement and sequence of the remains. This case begins with destruction and works backward toward the institution that was worth destroying.

Cerro Baúl is a mesa in the Moquegua region. During the first millennium CE it became a Wari center, near communities associated with the rival Tiwanaku polity. These were states that preceded the Inca; calling the site an Inca brewery would collapse distinct histories. Michael Moseley and his colleagues described the settlement as serving an embassy-like delegation, with monumental buildings and the agricultural support needed to maintain occupation. Their account places the brewery within a long-established frontier enclave, rather than treating it as an isolated place that happened to make a great deal of drink.[1]

The distinction changes the question. A large brewery at a political center can do more than supply a thirst that already exists. The occasions on which its product is served can help define relationships. Who hosts, who is invited, where people gather and which vessels they use can all be part of the transaction. The drink has to be understood alongside the setting that makes its distribution significant.

First, however, someone has to make it. The excavated building contained separate spaces interpreted as areas for milling, heating and fermentation, together with hearths, grinding equipment and large ceramic vessels. The original researchers estimated a capacity of roughly 1,800 liters per batch. That figure is

an archaeological reconstruction from the installation and vessels, not a surviving record of what every batch actually yielded.[1]

Even as an estimate, it forces a change of scale. Water weighs about a kilogram per liter. A volume in that range represents close to two tonnes of water before we consider ingredients, fuel and containers. The position of the brewery on a summit makes the apparently simple statement "beer was served" conceal a considerable amount of movement. A feast's spectacle depended on work that began before its guests arrived.

Archaeologist Donna Nash has emphasized this labor in explaining the site and its experimental investigation. Her account describes bringing water and materials up to the brewery, and the practical effort involved in producing chicha. She also points to *tupu* pins, associated with women's dress, among the archaeological evidence used to infer the involvement of elite women in brewing. These are clues about the people at work, not a list of names or a set of eyewitness portraits.[2]

The pins complicate a familiar division between rank and manual work: here the interpretation places high-status women within a demanding productive activity. Who was entitled to make a valued drink could matter alongside who received it.

The drink itself has another surprise. *Chicha* is a name applied to diverse fermented beverages, and the Cerro Baúl evidence does not reduce neatly to maize alone. Molle, the Peruvian pepper tree, matters here. Its small fruits can provide sugars for fermentation. The archaeological and experimental work has considered maize preparations, molle preparations and mixtures of the two. The proportions and exact practices of every ancient batch remain beyond recovery.[3]

For a reader used to barley and hops, the fruit can look like an exotic addition. That is a poor starting point. The relevant question is what locally available materials enabled brewers to do. A sugar-bearing fruit and a starchy grain present different processing problems. Molle need not be understood as a flavoring added to a presumed European beer. It can be central to the beverage and to the knowledge required to make it.

Research published in 2019 by Patrick Ryan Williams, Nash, Joshua Henkin and Ruth Ann Armitage examined both the ceramics and the traces within them. The team found evidence consistent with molle in the pores of brewing and serving vessels and argued for regional sources of ceramic raw materials. This brought two supply questions into one account: where the containers came from and what went into them. The authors connected local resources and a drought-tolerant ingredient with the resilience of the feasting institution.[4]

The important object is therefore not just a cup, considered as a finished piece of pottery. It belongs to a chain of decisions about clay, manufacture, imagery, contents and use. If replacements can be made from materials available in the region, the system depends less on uninterrupted shipments from a distant center. If a beverage can be produced from plants suited to the local environment, that may help the institution continue through some kinds of stress. These are practical advantages, not a guarantee that political relationships will survive indefinitely.

The popular shorthand is that beer held an empire together. It has the convenience of fitting on a label. The research asks a narrower and more interesting question: how could repeated occasions of making and sharing a particular drink help people enact political relationships? A gathering can be hospitable and unequal at the same time. Receiving something from a powerful host need not make the recipient powerless, but the circumstances of the gift can matter. A feast is an event whose social meaning is not measured by its volume alone.

This makes the closing act more legible. If brewing and serving helped sustain the enclave's relations, destroying the facility and its vessels could mark an ending to that arrangement. The excavators' ceremonial-closure interpretation draws force from the broader pattern of evacuation and the particular treatment of objects. We can follow that reasoning without claiming to know the exact belief or intention of each participant. Deliberate action can be recognizable even when its full explanation is not.

The fire also changes what survives for archaeology. An operating brewery is a place where containers are moved, repaired and replaced. The assemblage left at closure may be unusual precisely because the ending was unusual. It offers a concentrated view of equipment and objects, while inviting the mistake of treating a final episode as a normal working day. The most vivid evidence is not necessarily the most representative evidence.

Later research pushed the question beyond the monumental site. In a study published in 2024, Nash, Matthew Biwer and Emilee Witte compared excavated deposits with experiments guided by knowledgeable brewers. They considered what residues different chicha preparations leave and argued that, in the Wari-affiliated settings they examined in Moquegua, production was largely specialized and state-supported. Within those sampled settings, their evidence challenged the expectation of routine household brewing; it did not establish what every household across the Wari world did.[3]

Experiments of this kind depend on contemporary knowledge. A brewer who knows how to handle a fruit, judge a mixture or separate useful liquid from residue is contributing expertise that a laboratory instrument does not replace. At the same time, present-day practice should not be treated as an unchanged recording of the first millennium. Its value is practical and comparative. The people involved are collaborators in solving a problem, not exhibits demonstrating an ancient past.

Cerro Baúl has also inspired modern beer. The Field Museum described its collaboration with Off Color Brewing on a drink using purple maize and molle. The museum's account says that the brewers developed and selected among recipes. This is a legitimate way to explore and communicate an archaeological subject, provided "inspired by" keeps its meaning. A contemporary glass can invite interest in the site without posing as a sample of its last batch.[5]

The final service at the ancient brewery cannot be repeated. We have no guest list for the closing feast, no speech announcing departure, no agreed explanation for every object left behind. What we have is more material: an installation built for substantial work, vessels and ingredients linked to a political institution, and a destructive ending that the excavators interpret as planned.[6]

To destroy an ordinary cup is to lose a useful object. To destroy a cup that helps enact an affiliation may also be to terminate a relationship. At Cerro Baúl, the broken vessels make the institution visible from its far end. We meet the brewery at the moment it stopped being available for another gathering.

06

The Inventor Arrives Late

Samuel Whitbread bought a steam engine and kept the horses. The engine, ordered in 1784 and installed the following year at his London brewery, survives in the Powerhouse collection in Australia. Its gears and shafts drove equipment for crushing and moving malt, pumping, and stirring. Yet the brewery retained its horse wheel as insurance against mechanical failure. Modernization arrived with a backup plan.[1]

That practical arrangement is a useful entrance to the history of porter. A new machine could enter an existing brewery without creating the brewery, its market, or its beer. Its date of installation tells us something exact and important, but a great deal had already happened before someone signed the order. To begin the whole story with the engine would mistake one visible change for the system that made the change useful.

Porter's alleged invention presents a similar temptation. Here the memorable object is a man. In the familiar account, a London brewer named Harwood solves an irritating problem: publicans must draw three different drinks to produce a mixture called three-threads. He makes a single beer that supplies the combined taste from one cask. The result, associated with working porters, becomes porter. A chore disappears, a product appears, and the inventor has earned his place in history.

The trouble is the lateness of the complete story. Martyn Cornell's investigation traces its influential publication to John Feltham's *Picture of London* in 1802, decades after the supposed event. Cornell finds no contemporary support for the claim that porter was devised as a replacement for a three-cask mixture. Harwood was a real brewer; that does not make the narrated invention a documented occurrence.[2]

The distinction matters because a proper name gives an account an appearance of precision. We can picture a person conceiving an improvement more readily than we can picture an industry changing over several generations. The inventor story also flatters everyone involved. Customers know what they like, publicans struggle honorably to supply it, and a clever producer removes the

inconvenience. The transaction contains no argument about prices, no expensive warehouse, and no year of waiting for payment. It is an admirably portable story about a less portable commodity.

To get closer to the commodity, open a brewing book published much nearer the beginning of the eighteenth century. The second edition of *The London and Country Brewer*, dated 1736 and attributed to William Ellis, describes brown, amber, and pale malts. Its account of London brewing has workers moving the mash with oars and judging heated water by observation and touch. Brewing already demands coordinated labor, equipment, judgment, and distinctions between materials. Nothing in that picture resembles a world awaiting the discovery that beer could be brown.[3]

The book's author is an argumentative guide. He wants readers to judge drinks and improve brewing, and his explanations mix practical observations with ideas about health that should remain firmly in their historical setting. His preface prefers an experienced eye to a fixed boiling time. His chapter on keeping beer considers how storage conditions influence the drink after it enters a cask. The brewery door is therefore no boundary between making and merely waiting: what happens afterward still helps determine what the customer receives.

That is a different kind of invention problem. Suppose a brewer discovers that customers prefer a product after a longer rest. Even without changing the principal ingredients, the brewer must change the business. Space becomes necessary. Containers remain occupied. Money spent on a batch cannot immediately pay for the next one. A desirable flavor may be inseparable from an undesirable delay in getting paid.

An imaginary calculation makes the constraint clear without pretending to reconstruct any particular firm's books. A brewery selling one batch every month, with each batch stored for one month, needs room for roughly a month's production at a time. If the same sales require twelve months of storage, it needs roughly twelve batches awaiting sale. Real brewing accounts are more complicated, but the direction is unavoidable. Increasing the age of a product increases the resources committed before it earns its return.

This is why the origins of porter cannot be settled by asking only what went into a kettle. A technique can be widely understood and still difficult to afford. Two brewers can possess comparable knowledge while only one has the premises, cash, and credit to act on it. A recipe that omits those conditions can explain how a drink is made while failing to explain who gets to make it commercially.

Historian James Sumner follows that larger transformation. His study finds substantial London brewers before porter's conventional invention date, and

treats the drink's identity as something that developed alongside industrial concentration. He also follows the origin story through print: an account associated with the pseudonym Obadiah Poundage in 1760 was excerpted and circulated, while a more elaborate Harwood narrative appeared in 1802. Repetition eventually made borrowed accounts resemble independent agreement.[4]

One source copied into ten books remains one source for the copied claim. The ten books can, however, provide excellent evidence for something else: the success of the explanation. Its survival tells us that people wanted this sort of origin, and that publishers found it useful to supply one. A history of a drink and a history of what people say about the drink can occupy the same page while answering different questions.

Sumner's further argument concerns scale as a claim to authenticity. The great breweries' size became part of what made their porter seem special. His distinction is valuable: an enormous vessel can reduce costs or support a production method, but its impressive appearance can also persuade an observer that the product requires something unavailable elsewhere. Economic advantage becomes a story about the nature of the beer itself.

The argument should not flatten every producer into a calculating propagandist. A brewer might sincerely believe that a large vessel improved the drink. Customers might discern differences without understanding their causes. An explanation can be commercially convenient and honestly held at the same time. What needs testing is the step from an observation about one successful beer to a rule declaring that only a certain sort of establishment can produce the real thing.

Return now to Whitbread's engine. Unlike a moment of inspiration inside Harwood's head, it is a surviving object with a documented career. Even here, the material evidence needs interpretation. A photograph of a drawing, also held by the Powerhouse, preserves an inflated claim about the engine's horsepower. The museum's account explains that earlier figures of thirty-five horsepower, supposedly doubled to seventy, do not fit the other evidence. It gives a much smaller original output and a final rating around fifteen or sixteen horsepower.[5]

The important lesson is not that a museum caption defeats an old caption merely by being newer. The correction rests on comparing the claim with documents and the machine's physical characteristics. A confident number attached to an object is still a statement somebody made about it. The iron does not authenticate every sentence printed beneath its picture.

It also matters which date a description refers to. The engine was altered during its working life. Its later condition cannot simply be projected backward onto its arrival at the brewery. The same discipline applies to porter: a drink sold under an established name may change while the name continues. Continuity of the label cannot by itself establish continuity of color, ingredients, strength, aging, or the customer's understanding of what the label promises.

There is no need to replace the Harwood story with an equally tidy counterlegend in which one historian proves that no individual ever mattered. Brewing change depends on people trying things, deciding what to buy, persuading employers, managing failures, and teaching others. The absence of a secure single inventor does not dissolve their agency. It asks us to distribute attention more fairly among the kinds of work involved.

A porter history that includes storage makes room for the people who built and maintained containers. One that includes power makes room for the work the machine replaced, and the horses kept in reserve. One that includes publication makes room for writers whose explanations outlived the circumstances they described. These are connected histories; selecting only the most quotable one produces a narrower drink than the evidence supports.

Harwood can remain in the account as a brewer and as a figure in a remarkably successful origin story. He need not carry sole responsibility for a beer whose production and reputation changed over time. The more interesting question is how an industry came to be represented by one man's supposedly convenient idea. Once that question is asked, the late arrival of the inventor ceases to be an embarrassing gap. It becomes an event in its own right.[6]

PART II

Inside the Glass

Aroma, bitterness, light, time and foam turn the glass into a small laboratory. The explanations lead through things a brewer can control and things a drinker may never notice.

07

The Banana That Was Never Added

The accusation is easy to make. A wheat beer smells of banana; therefore someone has put banana in the beer. It has the neatness of a solved case, and the considerable advantage that the evidence can be inhaled. The ingredient list, which mentions no fruit, now looks evasive. Perhaps the brewer has found a loophole. Perhaps the taster has discovered a secret.

There is another suspect, too small to see without a microscope. Yeast can make the smell while working through a meal that contains no banana at all. The fruity impression is real. What has gone wrong is the inference that followed it.

Among the molecules associated with banana aroma is isoamyl acetate, an ester. An ester is a chemical arrangement, not a declaration of agricultural origin. The same molecular structure can occur in a fruit and in a fermentation tank. Once it reaches the nose, it does not present a passport. The nose encounters a stimulus; the mind supplies a familiar name. Banana is an excellent description of a resemblance and a rather poor audit of a brewery's purchases.[1]

The fruit itself offers a useful place to begin. In 1985, Miki Harada, Yoshinori Ueda and Takashi Iwata reported isolating an enzyme from banana pulp that helped produce isoamyl acetate. The enzyme joined isoamyl alcohol to an acetyl group supplied by acetyl-CoA, a molecule that carries chemical building material inside cells. Its activity increased as the fruit ripened. A banana was making its own characteristic aroma by biochemical work, rather than merely containing an essence called banana.[2]

This makes the apparent trick less mysterious. There is no rule reserving that piece of chemistry for a particular plant. Different living things can arrive at the same molecule. A recipe names what enters a process. An aroma may describe what the process has made. Confusing these two lists is like examining the bricks in a finished house and expecting to find a house among the builder's deliveries.

In 2003 Kevin Verstrepen and colleagues took the investigation inside yeast. They altered genes associated with enzymes that make acetate esters, then

measured the volatile compounds produced during fermentation. Deleting both *ATF1* and *ATF2* from the laboratory strain eliminated its detectable production of isoamyl acetate under the tested conditions. Other esters did not all disappear. Increasing expression of the relevant genes in a commercial brewing strain changed ester production, and different versions of the genes did not behave identically.[1]

That is stronger evidence than finding banana aroma in a beer and finding yeast nearby. The researchers changed something specific in the organism and watched a particular product change. They also discovered why a broad label such as fruity can conceal several mechanisms: removing two important pieces of the machinery did not silence the whole aroma-producing system. One familiar adjective had compressed a complicated chemical operation.

Notice what this experiment did not require. No banana had to be concealed in the fermenter. Nor did yeast have to know what a banana was. We describe the result using the most recognizable experience available to us; the cell proceeds without consulting our vocabulary. In the brewery, the useful question becomes which culture, working under which conditions, produces the desired balance. The presence of a species name on a packet cannot answer that by itself. The experiment's differences between genetic variants are already enough to defeat that shortcut.

The quantity of a compound is another part of the puzzle. An analytical instrument can detect a substance without establishing that it dominates the experience of drinking the sample. Imagine a room in which several people are speaking. Identifying one speaker does not tell you whether that person can be heard from the doorway. Likewise, a plausible aroma molecule is a candidate explanation. Its concentration, the other compounds present and the sensory test determine how persuasive that explanation becomes. A chemical list and a tasting description are complementary forms of evidence.

This distinction matters especially when bananas actually are involved. In Uganda, Tonto is made by fermenting banana juice. A 2026 study by Ambrose Atwine and colleagues examined samples from Ntungamo and Kalungu districts and connected chemical measurements with sensory assessments. The researchers identified thirty-eight volatile compounds, principally esters and alcohols, and reported differences among the sampled products. Banana-like and fruity descriptions were prominent in some samples, alongside a varied chemical profile.[3]

Here, an ingredient record and an aroma description can point in the same direction. But the drink's aroma still cannot be reduced to the smell of its original

juice. Fermentation adds its own work. The comparison with wheat beer is valuable precisely because the production histories differ. It would be absurd to conclude that a Ugandan banana ferment and a European wheat beer are secretly the same beverage because one adjective can describe both. Shared chemistry does not erase agricultural choices, inherited methods or the names by which makers know their drinks.

The Tonto research also illustrates how much work sits behind a credible sensory claim. A trained descriptive panel and an acceptability panel answer different questions. Describing an aroma does not establish that people prefer it, and preference in a particular group does not become a judgment binding on everyone else. Researchers have to keep these tasks apart even when casual conversation rolls them into a single verdict: that smells good. The distinction protects both the experiment and the drink from being made to say more than they do.[3]

Banana has an accomplice in many discussions of wheat beer: clove. This time the central molecule is commonly 4-vinylguaiacol, which can arise when ferulic acid from cereal material is transformed during brewing and fermentation. Stefan Coghe and colleagues followed those changes in beers made with differing proportions of barley malt, wheat and wheat malt. Their 2004 study found that strains able to produce phenolic aromas generated more of the compound during fermentation than strains lacking that behavior. Boiling also contributed to its formation.[4]

A clove jar is unnecessary to explain this route. Yet the lesson should not become a new reflex: banana always means yeast, clove always means yeast, case closed. Brewers can use fruit and spices, and the same sensory word can arise from a mixture of causes. The useful correction is to keep the alternatives open until evidence distinguishes them. A recipe may settle an ingredient question. A fermentation trial may establish a biological pathway. A tasting note alone usually cannot do both jobs.

There is a further question hiding inside the banana case. Why make these volatile compounds at all? The obvious answer from the drinker's side is that they improve the beer. This is a flattering explanation of evolution, with humanity installed as its intended customer. It also leaves yeast waiting through a very long prehistory for a brewery to open.

Joaquin Christiaens and colleagues tested a different possibility in research published in 2014. They compared yeast with a functioning *ATF1* gene and yeast from which it had been deleted. The altered aroma changed the response of fruit flies: the insects were less attracted to the mutant yeast, and they dispersed its cells

less effectively. The research connected a molecular change, an insect's sensory response and the movement of microbes.[5]

The experiment gives the aroma an ecological setting. Yeast cannot fly, but some of the creatures that encounter it can. A volatile signal may help bring an insect close enough for cells to be carried elsewhere. This does not establish that every yeast ester arose solely to advertise to flies, or that every fruity fermentation repeats the tested relationship. It does make the human-centered explanation seem provincial. A smell we enjoy may participate in interactions that have nothing to do with us.

The flies return us to the smell with a different question. A volatile compound has left a cell and reached another organism. What happens next depends on the receiver: an insect may approach, a drinker may think of fruit, an analyst may identify a chemical structure. The cell's product can participate in all these encounters without acquiring one exclusive meaning.

There is no need to abandon ordinary language at the end of this journey. A drinker should not have to announce the detection of an acetate ester when banana will do. Tasting vocabulary works because it borrows from experience. The discipline lies in knowing when the description has been asked to become something else. Saying banana-like records an impression. Saying bananas were added makes a claim about events outside the glass.

The beer has not deceived anyone. It contains the chemistry it contains, and the aroma arrives without commentary. The small fraud was supplied by an explanation that seemed so natural it escaped inspection. Once that explanation is removed, the pleasure remains. The banana is still there to smell. Only the banana itself has gone missing.[6]

08

Two Beers, One Number

One beer measured 47.5 bitterness units. Another measured 49.9. To someone choosing between them, the figures suggest near neighbors: roughly fifty, both of them. Then the chemists measured a major family of bitter compounds, the iso-alpha acids. The first beer contained 39.9 parts per million. The second contained 2.2.

The figures come from research by Toru Kishimoto and colleagues at Japan's National Research Institute of Brewing, published online in 2021 and in a 2022 journal issue. They are an awkward opening for a statistic that appears so confidently on beer menus. An almost matching pair of numbers had concealed very different chemistry.[1]

The tempting verdict is that bitterness units are nonsense. That would be another mistake, and a less interesting one. The number did not fail to count what it was supposed to count. People had asked it an additional question without noticing: how will this beer feel in my mouth? To understand the difference, we have to follow the sample into the laboratory, where no tongue is involved in the first measurement.

The American Society of Brewing Chemists describes the method in a public summary. Bittering substances are extracted from a sample, and the resulting solvent layer is examined by a spectrophotometer. The instrument measures how much light the extract absorbs at a specified ultraviolet wavelength, 275 nanometers. A calculation turns that measurement into bitterness units. The procedure is useful for comparing production runs and checking consistency. It is an analytical estimate, with a defined way of obtaining it.[2]

An absorbance measurement is not an inventory of individual molecules. Different mixtures can give similar totals. A useful analogy is the weight of a shopping bag: two bags can balance on a scale while containing quite different dinners. The scale has performed honestly. The error begins if we treat the weight as a recipe. In beer, the extraction and optical response collect contributions from more than one kind of compound.

Kishimoto's team examined those contributions. Their abstract identifies iso-alpha acids, unconverted alpha acids, oxidized polar compounds and malt-derived material among the components affecting the reading. In beers made with deliberately deteriorated hops, the relationship changed markedly. Samples could have matching bitterness-unit values while differing in perceived bitterness intensity. The study provides a counterexample, rather than a new universal formula: a particular analytical total can be reached through different chemical routes.[1]

Those beers defeat the rule that equal units guarantee equal sensation. They do not supply a correction factor for the next two beers on a menu. To build one, we would need to know what contributed to each reading and how the resulting mixtures were perceived. The next stage of the investigation is therefore to separate some of the contributors.

The bitter compounds themselves complicate matters further. Hops contain chemical precursors that change during processing. One familiar transformation during wort boiling converts alpha acids into iso-alpha acids. Oxidation can produce other bitter substances, including humulinones and hulupones. These names are unlikely to improve conversation at a crowded bar, but they help explain why a single total may disguise differences that a mouth notices.

Victor Algazzali and Thomas Shellhammer investigated the bitterness of those oxidized compounds by preparing extracts and adding them to unhopped lager. A trained panel compared the resulting samples with beers containing iso-alpha acids. In that experiment, humulinones and hulupones were bitter, but their measured relative intensities were lower than those of the iso-alpha acids. The researchers reported estimates, with uncertainty, for their specific sensory comparison.[3]

There is a valuable restraint in leaving the numbers in the paper rather than turning them into a pocket conversion chart. The experiment separated selected compounds and tested them in a particular beer. A commercial drink is a more crowded place. The result establishes that different bitter substances cannot simply be assumed to contribute identical sensory effects at identical concentrations. It does not authorize us to add up every component of a finished beer and announce exactly what every drinker will experience.

So far, our problem has concerned the amount of bitterness. But bitterness also has a shape in time. A taste may arrive promptly or build after the sip. It may recede or persist. Calling two experiences equally intense at one moment does not make their whole courses identical, any more than two roads reaching the same

height must have the same gradient. This gives investigators another question: can people distinguish bitter experiences after their intensity has been matched?

Molly Higgins and John Hayes tested that question in a beer model system. Their 2020 paper describes three experiments using a commercial nonalcoholic beer to which selected bitterants were added. The stimuli included a hop extract, quinine and sucrose octaacetate. The researchers first matched perceived intensity, then examined discrimination and liking. Participants could distinguish the tested bitter stimuli even when the samples were comparably intense. Yet the experiments found no significant difference in liking or ranking, and the descriptive checklist did not yield a clear verbal separation.[4]

That is a wonderfully inconvenient result for anyone who expects a sensation, a word and a preference to move together. People could notice a difference without reliably explaining it through the supplied vocabulary, and noticing did not establish a favorite. The study did not show that all bitterness is interchangeable. Nor did it reveal a universal hierarchy in which one bitter substance always tastes better. It separated three questions that ordinary tasting language often ties into a knot.

There is no contradiction in perceiving more than one can readily name. A person may recognize a friend's handwriting without being able to describe the exact curve that makes it familiar. Similarly, failure to select a useful adjective does not prove failure to detect a sensory distinction. For the researcher, the task given to the participant matters. Asking whether two drinks differ, how they differ and which one is preferred produces three kinds of evidence.

Research on commercial beers has approached the problem from another direction. In 2016, Olayide Oladokun and colleagues analyzed thirty-four lagers, then selected ten with contrasting chemical profiles for sensory assessment. The bitter character differed among beers associated with differing hop-acid and polyphenol profiles. The authors used terms concerning both quality and progression of bitterness, rather than treating every response as a point on one intensity scale.[5]

This is closer to the crowded reality of a shop shelf, but it creates a different limitation. Commercial beers vary in several respects at once. Their comparison can reveal relationships worth investigating without proving that one isolated molecule caused every sensory difference. The laboratory model offers control; the commercial selection offers variety. Neither needs to impersonate the other. Together they make the original menu number look useful but incomplete.

Even the nose enters what initially seemed to be the tongue's business. In a 2017 experiment, Oladokun and colleagues investigated beers made with three

hop varieties and examined the effect of hop aroma on perceived bitterness. Their findings linked aroma to changes in bitterness intensity and character. The experience called bitter was responding to more than a chemical stimulus encountered by taste receptors in isolation.[6]

This is not evidence that the senses are broken. A drink arrives as a combined event: smell, taste and physical sensation occur together. Experimental separation helps us understand the contributions. It does not oblige ordinary experience to remain separated afterward. The surprising feature is that a number with a scientific appearance can persuade us to forget this complexity even when we would never make the same demand of other foods. Few people expect two soups with the same sodium concentration to taste alike in every respect.

The brewer still needs measurements. A repeatable method can help detect a change between batches, support a specification and prompt an investigation. Discarding it because it does not reproduce a human experience would be like throwing away a thermometer because it cannot describe a room. The method's usefulness depends on remembering its job. It becomes misleading when a convenient estimate is promoted into a complete account of perception.

Nor should the taster claim automatic superiority over the instrument. If two beers feel different, that observation deserves attention; it does not reveal their exact chemistry. A persuasive investigation uses the disagreement to ask better questions. Were the same compounds present? Were their proportions similar? Was aroma different? Did the tasting task separate intensity from preference? Evidence accumulates when the competing descriptions are allowed to coexist long enough to be examined.

Return to the two Japanese samples. Their bitterness-unit readings remain 47.5 and 49.9. There is no need to cross either figure out. Beside them sit the strikingly different iso-alpha-acid concentrations and the larger story of what an extract, an instrument and a person can each reveal. The mystery was never that one of them had to be lying. It was that the same word, bitterness, had been assigned several jobs.[7]

09

The Culprit Was Light

The decisive witness in one beer investigation was a particle that would never survive long enough to be bottled. In a 2001 experiment, Colin Burns and colleagues used laser pulses and electron paramagnetic resonance to examine what happened when light struck hop-derived bittering compounds. Their subject was the route toward an aroma politely described as lightstruck and less politely described as skunky. The apparatus was listening, in effect, for the fleeting chemistry between the beer as made and the beer as smelled.[1]

There was no skunk in the experiment. There need not be one in the explanation. The animal lends its name to a sensory resemblance, just as banana does in a wheat beer. But this case differs from the banana case in a crucial respect. A desirable aroma may be produced during a planned fermentation. Here, an exposure after brewing can create a smell that the brewer did not intend. The ingredient list is unchanged while the contents cease to be quite the same.

The best-known offender has an uninviting name: 3-methyl-2-butene-1-thiol, usually shortened to MBT. The thiol part identifies a sulfur-containing chemical group. Small amounts can have a conspicuous sensory effect. A 1993 study by A. J. Irwin, L. Bordeleau and R. L. Barker estimated a threshold of seven nanograms per liter in the beer and test conditions they used. A nanogram is a billionth of a gram. This is a drama in which the central character occupies very little space.[2]

That threshold is a measured result, not an immutable boundary built into every nose and every beer. It nevertheless explains why the damage can be disproportionate to the quantity formed. A beer does not need to turn visibly cloudy, change color or lose a measurable portion of its bulk before a new odor becomes noticeable. The sensory world has no obligation to allocate attention in proportion to mass.

To understand how MBT appears, start with a useful ingredient. Hop-derived iso-alpha acids contribute bitterness. Their structures also contain portions capable of taking part in photochemical breakdown. Light is energy, and a molecule that absorbs it may enter an excited state from which reactions become possible. The relevant question is therefore not simply whether a bottle has

become warm. A beam of light can do chemical work that a thermometer fails to describe.

Burns's team examined a direct light-driven pathway in a model solvent system. Their laser operated at a specific ultraviolet wavelength, and the researchers analyzed short-lived radicals, reactive intermediates with unpaired electrons. The method helped distinguish where energy was absorbed from where a bond subsequently broke. Their findings were a way into the mechanism, not a simulation of every bottle on every shop shelf.[1]

The distinction between a model and a bottle is where the next piece enters. Beer contains riboflavin, also known as vitamin B2, which can absorb light and participate in reactions involving other molecules. A substance familiar from nutrition becomes a participant in flavor damage. Chemistry is indifferent to whether we have placed a compound in the good column of a dietary chart. Its behavior depends on the company it keeps and the energy it encounters.

In research published in 2005, Kevin Huvaere and colleagues investigated interactions between light-excited riboflavin and hop bittering compounds. They trapped reactive intermediates and examined their signals, supporting a pathway from isohumulones toward precursors of MBT. Reduced hop derivatives behaved differently; their reactions could generate precursors associated with other flavor changes. Preventing one particular route was not equivalent to granting the beer immunity from all photochemistry.[3]

The missing sulfur had to be accounted for as well. In a 2006 paper, Huvaere and colleagues followed the light-driven degradation of sulfur-containing amino acids and related substances in the presence of flavins. They combined kinetic observations, radical detection and identification of reaction products. Their findings connected sulfur-bearing intermediates with a carbon-containing fragment derived from the hop compounds, supplying the pieces needed for the characteristic lightstruck molecule.[4]

The light did not have to be ultraviolet. Visible light drove the relevant chemistry in the flavin experiments. This widens the question a protective package must answer. Saying that a material blocks ultraviolet rays does not describe everything it lets through. The useful boundary is drawn by the molecules' response to the transmitted light, rather than by a familiar division in the spectrum.[4]

The explanation is more satisfying than the vague idea that sunlight makes beer go bad. It specifies materials already present, energy arriving from outside and a pathway by which new molecules can form. It also shows why an unopened package is not necessarily a protected package. A seal controls the

passage of material through a closure; transparent walls admit energy without opening a hole. Nothing needs to crawl under the cap.

Following the mechanism into finished beer brings a practical question: what can investigators measure in the more crowded liquid as light exposure proceeds?

Richard Pozdrik and colleagues took a useful step toward the actual drink in 2006. They examined mainstream lager and model beer, tracking the disappearance of riboflavin's absorbance at 445 nanometers during light exposure. That change was associated with the development of lightstruck character. Several tested additions reduced the changes but did not stop them. In the model system, increasing isohumulone could diminish the protective effect of some additions.[5]

There is a modest but important lesson in that last complication. Adding a substance called an antioxidant does not automatically stop every undesirable reaction under every condition. A brewer needs an effect demonstrated in the relevant formulation, exposure and sensory outcome. The experiment offers evidence of interacting variables, not a kitchen remedy. A promising ingredient cannot be promoted into a guarantee merely because its general chemical description sounds reassuring.

The bottle then becomes part of the investigation. It is tempting to arrange glass colors into a simple league table and finish the chapter there. Amber is usually a sensible clue, but a color name conceals a spectrum. Two objects described as green may transmit rather different amounts of light at the wavelengths that matter. Thickness also changes the route light takes through the material. The apparent color is what our eyes receive, not a complete report on what the contents receive.

Barry Hucker and colleagues measured transmission through bottle glass of several colors and thicknesses in work published in 2016. Their amber samples strongly restricted relevant light, while clear and bright green glass transmitted more. Darker green glass behaved differently from bright green. Within the amber group, thicker glass obstructed more light than thinner glass. These measurements make a more useful argument for protection than the claim that every bottle of one color must behave identically.[6]

They also defeat a small piece of bottle folklore. Green glass cannot tell you how old a beer is. It gives information about a container, and even that information is incomplete without its actual transmission properties. A bottle kept in darkness and a bottle placed under light may look alike before opening. Their exposure histories can differ. Age and light damage are separate variables, although real storage may combine them.

From this perspective, the package is neither an innocent wrapper nor a universal villain. It mediates an encounter between the surroundings and the drink. That is true even when its surface advertises a tradition or displays an attractive color. The same object can perform a cultural job, a commercial job and a physical job, with no requirement that those jobs agree. A beautiful view of the beer may be exactly the route by which unwanted light reaches it.

This does not mean that every sulfurous smell is MBT, or that a familiar aromatic impression proves one mechanism without further evidence. It means that a particular explanation has earned its place through experiments. If the smell is suspected to be lightstruck, the exposure, formulation and analytical result matter. If it is ordinary aging, another investigation is needed. Placing everything under a single heading such as stale would hide the useful differences.

The glass measurements complete the route from the surroundings to the smell. Exposure supplies energy; the package controls how much reaches the beer; molecules inside participate in reactions; a small amount of their product can become perceptible. Each link gives investigators something specific to test when a bottle develops the suspected aroma.

The cap, meanwhile, can remain perfectly tight. That is what gives this case its pleasingly unfair conclusion. The brewer may have guarded the liquid against a host of material intruders while the culprit entered through the walls, carrying no substance of its own. It was light, doing what light does when an accommodating molecule is waiting on the other side.[7]

10

The Bottle Kept Changing

An unopened bottle is a persuasive object. Its cap promises that nobody has interfered with the contents. Put it somewhere dark and another obvious suspect, light, is removed. When the beer later smells tired, papery or simply less alive, the temptation is to look for a breach. Something must have got in.

Perhaps it did. But a package can be closed without its chemistry being finished. The brewery has sealed a mixture of substances capable of reacting, binding, separating and disappearing. The bottle is a container, not a pause button. To investigate what changed during storage, researchers have had to follow clues backward into the brewery and sometimes all the way to the malt.

Oxygen deserves its reputation as a suspect. In 1991, A. J. Irwin, R. L. Barker and P. Pipasts reported that traces of copper accelerated flavor staling in their beer experiments. Their work examined oxidation reactions and the participation of other compounds that allowed the metal to cycle through different chemical states. The effect was not captured by the crude picture of a lump of copper spoiling a bottle; small quantities could influence a chain of reactions.[1]

This is the difference between a reactant and a catalyst. Some substances supply material for a product. Others alter the route or speed by which products form. In a complicated liquid, asking how much oxygen is present is consequently important but insufficient. What else is there, what has already happened and what conditions the package experiences can change the outcome. Two bottles with the same present oxygen measurement need not have the same chemical past.

Oxygen also need not have entered after the cap was fitted. A sealed bottle may already contain it in the liquid or in the space above it. As reactions consume available oxygen, a later measurement cannot be assumed to reproduce the amount present earlier. The cap's integrity therefore answers one question while leaving others open. There is no contradiction between good sealing and a changing flavor, just as a closed cupboard can contain food that continues to change.

Brewing analysis recognizes this distinction in its vocabulary. The European Brewery Convention has a method for total package oxygen that includes both the dissolved portion and the headspace portion. Measuring the liquid alone does not automatically count everything in the container. The method is another reminder that the apparently empty space above the beer belongs to the sample too. A laboratory definition can reveal a missing piece of the object that ordinary inspection treats as complete.[2]

Many investigations focus on aldehydes, a chemical family that includes compounds associated with several aging aromas. The word does not mean cardboard. Different aldehydes have different sensory properties, and the experience depends on concentration and surroundings. Researchers must identify which compounds change, not merely announce that the beer contains a family with a suspicious reputation.

Alexandr Mikyška and Karel Štěrba followed fifteen carbonyl compounds through 200-liter pilot brews of pale lager in a study published in 2023. They compared boiling systems and hopping regimes. Many measured compounds decreased during atmospheric boiling but increased under the tested pressurized systems. Fermentation then substantially reduced the free carbonyl concentrations and largely erased the differences among the resulting fresh beers. The authors proposed that hidden differences could still affect later stability, while explicitly saying that this required further investigation.[3]

The surprise is temporal. A fresh beer can seem similar to another fresh beer while carrying a different potential for change. The experiment does not prove that one boiling system invariably makes inferior beer; too many variables stand between a pilot trial and that sweeping verdict. It does show why examining the final package alone may be like reading the last page of a witness statement. Some relevant events happened before the object under investigation existed.

There is another difficulty: an aldehyde can be present in a form that does not behave like a freely available aroma compound. Chemical interactions can bind it into a less volatile state. If that equilibrium changes, a free concentration may rise without all of the newly detectable material having been manufactured at that moment. Detecting more after storage does not, by itself, distinguish formation from release.

This distinction has inspired plausible explanations that experiments have then had to prune. Paula Bustillo Trueba and colleagues tracked free aldehydes and a particular family of cysteine-bound aldehydes from malt through brewing and storage. Their 2021 study supported the importance of considering bound forms, but its results did not support those specific cysteinylated compounds as

the source of the rising aldehydes in the aged beers. A convincing general idea had not rescued a particular proposed mechanism.[4]

The negative finding left a sharper question. Which bound forms, in what amounts, could actually survive the process and contribute to the later aroma? It was no longer enough to point to a possible attachment. The investigation had to follow that attachment through brewing and establish whether its release could account for the change in the beer.

Carolina Maia and David Cook approached binding differently in work published online in 2025. They freeze-dried three commercial lagers and reconstituted concentrated material from each, then added selected aldehydes and measured responses in the space above the liquid. The resulting matrices differed in how they retained compounds. Adding one aldehyde could displace others. The researchers had evidence for competition, while the full identity and range of binding sites remained unresolved.[5]

These were deliberately altered analytical systems, not three untouched bottles watched on a shelf. Concentrating the beer-derived material made interactions easier to investigate. It also means the findings should not be turned into a universal expiration clock. The study's value lies in making the beer itself an active participant. The surrounding liquid is not merely water carrying independent odor molecules toward a nose; its constituents help determine which molecules are available to leave it.

New formation and release can operate together. Arndt Nobis and colleagues examined six pale lagers made with differing degrees of malt modification in research published in 2021. They followed aging and the compounds capable of giving rise to aldehydes. Their results linked the starting material to subsequent changes and supported new formation alongside release from bound states. Within their experimental series, the balance changed as storage proceeded.[6]

One route involved reactions between amino acids and other carbonyl compounds. The name Strecker degradation sounds remote from a drink, but the practical idea is straightforward: substances left in the beer can become starting materials for later aroma chemistry. The beer arrives in its package with unfinished possibilities. Neither a low concentration of one free aldehyde nor the absence of an obvious defect on bottling day counts those possibilities completely.

The timeline now contains at least two routes toward a rising aldehyde concentration: release of material that was already bound, and production from other constituents. Their relative importance can change during storage. A fresh

sample and an aged sample give two readings; the intervening measurements help distinguish the processes that connect them.

Meanwhile, an old beer may change partly because something pleasant has gone away. Christian Schubert and colleagues studied eleven commercial hoppy ales and one lager under different storage conditions. Their 2022 report found differences among aroma compounds: some terpenoids, including linalool and geraniol, were relatively stable in the studied ales, while some ester-driven aromas were less stable. Compounds that persisted could counteract oxidized impressions in some samples.[7]

That result makes flavor loss harder to describe as a simple invasion. The balance can change when one component declines and another remains. Imagine lowering one voice in a conversation: the other speaker may seem newly prominent without speaking any louder. An aging note can similarly become more noticeable because the context around it has changed. Measuring the suspect alone may miss the disappearance of its former competition.

The storage study was not a census of every IPA, and its one lager was not a universal representative of lager brewing. Different beers start with different mixtures. A particularly aromatic fresh sample does not necessarily possess a single quantity called freshness that will decline at a fixed rate. Several trajectories can overlap: a new note appears, a familiar note fades, another is released, and the sensory balance shifts.

For that reason, the pursuit of one universal staling molecule has an obvious appeal and a practical limitation. It promises a convenient test, a single culprit and a definite remedy. The experiments instead supply a collection of mechanisms that matter under stated conditions. A brewer can use that knowledge to reduce particular risks, but the drinker's disappointment does not retrospectively identify the responsible reaction.

The cap's promise was narrower than it first appeared. It could help preserve the contents by controlling exchange with the surroundings. It could not require every molecule already inside to remain in its original state. The old beer therefore invites a different picture of storage: time spent inside a continuing chemical process. Some of the evidence is in the bottle. Some was laid down long before the bottle was filled.[8]

11

The Foam That Came from the Field

The barley experiment seemed to offer a straightforward route from cause to effect. Grow two cultivars. Expose them to different strains of a fungal pathogen. Harvest the grain, malt it, and measure what happens. If the fungus is associated with beer that gushes out of its container, perhaps more fungus will mean more gushing.

The results were less obliging. In field trials conducted from 2019 to 2021, the relationships among disease, fungal abundance, toxin measurements and gushing did not move as one. Some outcomes depended on cultivar or pathogen strain. Gushing varied among years, but the researchers did not find a significant relationship with the abundance of the particular fungus they were tracking. Their deliberately inoculated barley did not produce malt with greater gushing activity than the uninoculated comparison. Each malt supplied an extract added to beer from the same batch, allowing the extracts to be compared in a common liquid.[1]

It is a good place to enter the foam mystery because it prevents an easy answer from hardening too soon. A bottle that erupts is spectacularly informative about one thing: gas has escaped into bubbles rapidly enough to carry liquid with it. It is far less informative about the chain of events that made this happen. A wet table cannot identify a fungal species.

The gas is usually not a stranger to the beer. Carbon dioxide can remain dissolved under the conditions inside a sealed package. Opening changes the pressure conditions, and the liquid now contains more dissolved gas than it would retain at equilibrium with the surrounding air. Some of that gas escapes at the liquid's surface; some can enter bubbles. The amount of gas matters, but the route by which it leaves matters too.

Creating a bubble from nothing is not the easiest route. Tiny pre-existing gas pockets, including those held at small surface irregularities or particles, can provide starting places. The technical word is nucleation. In 2021 Gérard

Liger-Belair and Clara Cilindre combined experiments on a commercial lager with physical modeling to examine these starting places and the possible number of bubbles in a glass. Their calculations depended on the beer, temperature, gas content and assumed geometry of the nucleation sites.[2]

The important discovery for our purposes is not a universal bubble count. There cannot be one without specifying the system being counted. It is that dissolved gas and the visible release of that gas are different observations. The same quantity in a liquid need not announce itself at the same pace under different surface conditions. A glass quietly streaming bubbles and a package abruptly overflowing may both involve carbon dioxide while presenting different physical problems.

Excess carbonation is one possible cause of excessive foaming. But brewing researchers distinguish a problem often called primary gushing, associated with substances carried from grain and malt. The surprising suspects are hydrophobins, small proteins made by filamentous fungi. They have both water-attracting and water-repelling regions, an arrangement that lets them assemble at boundaries between water and another phase. In the right circumstances, that surface behavior can have consequences much larger than their size suggests.

Tuija Sarlin and colleagues investigated gushing-active hydrophobins from *Fusarium graminearum* and related fungi. They identified candidate genes, produced selected proteins in another fungal host, purified them and tested their behavior. Their work, published online in 2011 and in a 2012 journal issue, reported vigorous gushing at very low concentrations of the purified proteins. This was direct experimental evidence that particular fungal proteins could induce the phenomenon.[3]

The route from a gene to an overflowing beer is an impressive piece of detective work. It moves beyond the observation that troublesome grain and troublesome foam sometimes occur together. Isolating an active substance helps distinguish the organism's many activities from the specific effect under investigation. The result does not require the fungus itself to be growing visibly in the bottle when it is opened. A material produced earlier can survive as the relevant participant.

How could the protein accomplish so much? Sylvie Deckers and colleagues proposed a mechanism involving very small gas structures stabilized by hydrophobins. Their 2011 research used dynamic light scattering, a method that infers information about small particles from fluctuations in scattered light. The signal indicated particles approximately a hundred nanometers across in the

tested gushing beer; the researchers proposed that these were hydrophobin-stabilized nanobubbles. Extracts containing hydrophobin induced gushing in carbonated test beverages, while the still-water comparison did not gush.[4]

The contrast is revealing: an active protein is not an unlimited source of gas. The gas and the surface-active material have to participate together. At the same time, the instrument's particle-size signal is not a photograph of every proposed structure. The researchers' model offers a mechanism supported by their experiments; its components should retain the distinction between what was measured and how the measurement was interpreted.

At this scale, a bottle can contain an enormous amount of surface without containing an obvious visible object. That is one reason microscopic structures can matter to a macroscopic event. The dramatic rush of foam is the end of the story, when many small events become visible together. Judging solely from the size of the eruption encourages us to imagine a correspondingly large intruder. The experiments direct attention toward interfaces and tiny quantities instead.

Now return to the Canadian barley trials. Anuradha Jayathissa and colleagues had good reason to investigate *Fusarium*: it is an important grain pathogen, and its products can cause problems during malting and brewing. Yet their different measurements remained different. The authors discussed the possibility that other fungi, also capable of producing surface-active proteins, influenced the gushing tests. That was an explanation to investigate, rather than proof that those other organisms caused the result.[1]

The control grain helps expose a common misunderstanding of experiments. Uninoculated does not mean biologically empty. It means the researchers did not add the specified inoculum. Field-grown material can still encounter organisms and conditions outside that intervention. When a comparison surprises us, the temptation is to treat it as a nuisance. Here it prevents the elegant but unsupported equation of more measured *Fusarium* with more overflowing beer.

The toxin measurements belong to a separate investigation as well. A fungal species can make several substances, and the conditions that influence one output need not predict another. Detecting fungal DNA, measuring a toxin and observing foam are different analytical endpoints. The presence or absence of gushing cannot therefore replace the chemical measurements required to establish a toxin concentration. A quiet bottle is not a certificate; an erupting one is not a chemical identification.

Ordinary beer foam adds another layer. A brewer may want a stable head after pouring while emphatically not wanting liquid to escape from the package. Foam

formation and foam persistence are related but distinct. Making a bubble requires one set of circumstances; keeping the films between bubbles from collapsing requires another. The word foamy can conceal the difference between gas release, ease of foam formation, durability and texture.

Chen Xu and colleagues investigated one of those relationships using protein Z, a barley-derived protein associated with beer foam, and humulinone, a hop-derived compound. Their laboratory study found that the interaction altered properties of the protein solution and improved its measured foamability and stability. This was work on a selected protein system, not evidence that every beer made with older hops would have a superior head.[5]

The contrast with gushing is useful. In one investigation, surface-active material helps explain unwanted gas release. In the other, interactions involving a barley protein help explain the formation and persistence of foam. A gas alone does not determine the character of a foam; the behavior of its surfaces helps determine what the bubbles do.

The practical detective work consequently begins by locating the event. Did foam rush from a package as it was opened? Did a poured glass lose its head unusually fast? Were bubbles attached to particular spots on the glass? These observations deserve different follow-up questions. Treating them all as too much or too little carbonation can hide the information needed to understand them. The spectacle is a starting point, however confidently it seems to announce its cause.

The trail can lead from that spectacle back through bottling, brewing, malting and farming. Yet it does not run backward automatically. Finding a plausible mechanism in a paper cannot diagnose an individual bottle any more than recognizing a footprint establishes the identity of every person who has walked across a field. The relevant tests must connect the mechanism to the actual materials and conditions.

What makes the case memorable is the scale of the connection. Grain can carry the consequences of interactions with fungi into a process designed around another fungus, yeast. Proteins associated with those earlier interactions can affect how gas behaves much later. The brewery is an impressive transformation, but it does not begin the story of everything it handles. Sometimes the foam remembers the field. The difficult part is determining what, exactly, it remembers.[6]

12

The Missing Parent

The two soil samples were collected about seventeen meters apart. One came from beneath sycamore leaf litter; the other from ground near oak, beech and cherry laurel. The location was the Belfield campus of University College Dublin, and the collection took place in September 2021 as part of undergraduate research. Out of those samples came two strains of a yeast species that had repeatedly failed to appear in European collections of wild isolates.[1]

For a discovery connected with lager, the setting is agreeably ungrand. No famous cellar was opened. No centuries-old bottle surrendered a surviving brewer's secret. The evidence came from ordinary soil on a campus. It would help revise a story that had already crossed from European breweries to Patagonian forests and the Tibetan Plateau. The object of the search was a parent.

Lager yeast, *Saccharomyces pastorianus*, carries genetic ancestry from two species. One is *S. cerevisiae*, familiar from many alcoholic fermentations. The other is the cold-tolerant *S. eubayanus*. A hybrid brings genetic material from distinct parental lineages into the same organism. That makes its family history more complicated than a simple line of descent, particularly when chromosomes have been lost, copied or rearranged after the original encounter.

Brewers had been using lager cultures long before scientists could identify those ancestral contributions. A culture could perform usefully without providing a legible account of its ancestry. This is an important distinction in the history of fermentation: the discovery of what an organism is does not mark the beginning of people's ability to work with it. It marks a change in what investigators can explain.

The major breakthrough in identifying the missing parent was reported in 2011 by Diego Libkind and colleagues. Working with yeasts from Patagonian southern-beech forests, they described *S. eubayanus* as a species and compared its genome with lager yeast. Their draft sequence was approximately 99.5 percent identical to the non-*cerevisiae* portion of the lager yeast genome used in the comparison. The unknown ancestral component now had a close wild relative with a name.[2]

The geographical result was irresistible. A yeast associated with the history of European lager had a newly identified relative in South America. A journey seemed to demand invention: some vessel, cargo or traveler must have carried it across an ocean. Such a route might make a splendid story. The genome comparison alone did not supply the ship, the date, the cargo or even proof that the relevant historical movement began where the first modern investigators found the organism.

Discovery location and origin are different coordinates. The distinction is easy to state and remarkably easy to forget. A species found first in one region may also live elsewhere, overlooked because it was rare, because samples were sparse or because previous methods did not recover it. An organism's distribution does not wait for a journal to print its name. The initial discovery changes the research map; it does not finish drawing it.

Jian Bing and colleagues expanded that map in 2014. They reported Asian populations of *S. eubayanus*, including a Tibetan population with a closer genomic affinity to the lager-yeast component than the Patagonian population in their comparison. They argued for a Far East Asian origin hypothesis. The finding challenged an easy identification of the first discovery site with the immediate source of the lager parent.[3]

Yet replacing Patagonia with Tibet in a finished migration tale would repeat the same mistake. Genetic resemblance can tell us about relationships among the organisms sampled. A historical itinerary requires more. It also matters which part of a genome is compared. Different stretches can preserve different pieces of an ancestral population's variation, especially when mating and recombination have repeatedly reshuffled that variation before the lineage of interest was formed.

David Peris and colleagues made this point explicit in a 2016 population-genomic study. Comparing wild isolates across regions with domesticated lager hybrids, they found that no single sampled wild strain was the sole closest relative across the relevant genome. Different regions of DNA carried different relationships. They interpreted the result in terms of pre-existing variation within wild populations and a more complicated biogeographical history than a one-isolate ancestry story would allow.[4]

Think of a family collection of photographs. One cousin may resemble a grandparent around the eyes, another in the shape of the face. Naming the cousin with the strongest resemblance does not make that person the grandparent. In genetic work, the comparison is far more exact, but the warning still helps: a close

living relative is not automatically the exact historical ancestor, and a relative's present address is not automatically the ancestor's address.

Even the number of hybridization events has required argument. Modern lager strains fall into major groups with differences in chromosome content. A 2015 genomic study by EmilyClare Baker and colleagues supported at least two independent origins. In 2019, Alex Salazar and colleagues used a more complete chromosome-level assembly and comparative analysis to argue for a single hybridization followed by different evolutionary trajectories. The dispute turned partly on how differences in complicated genomes should be read.[5]

This is not a reason to describe the whole ancestry as guesswork. The broad hybrid relationship is well supported. The reconstruction of particular events is a finer-grained problem. A chromosome can be present in several copies; different copies need not be identical; subsequent losses can make related descendants look surprisingly different. A complete account has to separate changes inherited from the parents from changes accumulated afterward. Better assemblies alter the evidence available for that task.

Against this background, the Irish soil samples became especially interesting. Sean Bergin and colleagues reported the two isolates, UCD646 and UCD650, in 2022. Genome analysis placed them in the broad northern lineage that includes relatives of the lager parent. Some of the lager ancestry resembled the Irish strains, while other portions were closer to Tibetan samples. Europe could no longer be described as a continent from which wild *S. eubayanus* had never been isolated.[1]

The discovery did not prove that medieval brewers obtained yeast from Ireland. Its power was more precise: an apparent absence had become an observed presence. Wild isolates collected in Europe were now available for comparison, rather than merely proposed to fill a gap. Those samples could not establish when the species had arrived or where it had lived centuries earlier. The historical questions remained, but investigators now had living material with which to pursue them.

Meanwhile, historians and microbiologists were asking where the parental lineages might have met in a brewery. In 2023 Mathias Hutzler and colleagues proposed a location and interval: Munich's Hofbräuhaus between 1602 and 1615, when top-fermented wheat beer and bottom-fermented beer were being produced in the same institution. Their historical argument reversed a familiar assumption by suggesting that *S. cerevisiae* could have entered an environment where *S. eubayanus* was already present.[6]

This is an archival hypothesis, not an observed mating event. It connects documented brewing arrangements with a biologically plausible encounter. Its

strength lies in the specificity of the proposed circumstances; its limit is that brewery records cannot ordinarily identify an invisible hybridization as it occurred. A shipment of yeast can be documented without its genome. A genome can preserve ancestry without preserving a shipment label.

The hypothesis also helps separate the history of cold brewing from the origin of the modern hybrid. Brewing practices can precede the particular organism now associated with them. If older brewers used mixed cultures or different cold-tolerant yeasts, a record of bottom fermentation would not automatically prove the presence of present-day *S. pastorianus*. A method's continuity and a microorganism's continuity must be investigated separately, even when both travel under the word lager.

Human activity nevertheless belongs inside the biological story. Reusing cultures gives some organisms repeated access to wort. Changing temperatures, ingredients and handling alters the conditions under which those organisms succeed. The surviving culture is therefore neither a wholly wild visitor nor a machine designed from a blank sheet. It is a living result of encounters, selection and repeated practical decisions, many made before anyone could inspect a chromosome.

The search has produced a better answer than a single dramatic arrival. It has identified a parental species, exposed geographic diversity, tested competing accounts of hybrid ancestry and connected genetic questions to particular brewing institutions. It has also shown that the next useful sample need not come from a place already famous for beer. A patch of soil can matter because the organism is there, regardless of whether anyone thought to look.

The two Irish isolates are not the original parent returned from the past. They are living relatives that make the past more intelligible. Seventeen meters of ordinary university ground widened the search. Somewhere among the samples not yet collected may be another relative, one that will force a line on the family tree to be drawn again.[7]

PART III

A Living Trade

The work moves between households, shared buildings, fields and places of sale. Following the drink makes its makers harder to leave out of the story.

13

The Women in the Fine Book

In Exeter in 1317, women who sold ale took down their signs. The price at which they had been ordered to sell was, in their judgment, unacceptable. Removing the signs withdrew the invitation to buy. A business that could be regulated could also refuse to do business.

The episode reaches us through the historian Judith Bennett, whose account of women in English brewing follows records made for purposes very different from celebrating women's work. An earlier case, from Wakefield manor in 1275, records Gillian, wife of Richard Pykard, resisting the officers who inspected ale. Bennett emphasizes that such protests were uncommon. These particular disputes nevertheless interrupt a familiar picture of the medieval brewer: anonymous, compliant, stirring something in the background while history happens elsewhere.[1]

A sign coming down is a small action. It becomes substantial when one asks what had to exist before it could happen. There was a product to sell, an expected customer, a price, and someone who considered the return insufficient. Brewing was work over which people bargained. The women in these records did not require a modern word for entrepreneurship before they could have interests of their own.

Yet the surviving evidence creates an awkward problem. How much of a working life can be reconstructed from the occasions when that life annoyed an official?

Imagine a history of modern restaurants assembled almost entirely from inspection reports, tax disputes and trading infringements. Even if every entry were accurately transcribed, the resulting book would be peculiar. Its cooks would seldom appear making a good meal. Customers would arrive mainly when they had cause to complain. A proprietor might work successfully for twenty years and receive a paragraph devoted to a disputed weight. The bias would lie less in any single false statement than in the selection of events that had become statements at all.

This is a problem for reading an archive, not a reason to discard it. The irritating official may preserve information that no admirer thought worth writing down. His records can disclose names, prices and repeated appearances in a trade. But the researcher has to keep two questions separate: what was the person doing, and why did the record maker notice?

Ale was well suited to being noticed. The medieval assize linked its permitted sale to the price of grain; a surviving statutory text also prescribes escalating penalties for brewers who persistently infringed the rules. This was a supervised market in a consumable staple, with obligations stated in quantities and money. The law documents what authorities wanted enforced. Its existence does not prove that enforcement was uniform or that every listed punishment occurred whenever someone charged too much.[2]

Once this distinction is clear, a fine stops being a complete character reference. It is evidence of an encounter within a regulated business. We still need to know the rule, the local practice and the circumstances before treating the brewer as a cheat. And even a genuine act of cheating would describe that act, not establish that the brewer's sex explained it.

Oxford provides a more local way into the difficulty. The Museum of Oxford names Maud London, an unmarried fourteenth-century brewer, among the women visible in the town's records. Its account also points out that poll-tax documentation represents substantial enterprises more readily than the smallest occasional sellers. A name survives; an unknown number of less conspicuous working lives does not.[3]

Maud's name gives commercial brewing a participant whom an account centered on male proprietors could easily obscure. The surviving description is brief, but it interrupts a history in which women appear only as helpers or exceptions.

The missing sellers complicate counting. Suppose a list contains the heads of taxed households. It cannot automatically tell us how many people carried water, handled malt, brewed, sold, collected debts or kept the household functioning while brewing went on. Those activities could involve different people. A named man does not prove that no woman worked in his enterprise; a named woman does not prove that she worked alone. Before turning a list into a chart of occupations, one must ask what qualified someone for a line on the page.

Bennett's broader argument follows a gradual change between 1300 and 1600: women were prominent in medieval ale production, while male-led, more specialized enterprises became increasingly important. Household work remained central to the story. The relevant unit could include spouses, children,

servants and hired labor, not simply one solitary brewer. Her account resists the convenience of a single takeover date.[4]

That matters because the phrase *domestic brewing* can quietly diminish what it describes. A household selling a product is participating in a market. Its front door does not cancel the transaction. Conversely, calling an establishment commercial tells us little about whether the work was fairly divided or who controlled the proceeds. Location, ownership, labor and authority are separate matters, even when one person appears under the occupational heading.

An investigation of women's brewing therefore needs a better question than whether women were once allowed to make beer. Allowed by whom, in which town, on what terms, and with what prospects of continuing? Selling a little surplus and directing an enterprise large enough to survive changing competition are not interchangeable opportunities. Nor can one assume that a widow's position, a married woman's position and an unmarried woman's position followed identical paths.

A change in the name recorded against a business can therefore conceal several different changes. Perhaps the person directing the work has changed. Perhaps its legal representative has changed while much of the workforce has remained. Perhaps a small operation has become large enough to attract a different kind of record. These possibilities demand different evidence. Following a household across documents is harder than counting names once, but it can reveal whether an apparent disappearance concerns labor, authority or simply visibility. The distinction matters most when the people becoming less visible are the people whose opportunities we are trying to understand.

The temptation is to find one picturesque cause for a slow economic transformation. A new ingredient arrives; men seize the kettles; the story is complete. Such an explanation travels well because its cast is small. It also asks one development to account for ownership, regulation, scale, household organization and reputation all at once. The sources here support a less obliging history, in which changes must be followed locally and over time.

There is another reason to be careful with the pictures. The alewife did not remain safely in the Middle Ages. She reappeared, astonishingly, in the United States Supreme Court.

In December 1948, the Court decided *Goesaert v. Cleary*, a challenge to a Michigan restriction on women working as bartenders. The law permitted an exception for the wife or daughter of a male proprietor. Writing for the majority, Justice Felix Frankfurter invoked the old English alewife as "sprightly and ribald." The Court upheld the distinction. Justice Wiley Rutledge's dissent pointed to its

practical asymmetry: a female owner could not do what a male owner's wife or daughter could.[5]

The medieval woman in that opinion was a literary recollection, not Maud London traced through a town's records. Her appearance did rhetorical work. A familiar type made an exclusion seem comfortably historical. One can disagree with a judgment on its own legal terms; one can also notice what happened to the history inside it. Centuries of women's labor had been compressed into an attitude.

The distance between Exeter's signs and the Michigan bar is not evidence of an unbroken, identical system. The laws, places and businesses were different. It is evidence that a representation can outlive the setting that supposedly explains it. A caricature may remain available long after the people it caricatured have become hard to retrieve.

This is why recovering a name is only the beginning. Merely swapping a comic alewife for a heroic one leaves the structure of the story unchanged: a representative woman stands in for everybody. The harder task is to restore distinctions. Some brewers had money; some struggled. Some worked occasionally; others made a living from the trade. A person could exercise initiative within severe constraints, and suffer an injustice without being an exemplary business owner.[6]

Return, then, to the signs. We cannot hear what every Exeter brewster said in 1317, or know how long each household could afford to forgo a sale. Bennett's account preserves the refusal. For a moment, the regulators' record shows people using the terms of a market against those who set them.

The sign had ordinarily announced that ale was available. Taken down, it announced a different fact: the person who made it had a say.

14

The Brewery with Many Front Doors

Follow a batch of Zoigl through the account published by its community brewers and the brewery appears to come apart. Grain and water meet in a communal building. The wort, the sweet liquid that will become beer, is heated, hopped and cooled there. Then it leaves. Fermentation takes place in the brewer's own premises, and the finished drink may eventually be served in a room belonging to the household.

There is no single front door behind which all the important things happen. The arrangement is distributed across a town.[1]

This can be confusing if one expects a brewery to be a building with a name above it, or expects every beer name to describe a recipe. Zoigl, in the Upper Palatinate of Bavaria, asks the visitor to understand a different arrangement: shared production facilities, distinct households, rights attached to property, and a changing invitation to drink. The most useful clue is outside the house.

The Zoigl star announces service. Windischeschenbach's municipal account describes the six-pointed sign hanging from the facade and a calendar that sets out which household is opening. It links *Zoigl* to a local form of the word for a sign or indicator. The point is practical. A passer-by needs to know which door is temporarily a public one.[2]

A permanent pub can advertise an address. A rotating household business must also advertise a moment. The sign communicates the conjunction: here, now. Once that is understood, the star becomes less mysterious and more interesting. It solves a small problem of coordination in a place where the appearance of a building may not disclose its occasional commercial life.

It is tempting to ask what each point of the star has always meant. The practitioners' website offers a symbolic explanation involving brewing materials and elements. Such explanations are part of how a tradition presents itself, but they are not, by themselves, dated proof of the symbol's origin. For this

investigation, the secure clue is its documented use. A sign can do important work without an agreed prehistoric biography.

Germany's national intangible-heritage inventory recognized Upper Palatinate Zoigl culture in 2018. Its account emphasizes shared maintenance, oral transmission and the social life around service: conversation, local affairs and encounters among residents and newcomers. It identifies five places where the tradition survives: Eslarn, Falkenberg, Mitterteich, Neuhaus and Windischeschenbach. The recognition concerns a living practice, not merely the chemical contents of a tank.[3]

The word *national* matters. This was entry in Germany's inventory, not a separate inscription on UNESCO's international Representative List. The distinction is easy to lose when a badge is reduced to a sentence on a travel page. Keeping it restores the scale of the claim without making the tradition less valuable. Recognition is evidence of an institution's assessment; the people doing the work remain the reason there is anything to assess.

Those people also have bills. The brewers' English-language account explains that communal buildings are municipally owned in some of the towns and jointly owned by brewers in others. Users contribute *Kesselgeld*, literally kettle money, toward upkeep. Brewing rights are associated with specified properties. Windischeschenbach's own description points to their registration in the land register.[4]

This is the first complication in the inviting word *communal*. Sharing an expensive installation does not necessarily mean that anybody arriving with a sack of malt can use it. A system can pool equipment and still have defined membership, inherited rights and rules of access. Nor does sharing prove that every participant has equal influence. Those questions require local knowledge beyond the attractive view of people drinking together.

The distinction is useful because it keeps a communal brewhouse from becoming a fantasy about a world without administration. The structure survives through administration: someone must maintain the equipment, arrange access and account for expenses. Kettle money is a modest phrase for one of the conditions of continuity. A tradition that cannot repair its kettle is in trouble, however stirring its history sounds.

Another clue lies in the brewers' own effort to distinguish their product from beers merely marketed under the same name. Their shared logo has stated conditions: the beer must be made in the communal brewhouse according to the recognized practice, and it must be served in the same town. This is a claim about

the connection between production and place. A remotely manufactured beer might resemble its flavor and still fail that test.[5]

Consider what would happen if someone reproduced a household's recipe perfectly in a distant industrial brewery. That is a thought experiment, not a report of a particular imitation. A blind tasting might find the drinks indistinguishable. The communal brewers' stated conditions would still distinguish them, because the second batch had not helped sustain the same local arrangement. Conversely, two household beers could taste different while both satisfying those conditions. The apparent paradox disappears once one recognizes that a recipe is only one of the things being preserved. The other is an organized way of making and offering the drink.

Variation already exists within the shared practice. The association explicitly describes differences among households' recipes and among batches. Its account presents Zoigl as a bottom-fermented beer, but not as an exercise in making all household production converge on one flavor. Calling that variation a defect would import an objective the participants have not chosen.

There are practical reasons why sameness is a demanding goal. A brewer can share a kettle without sharing all later decisions. Recipe proportions, fermentation and maturation remain parts of the work. The town is not one enormous production line divided by decorative walls. Each household's part in the process helps explain why a common institution need not erase individual products.

Notice, too, how much time the invitation outside the door conceals. By the time a sign announces service, grain has been handled, liquid transformed and beer given time to mature. The visitor sees the available result. The producer has had to coordinate the work beforehand. The calendar is therefore more than a visitor convenience: it is the visible edge of a sequence of commitments.

That observation does not tell us exactly how every household divides its labor. An account centered on the named brewer may leave preparation, food, cleaning and serving less conspicuous. The heritage description recognizes household hospitality, but the sources consulted here do not provide a systematic study of who performs every task or who receives its income. Those are questions worth bringing to the practice, particularly when its public image concentrates on the brew kettle and the glass.

There is no need to spoil conviviality to notice its work. Quite the opposite: a cheerful room becomes more intelligible when somebody's effort is allowed into the picture. Shared equipment may reduce duplication; it does not make the

remaining tasks do themselves. A book about beer should be able to admire a hospitable institution and still ask who opens the room and cleans it afterward.

The national inventory also describes adaptation, including the communication of service dates through digital tools. A phone displaying a calendar need not be the enemy of an old sign. Both can perform the same practical work for different audiences. The test of continuity cannot simply be whether a tool has been replaced. One must ask which relationships the replacement sustains, changes or abandons.

This is why Zoigl is a poor exhibit for the claim that a tradition survives by staying motionless. Even the decision to present a custom to outsiders changes its setting. Recognition, visitors, publicity and commercial imitation give practitioners new things to negotiate. The sources do not settle every consequence of those changes, but they show that the subject is active enough to have consequences.

A visitor need not master the land register before ordering. But knowing it exists changes what the visitor sees. The brewhouse becomes an arrangement among people with defined responsibilities; the serving room becomes one part of a larger sequence. What first looked like an unusually charming pub begins to disclose the work required to make that sort of pub possible.

The best final clue is therefore the least exotic one: a date next to an address. It connects a batch made through shared facilities to a household ready to receive customers. The star makes that arrangement visible in the street; the calendar carries it beyond the street. Neither tells the entire story, and both tell something a list of ingredients cannot.[6]

A glass can be carried away. A shared brewhouse, a right to use it and a succession of open doors are harder to bottle. That is the part of Zoigl that the sign points toward.

15

The Ingredient That Had to Be Carried

A twenty-nine-year-old pito brewer in Nagbo, northern Ghana, explained the problem in terms of the working week. When water was available, she could brew three or four times. When it was scarce, that might fall to once, or not at all. In the account published by Joseph Alhassan and colleagues in 2024, she connected the lost production to money needed for her household and her children's schooling.[1]

The difficulty was not a mysterious failure of fermentation. It began before fermentation could happen. The brewer knew how to make the drink; she could not reliably obtain one of the things that made the work possible.

A recipe would have written *water* on a line and moved on. Her account restores the distance between naming an ingredient and having it.

The study concerned women's livelihoods in the East and West Mamprusi municipalities. It combined a survey of 394 women with group discussions and other evidence. Pito brewing was one of the businesses discussed, alongside food preparation and trading. The Nagbo brewer's description was an individual account within that investigation, not the measured weekly output of all northern Ghanaian brewers. It is valuable because it shows how a shortage enters an ordinary decision: whether another batch can be made.

Once that decision is visible, a different kind of brewing history becomes possible. It follows the conditions of production through time, cash and work, instead of treating the brew vessel as the boundary of the subject. A missing supply can make every skill inside that boundary temporarily useless.

Consider the sequence implied by a commercial batch. Materials must be obtained before the drink can be sold. Work happens before revenue arrives. Containers and a place to serve need to be ready when customers want the product. If a producer cannot safely hold finished stock for long, selling time matters as much as making time. The precise arrangements vary; the ordering problem does not disappear merely because the enterprise is small.

That is why a questionnaire about pito can ask questions that would look quite ordinary in a study of any small business. In 2023, Aliata Issahaq Mumuni, Musah Dumah and Hawa Awuro Sam reported responses from 362 women brewers. Their methods describe purposive and convenience sampling in the Wa municipality, within the Upper West Region. The article foregrounds difficulties with finance and with packaging and storage. Its subject is an enterprise trying to keep operating, not an ornamental survival of a distant past.[2]

It would be easy to turn those findings into a heroic tale: resilient women overcome everything through determination. The problem with that tale is the word *everything*. A water supply cannot be replaced by an admirable personality. Nor does skill automatically provide affordable credit or a suitable container. Praising persistence can become a convenient way of looking away from the obstruction that required it.

The opposite story is no better: a supposedly backward technique waiting for someone else to arrive with modernity. It treats the producer's existing decisions as a blank space. A brewer deciding how much to make, when to buy materials and which customers to serve is already managing an uncertain business. An intervention should begin by understanding those decisions, including the constraints that make an apparently obvious change impractical.

There is a revealing distinction between buying equipment and financing a cycle of production. Equipment can remain in place while the business lacks money for its next batch. *Working capital* is the money needed to bridge that interval. A pot and a recipe do not fill the interval by themselves. Likewise, a larger vessel might increase potential output without increasing sales, available water or time. Capacity and livelihood are related, but they are not synonyms.

The Wa paper is not a reliable warrant for calculating a national number from its sample. Its tables contain reporting inconsistencies, and its statistical associations cannot establish that a particular constraint caused another. Those limitations matter when choosing how to use it. The descriptive finding that participants reported practical business difficulties is more defensible than a confident prescription based on a regression coefficient. The women deserve attention without having doubtful precision attached to their circumstances.

A separate 2024 paper from Wa discusses profits and sources of finance, but published figures from one local study should not be turned into the typical income of a Ghanaian brewer. A profit figure also needs a definition: which expenses were counted, over what period, and how was household labor treated? We cannot settle those questions by admiring the apparent exactness of a number.[3]

For a household enterprise, work can be economically consequential even when no wage changes hands. If the same person must collect water and prepare a batch, an extra hour spent on one task is unavailable for the other. If somebody else takes over the task, the cost has moved rather than vanished. This is why time belongs in an account of production alongside grain and fuel. Leaving it out can make a drink look cheaper to produce than its maker experiences it to be.

The Mamprusi study makes that connection concrete. Respondents described non-agricultural work as especially important when seasonal farming opportunities were limited. Water scarcity could then interfere with the activities on which they relied. That combination is more informative than treating brewing as a fixed occupation performed in identical conditions all year. A household's work changes with its opportunities; a shortage may arrive precisely when one source of income has become more necessary.

The geography must remain visible. Wa and the Mamprusi municipalities are not the same study location. Their participants are not one combined sample. The useful connection is a question crossing the studies: what must a woman secure, besides brewing knowledge, to turn that knowledge into income? Each investigation supplies a different portion of an answer.

To follow the drink beyond production, Isidore Lobnibe investigated pito bars and migration. His study, published in the 2018 volume of *African Geographical Review*, connects northern Ghana with the Brong Ahafo region, using fieldwork and the social life of two bars. The accessible abstract frames pito's movement through changes in agriculture, settlement and identity. A beverage can travel with people and become a way of understanding their changing surroundings.[4]

The view from a bar and the view from a brewing household need not coincide. A customer may value conversation, familiarity or a place to meet. The producer must also attend to what that visit costs and what it pays. A convincing history has room for both. Treating conviviality as the whole story risks making the work that sustains it disappear just as people begin enjoying themselves.

Nor should *pito* become a convenient replacement word for every West African cereal beer. Across the border, research on dolo in Burkina Faso describes women brewers often called *dolotières*, and an economy involving grain suppliers, malt preparation, fuel, transport and sales. A 2021 review led by Hagrétou Sawadogo-Lingani brings those connections together. It also identifies the difficulty of distributing a changing, short-lived product and the practical use of brewing by-products. This is a comparison of linked kinds of work, not a claim that the drinks or institutions are interchangeable.[5]

What the comparison offers is a wider cast. The person near the vessel may be the easiest participant to photograph, but the batch depends on people outside the frame. An account that stops at the pot is as incomplete economically as a chemical explanation that stops before fermentation. Grain has to arrive. Fuel has to arrive. The finished drink has to find its drinkers.

A proposal to change the process consequently has more than one outcome to consider. It might reduce fuel use but require a larger payment up front. It might extend storage but change the product customers expect. These are possible tradeoffs, not claims about a particular unexamined technology. They illustrate why the producer's own priorities belong at the center of the question. The most sophisticated apparatus is not automatically the most useful change.

The producer would also have to live with the new arrangement. Who would control an investment, collect its revenue or meet its repayments? Those questions belong beside the promise of a bigger batch.[6]

For the brewer in Nagbo, however, one obstacle was already specific enough. Water availability changed how often she could work. No grand theory is needed to recognize the consequence of a batch that never begins.

The ingredients on a page can look reassuringly complete. Somewhere outside that page, someone still has to bring them home.

16

A Flag Outside the House

The flag was white in the northern Peruvian account. In the southern account, the marker was red. Both announced that chicha was for sale.

It is a small difference, but it stops an easy mistake. There is no need to decide which researcher found the authentic color. Frances Hayashida was investigating production around Chulucanas, in Piura. Felipe Vargas Faulbaum was writing about communities in the Cusco region, including Huarocondo. The places were different. The signs belonged to local ways of making a household business visible.[1]

A stranger who knew only that chicha was a maize drink would have missed much of what the flag communicated. It announced availability. It pointed to a particular maker and a place where people could enter. It belonged to a timetable understood by customers, and to a relationship between a household and the street. It was a sign for a drink, but also an instruction about how to find it.

Hayashida's 2008 study interviewed women from ten chicherías, or household breweries, in Chulucanas and neighboring settlements. Producers had known sales days. The published methods give La Loba's Tuesdays and Wednesdays and La Cartola's Saturdays and Sundays as examples. Those modest calendar entries make the business concrete: the drink did not appear everywhere at once, as though a village had one communal tap.[2]

A regular customer could combine the day, the location and the visible signal. An archaeological excavation many centuries later would have a harder time. A pole might leave little behind; a flag might leave nothing recognizable. The timetable would be absent altogether. What had organized access to the drink could disappear while the vessels used to prepare it survived.

This was part of the reason for Hayashida's investigation. Archaeologists could recognize impressive drinking vessels and discuss feasts, yet production itself was harder to identify. Observing living brewers offered a way to connect tasks with the material traces they might leave. The exercise was not to find people unchanged since antiquity. It was to learn what questions to ask of a site whose workers could no longer explain it.

That difference matters. An analogy proposes that a feature might be explained by a certain activity. It becomes useful when the proposal can be checked against other evidence. It becomes dangerous when a present-day observation is silently converted into a photograph of the past. The same-shaped vessel might serve a different purpose; the same purpose might be achieved with different equipment. A good comparison creates possibilities and tests, rather than delivering instant ancestors.

The north-coast study supplies an especially useful clue about work done elsewhere. In the Chulucanas area, makers could purchase ground malted maize, called *pachucho*, and ordinary maize flour from vendors or mills. Brewing in a household therefore did not imply that every preparation happened inside it. A search confined to one house could miss part of the production chain.

This changes how to read an absent tool. If no milling equipment appears in a place used for brewing, the missing equipment need not prove that no grain was processed for the drink. The processing might have occurred elsewhere. Conversely, finding a mill does not prove that all its output became beer. The task is to reconstruct movements among places, not assemble a compulsory collection of objects beside every hearth.

The finished drink conceals these movements rather efficiently. A customer sees a serving vessel and a transaction. The grain may have passed through more hands and more decisions than the room reveals. The phrase *household brewery* identifies a center of activity; it does not mean a sealed little economy.

In Vargas Faulbaum's Cusco-region account, the work becomes personal through a named interlocutor, Engracia Surco Escobedo. His narrative follows activities around maize and chicha, including the use of an electric mill. In Huarocondo he describes red markers outside selling houses. The ethnographer's observations connect the serving room to agricultural work and to changes in the tools people use.[3]

The electric mill is an important detail because it refuses to cooperate with a picturesque idea of tradition. A visitor might hope for an ancient stone implement. A producer may have good reason to prefer a machine that does the necessary job. The replacement changes the material evidence, but does not settle by itself whether the knowledge, product or social setting has been abandoned.

That is a question to investigate with makers, rather than resolve by ranking tools according to their usefulness in a photograph. A new vessel may alter heat transfer or durability; a new mill may alter the organization of work. The meaningful inquiry is what changed for the people doing the task. *Modern* and *traditional* are too blunt to answer it without further description.

Names also require care. *Chicha* can lead a reader toward a family resemblance, but cannot provide a complete recipe for everything called by that name. *Chicha de jora* directs attention toward germinated maize. Germination helps make starch available for conversion into sugars, yet the grain's later treatment still matters. Knowing that a drink begins with maize is no more sufficient than knowing that bread begins with flour.[4]

The distinction between chichería and picantería brings the investigation into another set of documents. Peru's Ministry of Culture recognized the cultural spaces of Piura's picanterías and chicherías in a resolution dated 4 November 2015. Its account distinguishes establishments centered on preparing and selling food with chicha from those centered on making and selling chicha. A picantería need not produce the drink it serves. The official text also records the white flag as an announcement of availability.[5]

The legal document corroborates a practical detail in the field study, while looking at the subject for a different purpose. Hayashida wanted to understand production and its traces. The ministry wanted to identify and recognize cultural practices. Neither purpose is identical to a brewer's need to have a batch ready for customers. Read together, the accounts show several institutions looking at the same drink from different positions.

The recognition was deliberately regional. The resolution describes the decision to issue differentiated declarations because the places, utensils and practices were not identical across the country. That is a useful refusal of tidiness. A national appreciation need not require a national recipe. Even an official category can acknowledge that its meaning becomes clearer when the map is made more precise.

There is another detail in the ministry's announcement that broad histories can easily lose: the women representing these establishments had names. Its report of the recognition ceremony identifies, among others, Susana Victoria Yovera Siesquen and Ingly Quiroga Flores, representing Lambayeque and Piura respectively. They appear as participants in the public recognition of working institutions, rather than as anonymous figures decorating a general story about heritage.[6]

An award cannot tell us everything about a working day. Recognition does not specify income, guarantee the continuation of a business or settle who has authority over change. But a public document can still preserve which practices an institution chose to value and which people stood for them on that occasion. Those are historical facts worth keeping at their proper scale.

Across these accounts, women's work is repeatedly visible. That should neither disappear into an impersonal phrase such as *the production of chicha* nor expand into a claim that every task, everywhere, belongs to women. The proper unit is the documented setting: these makers, these households, this division of work where it is described. Generalization is least helpful when it takes the place of meeting the people who could complicate it.

A particularly revealing question is what a future observer would get wrong if only the serving vessels survived. Perhaps they would imagine a self-contained brewery where materials had actually been purchased. Perhaps they would assign one permanent function to a room that changed use during the week. Perhaps they would recognize consumption easily and underestimate all the labor before it. These are hypotheses prompted by the living evidence, not deductions about any specific ancient site.

We can also reverse the question. What would an account miss if it preserved only the recipe? It would lose the sales days, the route to the mill, the signals outside a door and the distinction between making a drink and serving someone else's. Much of the organization would vanish, even though every ingredient remained faithfully listed.

The white flag and the red marker help recover that organization because they belong to an exchange. Someone has prepared something and is ready to offer it; someone else knows how to read the invitation. Their significance depends on a place and its people. Detached from those relationships, the flag would be an oddly uninformative souvenir.[7]

No single color opens every door. The point of learning the sign is to find the right one.

17

The Cake That Would Not Be a Strain

The yeast appears to have gone missing.

In a survey published in 2013, researchers examined forty-two samples of nuruk, the Korean fermentation starter. They recovered *Saccharomyces cerevisiae*, the familiar alcoholic fermenter, from eighteen. Two years later, a study following makgeolli through fermentation reported that *Saccharomyces* dominated the fungal community. Here, apparently, was an organism absent from more than half the starters yet overwhelming in the drink. Before invoking a mystery of microbial succession, however, there is a less glamorous place to look: the second paper's methods.[1]

The investigators had added commercial yeast.

Their experiment used nuruk and a separate commercial yeast starter together. It was a legitimate study of how that particular fermentation developed. It was not an experiment showing what every nuruk cake would do unaided. Two accurate results had become an apparent contradiction because the experiments were being asked to answer the same question. They had never agreed to do that.[2]

This is an unusually useful mistake to catch. Beer writing often treats the word *starter* as if it meant a packet containing one selected strain. Nuruk makes that assumption awkward. The cake carries a material, a history of cultivation, enzymes, and a collection of organisms. Asking which yeast is inside it is sensible. Treating the answer as a complete description of what it does is much less so.

Start with the grain. Much of the carbohydrate in cereal is starch, an arrangement of linked sugar units. Alcoholic fermentation requires accessible sugars, not merely the promise of sugars bound into larger structures. A fermentation system therefore needs a way to cut those structures apart. It also needs organisms that can turn the resulting food into alcohol. The two jobs can overlap in time without being identical. A single label on a starter obscures that division of work.

The easy explanation assigns starch conversion to molds and alcohol to yeasts. It is a useful first sketch and a poor border fence. In 2016 Jin Ho Choo and colleagues investigated two *Saccharomycopsis fibuligera* isolates obtained from wheat-based nuruk, one from Jeju and one from Pohang. They selected the isolates for strong saccharification activity: these yeasts were good at helping release sugars from starch. Their genomic and physiological work examined an organism whose contribution could not be reduced to the job usually assigned to yeast in a brewing cartoon.[3]

For the brewer, functional cooperation need not follow our categories. An organism can matter because it prepares food for another organism. Its importance might be poorly represented by how much of it remains when someone samples the finished drink. The enzyme it produced earlier and the cell counted later are different objects of measurement. That distinction is elementary, but it changes what a laboratory list can tell us about a cake.

Return to the forty-two samples. Song and colleagues also recovered the mold *Aspergillus oryzae* from thirteen. Neither number licenses a declaration that the other cakes contained no useful mold or yeast. Isolation is an encounter between a sample and a method. Organisms must survive, grow under the chosen conditions, and be recognized. A failure to recover one species is not equivalent to proving that no cell of it was present. Nor does its recovery establish that it performed most of the work during a later brew.

A good detective asks how the witness came into the room. In this case the witness is a colony on a laboratory medium. The conditions that made it visible are part of the testimony.

The makgeolli time course asks a different question. Its researchers followed a controlled batch through successive days, measuring a changing fermentation rather than comparing many dry starters at one moment. Bacterial composition changed, especially over the first two days. The fungal community showed no significant change, and *Saccharomyces* remained dominant throughout. This was not a record of that yeast emerging from obscurity to displace its rivals. Its presence from the start makes the additional inoculation central to interpreting the result.

The attractive but unsupported version of the story would say that a diverse traditional cake naturally sorts itself until the familiar brewing yeast takes command. That may be a hypothesis to examine in an appropriate experiment. It is not the conclusion supplied by this particular comparison. The missing ingredient in our interpretation was printed in the method, available before any speculation began.

The next cake takes us to another set of places and another research question. Pooja Pradhan and Jyoti Prakash Tamang studied dry fermentation starters from the eastern Himalayas in work published in 2019. Their thirty-five samples included starters called marcha from Nepal, Darjeeling, Sikkim, and Bhutan, alongside separately named starters from Arunachal Pradesh and Bhutan. They obtained 201 bacterial isolates and selected sixty-eight representatives for genetic identification. This was an investigation of recoverable bacteria, not an inventory of every yeast, mold, and bacterium inhabiting every Himalayan starter.[4]

The distinction between a starter and a strain is especially clear in their account of making new cakes using material from an earlier starter. The practice passes something biological forward, but it also passes a technique: what material to prepare, how to incorporate the inoculum, when to dry it, and how to judge the result. A cake is both an ingredient for a future fermentation and evidence of work already done.

There is no reason to assume that this transmission preserves every organism at a fixed proportion. Each new preparation presents conditions in which some organisms may grow more readily than others. Drying and later use introduce further transitions. Continuity can therefore reside in a maintained practice without requiring an unchanging microbial membership. The human task is not made less skilled because the system is dynamic. Working with a changing material may be precisely what demands judgment.

Nor do the shared laboratory questions turn the named starters into interchangeable examples of one recipe. A researcher can investigate bacteria from marcha and phab in the same study while retaining the names, collection places, and boundaries of the samples. A reader should be able to do the same. A common analytical instrument does not create a common cultural identity.

This creates a problem for preservation. Imagine, as a thought experiment, conserving a single isolate from a starter. The culture may be scientifically valuable. It may make a repeatable product. But what exactly has been saved? One organism has been preserved. The relationships among organisms may not have been. The grain preparation has not necessarily been taught, and the reasons a maker accepts one cake and rejects another have not automatically entered the freezer.

Now imagine preserving only a written recipe. Quantities and operations survive, but the material may respond differently when ingredients, humidity, equipment, or available inoculum change. A recipe can guide practice without exhaustively containing it. Neither of these imagined preservation efforts is

useless. Each simply preserves a different portion of the thing we were calling a starter.

A public heritage document recognizes a wider frame. In 2021 Korea's Cultural Heritage Administration described the national designation of makgeolli making and sharing. Its account includes preparation and social practices of offering and sharing the beverage. That designation does not establish a microbial specification for nuruk, and it cannot speak for every maker's preferred technique. It does make visible a unit of continuity larger than a species name: the acts through which people produce a drink and bring it into relationships.[5]

There is a temptation to make this sound mystical, as if acknowledging knowledge beyond a recipe required surrendering to the unknowable. Nothing in the evidence requires that move. Temperatures can be measured. Organisms can be isolated and sequenced. Enzyme activity can be tested. Makers can explain choices, disagree with one another, and revise procedures. What changes is the scale of the inquiry. A chemical measurement and a conversation about acceptable results would answer related questions without becoming substitutes for each other.

The next useful study would not begin by announcing that the cake had finally been decoded. It would state what had been collected, who made it, what question the makers considered important, and which transformations the measurements followed. Those details would make a result more portable because they would show the conditions under which it was obtained. They would also make disagreement informative. A brewer's objection could reveal that the laboratory had measured a successful fermentation by a different standard from the one that mattered in use.

At the beginning, we seemed to be looking for a missing yeast. What we actually found was a missing distinction: between a dry starter and a fermenting drink, between recovery and absence, between a mixed preparation and an added selected culture. Once those distinctions return, the cake becomes more interesting. It is not waiting to be reduced to a strain name. It is a working arrangement whose parts become intelligible when we ask what they do together.[6]

18

The Beer Made in Two Directions

The revealing detail is the liquid set aside.

During preparation of mso, a finger-millet porridge, a brewer removes some warm, thin mixture before adding more flour. Later, that reserved liquid helps loosen the thickening contents. Cold water could thin them too. The brewer explains the reason for keeping the liquid warm: the temperature must not fall. Firewood is added or removed as the work proceeds.[1]

This is the sort of operation an ingredient list loses. It records water twice as the same substance. In the brewer's sequence, however, the liquid's thermal history matters. Water that has already been warmed can perform a different job from water fetched cold. The bowl or container holding it is, for a while, part of the heating system.

Ryosuke Kubo and Method Kilasara documented the process in Kilema kaskazini ward in northeastern Tanzania during field visits in 2015 and 2016. Their subject was mbege. Its two preparations followed different paths. The researchers interpreted fermented banana nyalu as a source of active yeast; sweet mso, made with germinated finger millet, supplied sugars. The preparations were combined, after which measured alcohol increased.

The apparent complication is the point. Why do separately what could be described as mixing banana, grain, and water in one vessel? To answer, follow what each preparation needs to become. An ingredient is not ready simply because it has arrived. Something may have to grow in it, something may have to be broken down, or something may have to cool before the next operation can work. The order carries information that the inventory does not.

For mso, the useful change involves starch. Finger millet is not valuable merely as additional bulk. Germination brings starch-converting enzymes into play. These catalysts help make the grain's stored carbohydrate available in forms a fermenter can use. But heat presents a bargain: conditions that encourage a

reaction can, if carried too far or maintained too long, damage the protein doing the work.

Dorothy Machunda Gimbi and Naofumi Kitabatake investigated that bargain in a laboratory study published in 2002. They followed alpha- and beta-amylase activity in germinating finger millet and tested the stability of malt extracts at different temperatures. The two activities responded differently to germination conditions and heating. Beta-amylase in their crude extract was completely inactivated after ten minutes at 70 degrees Celsius. Alpha-amylase had a different pattern of persistence and loss.[2]

A laboratory extract is not a barrel of porridge. The result does not supply one universal setting that a brewer should copy. It does establish why "make it hotter" cannot be an unlimited improvement. The task is to bring about a conversion while retaining useful enzymatic activity long enough for that conversion to occur. The reserved warm liquid now makes engineering sense without requiring us to pretend that the laboratory invented the brewer's procedure.

The experiment also complicates a familiar compliment. Calling a practiced brewer an instinctive scientist may sound generous, but it can quietly suggest that the knowledge only becomes respectable when translated into a laboratory vocabulary. A more direct account gives each contribution its proper owner. The brewer controls a process and explains a choice. The laboratory investigates mechanisms under specified conditions. Their findings can illuminate each other without one becoming a certificate of the other's intelligence.

Finger millet itself deserves another look. In a separate study published in 2016, Kubo combined fieldwork in Bupigu, in southern Tanzania, with comparative enzyme tests. The research found strong alpha-amylase activity in the finger millet tested and connected it with the grain's usefulness in brewing. Sorghum also performed strongly, a qualification easily lost if the title's question is turned into an advertisement for one cereal. Villagers' reported judgments about the drink and a laboratory measurement were brought into the same investigation.[3]

This helps explain a selection without exhausting it. Suppose one ingredient releases sugar efficiently but costs more fuel to prepare, or another produces an acceptable drink that customers prefer. Those are hypothetical choices, not additional findings from Bupigu. They show why maximizing a measured enzyme activity cannot, by itself, identify the best ingredient for every household. A crop enters a working economy of available land, labor, fuel, taste, and other uses. A useful biochemical property matters within that economy.

Kubo's earlier Bupigu research, published in 2014, provides a closer process comparison. It described three drinks, komoni, kimpumu, and kiambule, involving separately prepared fermented and sweet porridges. The names recorded there were nyambo and kikonde. Germinated finger millet again supplied starch-converting capacity, and temperature management mattered. The architecture resembles the mbege process, while the ingredients, drink names, and locality remain distinguishable.[4]

That resemblance is enough to generate a question about a shared technical problem. It is not enough to establish who taught whom. Similar arrangements can be transmitted, adapted, or independently developed; deciding among those histories requires evidence about contact, chronology, and learning. None follows simply from drawing two arrows into one mixing vessel. Comparison is strongest when it identifies exactly what is shared and leaves the route between places open to investigation.

The separation also makes us notice two different clocks. One preparation can be developing a fermenting population while another is being made suitable for that population to use. The final combination coordinates their readiness. A stopwatch started only at mixing omits the work that made the apparent beginning possible. This is why a brisk statement that a beer "ferments in a few hours" can distort a process even when a later measurement really was taken a few hours after mixing.

The same issue appears in accounts of labor. The visible moment when two preparations meet is easy to describe. Planning their readiness is harder to photograph or reduce to a recipe line. It includes knowing which operations can wait and which create the next deadline. Recognizing that coordination does not require inventing a brewer's private thoughts. It follows from the dependencies of the operations themselves.

There is another ingredient in the Kilema investigation. The researchers compared batches with and without msesewe, identified as *Rauvolfia caffra* stem bark, under a local brewer's direction. Their observations supported the reported acceleration of fermentation. The proposed microbial explanation remained an inference. That distinction matters: an observed effect, a mechanism, and a generally applicable quantity are three separate findings, not one.[5]

An experiment can give a local account a new kind of evidence while still leaving important work unfinished. Here the useful question is what the comparison isolates. A change associated with an addition deserves attention. It does not automatically tell us which compound acted, which organisms changed,

or how a different batch would respond. Treating those questions separately protects the value of the actual result from the burden of claims it never tested.

Step farther from the vessel and even the word *banana* acquires more context. Andreas Hemp's 2006 study examined sixty-two coffee–banana plantations on Kilimanjaro. Its abstract describes Chagga homegardens with layered vegetation and substantial plant diversity, including many uncultivated species. These were managed agricultural landscapes rather than a single crop occupying empty space. The paper also identified pressures involving wood demand and changes in coffee cultivation.[6]

Those observations are not a production ledger for the brewer in Kilema. They cannot tell us where every ingredient in the recorded batch grew. They do, however, challenge the convenience of imagining a raw material as something that simply appears at the brewery door. Brewing begins downstream of other decisions. Which plants are retained, what land supports, and which uses compete for fuel all belong to the conditions under which a technique can continue.

A different inquiry moves beyond manufacture. Anthropologist Knut Christian Myhre's 2015 article concerns Chagga-speaking people in Rombo District. Its abstract explores "showing": how substances including finger millet and beer can disclose relations to past and future. Beer enters the inquiry through what it helps people understand about their social world.[7]

Rombo is not simply another name for the Kilema research site. A technical description from one locality cannot be inserted into an ethnographic argument from another as if the two authors had watched the same event. Keeping the studies apart leaves room for both. Beer can be investigated as a sequence of transformations and as a substance through which relationships become intelligible. Neither inquiry cancels the other.

The warm liquid remains beside the work until it is needed. It has a purpose that becomes visible only when we stop reading the recipe as a list. Its importance lies in what happened before it was set aside and what must happen after it returns. To understand this beer, we have to recover those relationships. The two preparations make one drink because someone knows how to bring their different histories together.[8]

PART IV

The Unfinished Case

Laws, anniversaries and expectations shape what beer seems to be. Beyond them lie the practical questions of keeping it cold, growing its ingredients and deciding what knowledge to carry forward.

19

The Three-Ingredient Alibi

Barley, hops, water. It is a wonderfully compact declaration of innocence. Three familiar things go into the vessel; a respectable drink comes out. Nothing mysterious has been added. Nothing requires explanation. Put the year 1516 beside the list and even the calendar appears to vouch for it.

Then the beer starts fermenting.

Whatever a law says about ingredients, yeast is doing work. That awkward fourth presence makes the Bavarian beer ordinance an unusually good test of how we read the past. Is the missing name evidence that brewers lacked practical knowledge of fermentation? Or have we asked a legal restriction to do the job of a microbiology textbook? The first answer makes a neat anecdote. The second requires opening the document.

The Library of Congress reproduces the ingredient provision from the Bavarian ordinance of 1516. It names barley, hops and water, and follows the restriction with confiscation of an offending barrel. This is a command with a penalty. It is addressed to people who already know how to make beer, not to beginners seeking instructions. The dukes do not explain malting, mashing or fermentation. They do not specify how long to heat anything. Nobody would conclude from these omissions that the brewers had never warmed a vessel.[1]

A restriction works by selecting what its authors think needs restricting. Its silence is therefore different from the silence of an experimental report. If a town prohibits selling underweight loaves, the ordinance does not need to explain how dough rises. Likewise, a prohibition on certain brewing materials can take a body of working knowledge for granted. There may be further historical reasons for a particular omission, but the omission alone cannot supply them.

This matters because the three ingredients now carry a much larger burden. They are asked to guarantee quality, authenticity, continuity and sometimes the superiority of an entire brewing culture. Each is a different proposition. A drink can meet a short ingredient rule and still be badly made. A drink outside that rule can be carefully made. Neither observation decides whether a jurisdiction had good reasons to regulate its brewers.

The historian Klaus Rupprecht puts the famous provision back inside its larger setting. The 1516 ordinance concerned prices and public order as well as ingredients; similar restrictions preceded it. He also identifies a particularly inconvenient earlier document: a Bamberg brewers' oath of 1489 permitted malt, hops, water and yeast. Practical recognition of yeast therefore cannot be dated by its absence from the better-known Bavarian list.[2]

The Bamberg detail is more revealing than an easy joke at the expense of sixteenth-century science. It separates knowing a useful brewing material from understanding the organism involved. A brewer can preserve, exchange and employ a substance without possessing the later account of what it is. Modern people do this constantly. Most of us can use a medicine or a computer without explaining its operation at every level. Competent practice and theoretical explanation have their own histories.

There is another trap here. Once the missing yeast has been noticed, it is tempting to make the whole law ridiculous. That would be a remarkably poor reward for discovering more evidence. The ordinance's authors were not trying to anticipate every question a modern historian might ask. They were making a rule for a particular territory. Its restriction was real; its later reputation is a different object of study.

Even the name has a history. Rupprecht traces *Reinheitsgebot*, the familiar word for purity law, to the early twentieth century, with an occurrence in 1909. A much later vocabulary helped gather earlier provisions into a memorable national story. He also describes changing permissions and wheat-brewing privileges, including the ducal wheat-beer enterprise after 1602. The historical rule was administered, disputed and altered, rather than suspended outside time.[3]

This is not unusual behavior for a tradition. A tradition must be carried by people who make choices about it. Some elements are emphasized, others quietly recede, and a new name can make scattered practices look like one continuous thing. The question is whether the resulting account remains answerable to the records. The year on a label cannot do that work by itself.

Consider what happens when barley becomes a moral category. In the ordinance it is an authorized material. In a sweeping story about purity it may become the proper grain for beer everywhere. That leap would place the sorghum brewers of Wa, the maize brewers of Peru and makers using rice-based starters on the wrong side of an argument in which they were never participants. The problem is not that Bavaria regulated beer. It is the export of a local rule as a universal definition of honest fermentation.

The issue becomes clearer when we separate the questions a drink might answer. What was put into it? How was it made? Who had permission to make it? What did the buyer understand the name to mean? Is it safe? Does it taste good? A single word such as *pure* can slide between all six without announcing the change. Its persuasive power comes partly from this mobility.

In 1893, German judges encountered an ingredient that no elegant account of beer would want. As summarized by legal specialist Jenny Gesley from the reported court decision, a cat had entered a brewer's mash. The lower court's treatment of the resulting beer was overturned on appeal. The dispute turned on what counted as spoiled beer, and the higher court challenged the suggestion that animal remains could be treated as an ordinary brewing constituent. We know this case here through Gesley's account, not a fresh examination of the original court volume.[4]

The episode is grotesque, but its logical difficulty is useful. An ingredient list does not make every accident impossible. Nor does a successful fermentation turn every input into something fit to drink. Legal compliance, product quality and the events inside a particular vessel require different kinds of evidence. The court had a concrete case to decide. A slogan, however venerable, could not decide it unaided.

Nearly a century later, a dispute over the word *beer* brought the question into another courtroom. In 1987, the European Court of Justice considered German restrictions on beers lawfully made in other member states. The case distinguished rules governing the designation *Bier* from restrictions involving additives. It also considered consumer information and whether public-health arguments justified the barriers. The resulting judgment concerned the obligations of a state within a trading community, not the merits of anyone's favorite pint.[5]

That distinction is easy to lose. A decision about access to a market can be retold as a verdict that traditional brewing has been defeated, or that every restriction was foolish. Neither follows. A court can ask whether buyers could be adequately informed without excluding a product. A brewer can still prefer a particular ingredient discipline. A drinker can still value it. What changes is the claim that one country's definition must settle what its neighbors may call their beer.

The old and new disputes have a family resemblance. Both concern control over a name and the economic opportunities attached to it. But they are not the same legal event continuing unchanged across five centuries. Institutions,

territories, markets and rules have changed. To make the connection responsibly, we need to retain those changes rather than polish them away.

The most persuasive defense of an ingredient tradition is also the least extravagant. Constraints can be productive. A limited set of materials can direct attention toward distinctions in grain, water, fermentation and handling. Someone may take pleasure in mastering that discipline. None of this requires proving that other traditions are impure, that the same provision always operated everywhere, or that a brewery has reproduced an ancient flavor without alteration.

We can admire a choice more clearly once it is described as a choice. The reader who has met starter cakes and sorghum households no longer needs the entire fermented world to pass through a Bavarian doorway. The ordinance becomes more interesting at its actual scale: a territorial intervention in price, supply and manufacture, later recruited into a powerful account of identity.

Read the three-ingredient list again. It still says something. It identifies materials its authors chose to authorize in a specific legal setting. It does not supply a complete recipe, a taste judgment, a safety certificate or a universal history. Restoring those boundaries does not empty the document. It returns the human decisions to it.[6]

And the yeast, which has been waiting patiently through the argument, continues its work.

20

Thirty-Six Barrels and a Date

The beer in the square makes an excellent opening. Thirty-six barrels, declared unfit to drink, emptied in Pilsen in 1838: an institution admitting, in public, that its product has failed. A few years later comes the golden replacement. Waste, outrage, invention, triumph. The story seems almost to have edited itself.

It has certainly been edited by other people. In one museum account, angry townspeople storm breweries and seize the barrels. In a historical account translated from the brewery's anniversary volume, officials order the beer poured away. These are not interchangeable scenes. Both give a writer beer running through a square; only one gives the writer a mob. Before choosing the more cinematic version, we should ask where the action came from.[1]

Evan Rail did something especially useful for this case. Instead of simply repeating the brewery's retrospective story, he set its fiftieth-anniversary account beside the founding appeal of 1839. His 2012 article reproduces and discusses both. The contrast moves attention away from a moment of genius and toward an uncomfortable commercial problem. Pilsen's brewing-right holders feared for their income, and they had complaints about the arrangements under which their beer was being produced and sold.[2]

A right to brew is an asset. It can be valuable even when its holder is not personally standing at the mash tun. That distinction makes the phrase "the citizens founded a brewery" less simple than it sounds. Which citizens? With what rights? Who supplied the skilled work? Who controlled the income? A collective enterprise can still have boundaries, beneficiaries and disputes.

One beneficiary has a name. Rail identifies the innkeeper Václav Mirwald, whose tavern, U Zlatého Orla, became a meeting place for discussion of the proposed brewery. Mirwald held brewing rights as well as selling beer. His stake consequently had two sides: the income attached to those rights and the trade across his own counter. The institutional problem was also a proprietor's problem. A dependable local supply could matter to him twice over.

The 1839 appeal matters because its authors had not yet enjoyed the subsequent success. They could not write with the comfortable certainty that

their project would become famous. A document prepared to secure support is not free of persuasion, but its persuasion has a different target from an anniversary history. One asks people to invest in an uncertain remedy. The other explains why the remedy deserved to succeed.

Rail observes that price and competition figure prominently in the earlier evidence. Beer from outside Pilsen was competing with the local product, and the burghers wanted to recover their position. His discussion also corrects the idea that bottom fermentation arrived in Bohemia with the new brewery. The technique already had a history there. What emerged in Pilsen therefore needs a more exact description than "lager was invented."

This does not diminish the achievement. It changes its shape. Borrowing a technique, organizing production and making a successful product are substantial activities. They are just harder to commemorate with a single portrait. An inventor can be put on a label. A complicated settlement among people with brewing rights requires smaller type.

The date that survives this correction is 5 October 1842. Plzeňský Prazdroj's chronology identifies it as the day Josef Groll brewed the first batch at the new Burghers' Brewery. The company also records the subsequent movement of its beer beyond Pilsen. These are useful statements of institutional memory, though a producer's confidence about its own history does not remove the need to distinguish a first batch from a first-ever process.[3]

Nor was brewing day the same as drinking day. Grain had to be processed, wort fermented and the beer allowed time before service. A dated company account places the first tapping on 11 November, during St. Martin's Day festivities. Remembering both dates restores a little of the waiting that an invention story tends to erase. A fermentation cannot be hurried into a single afternoon merely because a chapter ending is approaching.[4]

Groll's name belongs in the account. It simply does not belong there alone. A brewmaster works within a material and institutional arrangement: a site, vessels, water, grain, labor, fuel, cooling, owners and customers. Each imposes constraints. The meaningful question is how a particular brewer and brewery brought those constraints together. "Genius" can be an invitation to investigate the combination, or a way of stopping before we have begun.

The familiar technical explanation emphasizes pale malt, local water and Saaz hops alongside bottom fermentation. Zuzana Lízcová's account for the Goethe-Institut presents that combination as a Central European story: a Bavarian brewer working with Pilsen's citizens and local materials. This is a better starting point than a contest over which modern nation owns the invention. It

recognizes movement across a region whose later political boundaries should not be projected backward without care.[5]

Yet even a plausible combination can be turned into a formula for inevitability. If we list the right water, malt, hops and yeast, does the famous result follow automatically? Of course not. An ingredient inventory leaves out processing and organization, just as a list of instruments leaves out a performance. Another brewer using similarly named materials could make a different beer. A difference in timing or handling could matter without leaving a place in the anniversary story.

The same caution applies to the claim that success happened by accident. An accidental outcome can be documented. But calling an unfamiliar result an accident is not evidence of one. To establish the case, we would need some record of what the brewer intended and how the batch departed from it. A later narrator's delight at a golden beer cannot supply Groll's private surprise. We can describe the product's importance without writing the expression on his face.

The German Patent and Trade Mark Office's retrospective account identifies another easily overlooked fact: Groll's employment in Pilsen lasted only three years. The brewery continued with other brewmasters from Bavaria. A durable enterprise therefore cannot be explained entirely by the continuing presence of its celebrated first brewer. The knowledge had to survive his departure, and the organization had to keep making and selling the product.[6]

That is the point at which a history of invention becomes a history of maintenance. Maintenance has fewer heroic verbs. It involves repeating tasks, correcting faults, replacing equipment and preserving the conditions under which a reliable result can happen. Its achievement is often that customers notice very little. A brewery which produces one remarkable batch has solved one problem. A brewery that keeps supplying people with beer they recognize has solved many more.

Distribution supplies another missing set of characters. The Prazdroj chronology describes railway wagons, sidings and later refrigerated transport. Such details are more than scenery around a successful drink. Beer that cannot reach a customer in acceptable condition has a limited commercial world. The enlargement of that world required containers, transport and cooling as well as a recipe. The popularity of a beverage is partly a history of the journeys it could survive.

We should therefore be wary of letting later reach rewrite early ambition. Once a beer has traveled widely, its founders can start to look as though they had planned the map in advance. The surviving appeal deserves to be read at the scale

of the problem it addressed. A local commercial remedy can acquire consequences far beyond the intentions recorded at its beginning. Those consequences are no less impressive for being unplanned.

The word *Pilsner* presents the final turn. A name connected to one place can become a term used for a kind of beer, while a producer seeks to distinguish its own product from the category. The company's chronology records the 1898 registration of the Pilsner Urquell name in response to the widening use of Pilsner. Success had created a naming problem: the place had become so influential that its name could no longer always point to one brewery.

A style name is useful precisely because it travels. It allows people to compare products, form expectations and attempt their own versions. A brand name has a different job: it directs the customer back to a particular commercial identity. Their tension does not mean one usage is fraudulent and the other pure. It explains why a triumphant origin story may continue to need careful wording long after the original beer has been sold.

Return to the thirty-six barrels. There is no need to fill the square with invented shouting. What survives is already interesting: a condemned product, competing accounts of its disposal, a group of rights holders confronting a threat to their income, a new organization, a named brewer and a dated batch. The decisive changes occurred across years, in documents and work as well as in a kettle.[7]

The date remains memorable. It now marks a point in a larger process, not a hole in history through which lager suddenly appeared. Groll can keep his place in the story. Around him there is room for everyone who made the brewery possible, and for the people who kept it working after he left.

21

The Water in the Bowl

The laboratory needed a car battery. In a Shawi community in the Peruvian Amazon, researchers used it to power a portable incubator while testing water and masato, a fermented cassava drink. The detail is tucked into the methods of a paper. It is also a useful reminder of what field research requires: equipment, local cooperation and a way to do the work in the place where the question matters.[1]

The result was striking. All twenty-six sampled waters, comprising nineteen household samples and seven outdoor sources, were positive for coliform bacteria. Two of eleven masato samples were positive. Nine were negative in the tests. One can almost see the headline forming: fermentation makes unsafe water safe.

But that sentence has traveled further than the experiment.

The difference between the sample groups deserves attention. So does the distance between a promising observation and a guarantee. The paper, by Alejandra Bussalleu and colleagues, appeared in 2020 and concerned testing conducted in November 2015. It was an exploratory investigation in one community. It did not establish that every bowl of masato, or every fermented drink, was safe under all the conditions in which people might prepare and consume it.

There is a simple numerical reason to slow down. The result compares the proportion of samples with detectable coliforms. It does not mean fermentation removed a particular percentage of all hazards. Nor can a comparison between water samples and finished drinks, on its own, isolate the effect of one step in preparation. To identify that effect, we would need to know what happened to particular contaminants through a controlled sequence, including everything added or done along the way.

The researchers could not confirm how long each sampled masato had fermented. That missing timing matters because a finished drink records the combined effects of preparation, fermentation and handling. The eleven samples cannot be arranged into a before-and-after experiment merely by calling one

group water and the other masato. The observation gives the next study a starting point, not a completed process map.

Coliforms are useful indicators in water investigation, but a test directed at one group is not a census of all biological and chemical hazards. A negative result answers a question defined by the method and its detection limits. It does not announce the absence of every possible danger. If that distinction seems fussy, imagine being told that a building is sound because one kind of defect was not found. You would want to know which part had been examined.

The fermentation itself is still relevant. Different processes can produce conditions that inhibit particular unwanted organisms. The difficulty lies in making the conditional statement disappear. "This organism did poorly under these conditions" is a result. "Fermentation kills germs" is a slogan. The slogan removes the identity of the organism, the conditions of the test and the difference between reducing a population and eliminating a hazard.

A separate experiment in contemporary beer makes the problem unusually clear. In 2023, Mario Çobo and colleagues reported a challenge study using low- and nonalcoholic beer matrices. They deliberately introduced selected foodborne pathogens and varied conditions including alcohol concentration and storage temperature. Some organisms could survive, and in certain nonalcoholic conditions they could grow. Their results differed among the tested organisms and conditions.[2]

The useful connection to masato is a question, not a transfer of the laboratory's results. Neither drink is an interchangeable substitute for the other. A controlled challenge study can ask what happens when a known organism is deliberately put into a defined matrix. Field sampling asks what is present in actual samples collected under actual conditions. Each method reveals something the other cannot. Neither becomes stronger by pretending to answer the other's question.

The distinction between survival and growth is especially important. A population may decline without disappearing. It may remain detectable while failing to multiply. Different statements about the same organism can therefore sound contradictory when they are not. A product that does not support growth has not necessarily destroyed whatever contamination entered it. A quiet microbial population can still matter.

Beer's ordinary obstacles to unwanted growth are likewise a combination. Acidity, alcohol, hop compounds, processing and storage can contribute in different ways. Remove or alter one component and the remaining system needs to be understood on its own terms. This is why reducing alcohol is not simply

subtracting a number from a label while leaving every other property untouched. The result can be an excellent beverage, but its stability has to be established rather than inherited by name.

Water treatment offers another way to see the same problem. The US Centers for Disease Control and Prevention describes how utilities account for germs entering water before treatment and during its journey through pipes. Some disinfectants are used partly because protection must persist through that distribution system. The general lesson is about a chain of events: what happened at an earlier stage does not settle what can happen afterward.[3]

The masato account contains just such a later step: fermented mash is mixed with water to prepare the drink. The sequence makes water quality relevant again at the point of dilution. It also sharpens the next investigation. What happens in the mash, what happens after water is added, and what happens during service are separate questions. They would deserve the same attention on an industrial filling line. A useful preparation step cannot be asked to guarantee everything that follows it.

At this point, however, a purely microbiological detective story would have missed half the investigation. Masato was not an anonymous liquid supplied for a test. A later photographic phase of the Shawi project documented the work and social setting of its preparation. Community members helped choose the messages and photographs, then validated the final selection at dissemination workshops in January 2018. The work placed the beverage within daily life rather than treating a bowl as an isolated laboratory container.[4]

That changes what a sensible research question looks like. If a drink is woven into work, food and hospitality, an intervention which ignores those relationships may solve a different problem from the one people experience. Knowing why a preparation matters is not a decorative supplement to knowing what grows in it. It affects which changes are meaningful, acceptable and possible.

It also changes who should be allowed to define success. An outside observer might prize one measurement while missing the time a new procedure requires, the labor it reallocates or the foods and relationships it displaces. The appropriate response is not to romanticize every established practice. It is to investigate with the people whose lives contain it. Respect does not require declining to measure risk. Measurement does not confer a right to disregard people's priorities.

The authors' discussion treats masato as a potentially valuable local practice deserving further investigation. This book's restraint about a safety guarantee should not be mistaken for rejecting that possibility. A finding can be hopeful

and incomplete at the same time. Indeed, the reason to do further research is that the preliminary observation is interesting enough to deserve a better answer.

The next questions would have to distinguish mechanisms. Which stages account for the observed differences? How much variation is there among preparations and occasions? Which organisms and other hazards remain unexamined? What measurements would actually connect a change in preparation with a change in health? Such questions require properly designed research and community participation; they cannot be answered by a visitor's confident impression of a bowl.

There is an older European story waiting nearby: people drank beer because water was unsafe. Its appeal comes from giving every historical drinker a sensible medical reason. But even before examining the documentary record, we should see how much the sentence assumes. It imagines uniform water quality, uniform beer, a dependable preparation method and a single motive for choosing a drink. The field study cannot retrospectively prove those things for another continent and century.

Nor should beer become the standard against which masato must explain itself. Cassava is a root crop, and the Shawi drink has its own vocabulary, work and cultural setting. It appears here because the investigation exposes an assumption shared across discussions of fermentation. Its relevance does not depend on reclassifying it as a European-style beer.[5]

The car battery gives this case a more fitting ending than the sweeping headline. People brought a question, instruments and a collaborative method to a specific place. They obtained a result worth pursuing. The work made something clearer without making every uncertainty vanish. That is often what discovery looks like before it is rewritten into a story about a miracle.

22

The Beer the Eyes Ordered

For three days in December 2016, visitors to the Musical Instruments Museum in Brussels could take part in an experiment involving beer. The researchers had prepared two color-adjusted versions of the same base blond ale, one pale and one dark. Before the main experiment, a small sensory test, with the colors hidden, had failed to show that tasters could distinguish the beers by flavor alone.[1]

Now the eyes were allowed back in.

The dark beer was expected to taste more bitter and stronger, with more body; participants expected to like the pale beer more. After tasting, those gaps narrowed, and the reported comparisons found no statistically significant difference between beers on those ratings. The first impression had not simply carried through unchanged. This is not the ending usually attached to a story about the senses being fooled.

One hundred and thirty-six people took part in the main experiment. Its authors, Felipe Reinoso Carvalho, Pieter Moors, Johan Wagemans and Charles Spence, reported the work in 2017. The result showed why expectation and experience need to be measured separately. People can anticipate a difference without continuing to report it after they have had the opportunity to taste.

A dramatic account might stop at the first questionnaire. Dark beer promises more body, strength and bitterness; therefore the eyes have rewritten the tongue. But the second questionnaire was part of the investigation too. Removing it would turn an experiment about the limits of an effect into a demonstration of its power. The missing evidence would be hidden in the edit rather than in the glass.

The experiment's preliminary test also deserves careful reading. It did not prove that the two liquids were chemically identical or that nobody could ever distinguish them. It failed to establish a detectable flavor difference under that test's conditions. This is a narrower and more defensible starting point. The researchers were trying to make visual appearance the salient contrast, not proclaim a universal equivalence between every aspect of the beers.

That modest difference in wording becomes important when the story leaves the laboratory. “Indistinguishable” can be read as a description of a test result, or as a permanent property of the world. The first interpretation keeps the participants and method in view. The second quietly removes them. Sensory experiments are especially vulnerable to the removal because their subject is an experience, and experiences vary.

The Brussels result does not settle the larger question in favor of an infallible tongue. Later beer experiments reported effects of cues on perception. In a 2021 paper, Helena Blackmore, Claire Hidrio and Martin Yeomans examined beer color and sensory descriptions as sources of expectation. Their published abstract and highlights report changes in expected and perceived properties, with expectations mediating the relationship. Here, some of what happened before tasting mattered afterward.[2]

In 2022, the same research group reported a study of eighty-seven regular beer drinkers that varied actual alcohol content, the alcohol information supplied to participants, beer color and sensory wording. Its results separated different judgments. A descriptor could influence the attribute to which it referred; liking did not simply follow every color or description manipulation. The effect was not an all-purpose dial that could be turned up by making a label more enthusiastic.[3]

The differences among these studies are the beginning of an explanation, not an embarrassment to be concealed. They used particular beers, participants, cues and tasks. Changing any of these can change what a person is being asked to do. Predicting a drink’s body from its appearance is one task. Rating its bitterness after tasting is another. Deciding whether to have a second glass is a third, even if a questionnaire places all the answers next to one another.

The word *expectation* can hide this variety. An expectation may concern strength, refreshment, sweetness, price or the likelihood of enjoyment. These ideas need not move together. A drink can taste as bitter as expected and be less enjoyable than expected. It can seem expensive without seeming strong. One experiment’s effect on a particular rating cannot be casually carried over to every other judgment.

This is why a better question than “Can color fool you?” is “What did the cue change, for whom, and under which circumstances?” The second question sounds less ready for a television demonstration, but it has much better prospects of an answer. It allows both a positive result and the Brussels convergence to belong in the same account.

It also makes room for learning. A dark appearance is not an arbitrary sign in all drinking experience. People may have encountered dark beers whose ingredients and processing gave them distinctive flavors. The trouble begins when an association is promoted into a necessary relationship. A clue that has often been useful can still mislead in a particular glass. The experiment changes the usual relationship and asks how the observer responds.

The observer is not being exposed as a fraud. The mind has to make sense of incomplete information, and prior experience can be useful. Refusing all expectations would mean refusing much of what makes familiar things recognizable. The scientific problem is to understand when a useful prediction guides attention, when it bends a report and when incoming evidence is strong enough to displace it.

In that respect, tasting has something in common with reading a historical document. A famous date, a respected brewery or the word *ancient* supplies an expectation before the evidence has been considered. A reader may begin looking for the explanation already promised by the title. The remedy is not to become blank. It is to give contrary details a chance to matter, as the second set of ratings mattered in Brussels.

The practical world of judging offers a different response to the problem. An educational guide by Edward Wolfe for the Beer Judge Certification Program discusses controlling distracting smells, the tasting environment and the way samples are presented. It also recognizes a tradeoff: revealing a bottle may influence a judge, while transferring beer before service can lose fleeting aromas. Even a controlled presentation requires choices.[4]

The guide is professional advice, not a randomized experiment, and much of it has been superseded within the judging program. Its value here is the question it makes visible. What are we trying to preserve, and what are we trying to conceal? Hiding the label can reduce one cue while leaving color, temperature and the social situation intact. A blind tasting is a designed condition, not an escape from having a body in a room.

A competition also needs to decide what its scores mean. It may reward a beer for meeting a stated style description. A consumer study may ask how much people like a sample. A discrimination test may ask which sample differs. These tasks cannot be exchanged simply because they all involve small glasses and concentration. The best-liked beer need not be the closest match to a competition category, and a detectable difference need not be an improvement.

There is a corresponding mistake on the other side: assuming that any contextual influence makes an experience unreal. Suppose a person enjoys a drink

more in one setting than another. The observation does not mean the enjoyment was counterfeit. It means the experience includes a setting. The narrower claim that a particular compound increased in the liquid would still require chemical evidence. Enjoyment and composition answer different questions.

This gives the detective a useful way to keep a notebook. Record the sensation separately from its proposed cause. “I perceive banana” is a report. “Someone added bananas” is an explanation. “This looks strong” is an expectation. “It contains more alcohol” is a claim about composition. The distance between those sentences is where the investigation happens. The chapters about esters and bitterness have already supplied reasons to keep it open.

The same discipline protects pleasure. One does not need to perform a controlled experiment at every meal or apologize for liking a label. The aim is to know which kind of statement is being made. A personal preference can remain personal; it becomes troublesome when presented as proof of a recipe, a chemical measurement or the inability of everyone else to taste properly.

Brussels leaves us with a small, satisfying surprise. An appearance made a prediction, but the prediction did not entirely dictate the later report. Other experiments show occasions when expectations do shape perception. Taken together, they describe an observer who is neither an impartial measuring instrument nor a helpless victim of packaging.[5]

The dark beer was allowed to make its case. Then the participants tasted it. The second part was essential.

23

The Borrowed Winter

Winter used to arrive at a brewery in pieces. Someone had cut it out of a frozen pond or lake, lifted it onto a cart and brought it to a building where it might survive the weather that made it. An ice house held an agreement between seasons: the cold months would lend something to the warm ones. The agreement was always vulnerable to a thaw.

The University of Chicago Library's exhibition on brewing technology places harvested ice at the beginning of its refrigeration story. Before mechanical systems, brewers stored lake and pond ice for later use. The cellar's temperature depended partly on an inventory collected elsewhere and earlier. What might look like a static building was the last stage of a supply operation.[1]

Ice did its work by accepting heat. As it melted, the stock diminished; the disappearance was part of the service being purchased. That gives the old trade a peculiar quality. A successful delivery arrived with as much of the commodity as possible still present, but successful use required that commodity to change. The brewer was paying to manage a loss.

Géraldine Lysser's history for the Swiss National Museum follows the labor behind that arrangement. Workers marked frozen surfaces, sawed blocks free and moved them by cart to railway stations. Insulating materials helped protect the cargo. Some breweries maintained their own ponds, but local supplies could fail to meet demand, sending buyers toward Alpine sources. The article records Hürlimann's installation of mechanical refrigeration in 1879, a step toward cooling that could be scheduled independently of an ice harvest.[2]

The geography matters. A cold place had something to sell to a warmer one, and transport could turn a local winter into a distant advantage. The brewery's apparent command over temperature was distributed across landscapes, workers, buildings and routes. A hillside cellar concealed those connections rather effectively. Once the ice was inside, its journey could disappear from view.

Norwegian beer exports offer another angle on the same network. Curator Ingebjørg Eidhammer describes research in the Ringnes brewery archive, which also holds records from other Oslo breweries. Her account connects

nineteenth-century exports with shipping contacts, steamships, railways and the availability of natural ice. The ice served production and helped keep beer cool during transport. Captains could carry samples to potential markets before a regular trade developed.[3]

That last detail is a reminder that distance is partly an organizational problem. A ship needs a cargo, a seller needs a customer and a perishable product needs conditions under which it can arrive in useful form. Cooling belongs in the commercial story because it changes what a journey can accomplish. The route is more than a line between two dots on a map.

These records do not describe every brewery in every country following the same sequence. Nor does the arrival of a machine imply that natural ice vanished immediately. A technological alternative can exist while older arrangements remain practical in particular places. The useful historical question is when a brewery could obtain dependable cooling on terms that suited its work. A patent date alone cannot answer it.

One influential answer came through Carl von Linde's ammonia-based vapor-compression equipment. The Chicago exhibition identifies his design among the successful early refrigeration systems and reproduces a patent drawing. It also points beyond the brewery: refrigeration in railway cars helped extend the distances over which beer could travel. Temperature control was becoming part of an expanding transport system.[1]

The achievement was not the manufacture of a substance called cold. A refrigeration machine moves heat. It uses energy to take heat from the place that should be cool and release it somewhere else. The bottle inside a refrigerator benefits from an exchange that continues outside the insulated compartment. Anyone explaining the machine only from the cold side has stopped halfway through its operation.[4]

An ice house and a mechanical refrigerator therefore solve related problems through different arrangements. One depends on stored material whose useful capacity is consumed. The other depends on machinery and a continuing energy supply. Each arrangement has its own forms of dependence. Replacing the harvest does not remove the brewery from the world; it changes the connections through which the world reaches it.

There is a freedom in being able to request cooling when it is needed. There is also a bill. Equipment must be acquired, operated and maintained. Production plans can now rely on a service that winter once provided, but they rely on the service continuing. A brewery capable of operating through warm months has escaped one constraint while accepting a new set of responsibilities.

Following the expelled heat leads to the next turn in the story. A factory can need cooling in one place and heating in another. If those needs are considered separately, heat may be rejected while energy is purchased to supply heat elsewhere. The interesting engineering question becomes whether some of those requirements can be connected. The useful resource may already be passing through the site.

A heat pump makes such connections possible by raising heat from a cooler source to a warmer destination. The International Energy Agency describes a system of refrigerant, compressor and heat exchangers that transfers energy between them. The heat delivered can exceed the electrical energy used to operate the equipment because part of the delivered energy has been collected from the source. Nothing has been created from nothing; the accounting includes more than the power cable.[4]

The destination still matters. A stream that is warm to the touch may be too cool for another process. Heat has both an amount and a temperature, and an engineer cannot match demands by adding quantities alone. The location and timing matter as well: a useful supply that appears after the demand has ended may need storage before it can help. Connecting a factory requires more than finding two pipes that feel different.

Roger Padullés and colleagues investigated this problem for breweries in a paper published in 2026. They assembled a generalized brewery from literature and case studies, then compared its heating requirements with opportunities to recover heat within the process. Their three proposed arrangements ranged from replacing a boiler with a centralized high-temperature heat pump to a more integrated system that also compressed process vapor to reuse its heat. The abstract reports differences in energy consumption and investment, with the more integrated designs trading greater complexity and capital cost for improved performance.[5]

A generalized brewery is a useful intellectual construction. It gives competing designs the same starting job, so their differences can be examined. It is also recognizably different from a particular plant with equipment already installed, limited space and a production schedule that customers expect it to maintain. The paper provides a method and comparative designs. It does not report that every brewery has already achieved the modeled result.

The resemblance to the older ice trade is unexpectedly close. Both require someone to connect a source, a destination and a moment of need. The nineteenth-century solution might send a cart toward a pond and reserve room in a cellar. The contemporary solution might redirect a heat stream and decide

whether a storage tank is worthwhile. In each case, the valuable thing is a controlled transfer through time and space.

This continuity should make us cautious about assigning environmental virtue by appearance. A block of natural ice looks innocent; a compressor looks industrial. Yet the first may have required extensive transport, and the second may recover energy that would otherwise be discarded. Their relative burdens cannot be established by the materials' visual character. The necessary evidence concerns what each system did to deliver a comparable service.

Modern emissions accounting makes an equivalent distinction. The Greenhouse Gas Protocol separates a company's direct emissions from those associated with purchased energy and other activities in its supply chain. A reduction inside a brewery does not automatically describe the effects of growing ingredients, manufacturing containers or transporting the finished drink. Those activities belong to the wider investigation, even when they happen beyond the brewery gate.[6]

A comparison between a historical ice house and a modern heat-pump installation would need still more care. What quantity of beer was being cooled? To what temperature, for how long, and with what losses? Where did the ice or electricity come from? The appealing story of technological progress supplies none of those missing measurements. Equally, nostalgia cannot turn an undocumented older supply chain into a zero-emission system.

The history records a change in when and where brewers could obtain cooling. They once purchased a season in solid form. They later bought machines that could maintain conditions on demand. Engineers now investigate how a brewery's heating and cooling requirements might help satisfy one another. The ingenuity lies in recognizing a relationship that had previously been treated as separate jobs.

The machinery has also made the dependencies less visible. An ice block declares where the cold is stored. An electrical connection and a quiet heat exchanger reveal little about the wider system sustaining them. To understand the beer, the investigation has to move beyond the object immediately at hand. The more effortless the result appears, the more tempting it is to forget the work that makes it possible.

One dependence remains stubbornly outside the insulated walls. A brewery can control a tank more precisely than a field. The barley and hops still have to grow under weather that no compressor can rearrange for an entire agricultural region. Following the borrowed winter to its modern successors finally leads out of the building again, toward the plants from which the next batch must begin.[7]

24

The Field Ahead

At a field site on Burray, in Orkney, the barley harvest of 2017 contained a sharp disagreement. Several locally adapted Bere lines produced grain. The modern varieties beside them developed ears that were largely empty. At a second site near Dundee, with more favorable nutrient conditions, the modern varieties yielded much better. Sidsel Birkelund Schmidt and colleagues reported both outcomes in the same study.[1]

It would be easy to choose one of those fields and make it the whole story. Orkney could prove that the old crop possessed wisdom modern breeding had lost. Dundee could prove that the old crop belonged in a museum. Together, the fields ask a more interesting question: useful under which conditions? The answer matters whenever the conditions themselves are changing.

Beer depends on plants that develop through seasons. Its ingredients are living crops before they become materials with specifications. A brewer can adjust quantities and methods, but those choices begin after weather, soils and agricultural decisions have already shaped what is available. The familiar glass stands at the far end of an arrangement that cannot be confined to the brewery.

In 2023 Martin Mozny and colleagues compared hop records from five growing regions in Germany, Czechia and Slovenia across two periods, 1971–1994 and 1995–2018. Yield declined in four regions while remaining comparatively stable in Žatec. Alpha-acid content fell in all five, and cone development began earlier. Their projections, using three climate models and a specified emissions pathway, indicated further risks to yield and alpha content over 2021–2050.[2]

These measurements describe different properties of a crop. A field can deliver fewer cones, or cones containing a different concentration of a useful compound, or both. Neither result translates automatically into a tasting note for a future beer. Brewers buy, blend and process ingredients; they do not simply bottle a weather record. Yet adjustments have material consequences. Keeping a familiar result may require a different quantity or source of ingredients.

The dates distinguish two kinds of evidence. Harvest records concern things that happened. A projection asks how a system may behave under specified

future conditions. The second is useful precisely because the future cannot be harvested early and measured directly. Treating a projection as a scheduled event removes its assumptions; dismissing it because it is conditional removes its purpose. It is an organized way of examining a risk while there is still time to respond.

Responses have to survive contact with farming. Prossie Nakawuka and colleagues tested four hop cultivars under different irrigation levels near Prosser, Washington, during 2011 and 2012. Reducing water generally reduced cone yield, while the measured alpha- and beta-acid concentrations remained broadly similar. Under the study's economic assumptions, the lost revenue outweighed savings in pumping costs. The effect also differed among cultivars and irrigation treatments.[3]

This makes adaptation a set of choices rather than a reassuring word. Saving water, maintaining output and earning enough to continue farming are connected goals, but they need not improve together in a particular experiment. Water that is scarce may have to be rationed even when a crop would yield more with additional irrigation. The farmer is working within a landscape of competing uses and constraints, not solving a single equation labelled more hops.

Barley presents a related problem on a wider scale. Wei Xie and colleagues linked climate, crop and economic models in a 2018 study of concurrent drought and heat extremes. Their scenarios produced losses in barley supply and consequent changes in beer availability and price, with effects that differed among regions. The work examined extreme events under a range of possible climates, rather than forecasting a particular year's bar bill.[4]

The economic step changes the question. A crop loss is a physical event; its consequences for buyers depend on allocation, trade and demand. A percentage reduction in barley does not have to become the same percentage reduction in every product made from it. The chain contains decisions as well as plants. A model connecting the two makes those relationships available for inspection, even while depending on assumptions about how the decisions are made.

It is also a reminder of scale. Beer is one use of agricultural resources among many. Its vulnerability can make an abstract problem tangible to a drinker, but the crop and the people who grow it deserve attention beyond the fate of a favorite label. The investigation becomes more useful when the glass opens a view onto those dependencies instead of obscuring them.

Return, then, to the two Scottish fields. Bere is a six-row barley maintained in the Scottish islands, with variation among the seed lines preserved from different places. The important inheritance is more than a name or a picturesque association with the past. It includes traits that developed under particular soils,

seasons and ways of farming. A review by Peter Martin and colleagues describes that diversity and the crop's continuing uses.[5]

The Orkney trial turned on manganese, a micronutrient needed for photosynthesis. In the alkaline sandy soil, several Bere lines obtained and used it effectively enough to maintain growth. Another Bere line lacked the same advantage. The study's successful plants therefore could not stand for every seed carrying the name. In Dundee, the modern varieties' stronger yield and resistance to falling over supplied the other half of the comparison.[1]

A trait is an ability expressed in circumstances. Calling a variety superior without naming the circumstances conceals the most useful information. The two fields are valuable together because neither result has to be discarded. Researchers can ask how to retain one plant's performance under nutrient limitation while drawing on other traits that make a crop productive and manageable elsewhere. The aim is a fuller set of possibilities.

J. E. Cope and colleagues investigated that variation further in research published in 2020. They screened barley lines in a hydroponic system, measured responses to manganese limitation and compared the results with genetic variation. Some Bere lines performed particularly well. The analysis identified genomic regions associated with the trait, but those regions required further examination and validation before particular genes could be confidently recruited for breeding.[6]

That interval between finding an association and producing a dependable crop is where much of the work lies. A useful line in an experiment is a starting material. A candidate gene is a lead. A variety for farmers must perform across the conditions in which it will be grown, and a brewing ingredient must also meet the demands of processing. The steps connect, but no single promising result performs all of them at once.

The plants were available to investigate because someone had continued growing them. Niamh Mahon, Shawn McGuire and Mofakkarul Islam studied the reasons through interviews with Bere growers and food producers in Orkney and Shetland. Their 2016 paper identified a mixture of motives: markets, cultural attachment, adaptation to local conditions and the crop's suitability for relatively low-input cultivation. Their importance differed between the islands.[7]

Orkney's Barony Mill gives that chain a place and some names. The Birsay Heritage Trust says it was founded in 1998 to bring the abandoned mill back into operation. The mill's history describes Rae Phillips, whose father and grandfather had worked there, passing his skills to Alister Harcus. Grinding Bere into flour, or beremeal, serves a food market as well as a heritage project. A crop's future can depend on someone learning to process and sell its harvest.[8]

The crop's survival also depends on ordinary transactions. A 2023 study by Martin and colleagues traced recent cultivation across the Scottish islands. It reported an increase in Orkney's Bere area associated with milling and malting supply chains, alongside a much more precarious situation in Shetland. In the Outer Hebrides, mixed landrace crops continued to serve animal feeding, with threats to the fields used to produce seed.[8]

There is no single conservation story hiding behind the shared crop name. A growing market in one place does not cancel the loss of growers in another. A seed collection can preserve valuable material, while continued cultivation preserves a different set of relationships among plants, soils and practical knowledge. Protecting both kinds of continuity requires noticing where they are strong and where they are beginning to break.

Nor does a successful old barley remove the need to reduce the pressures producing new risks. Tolerance to one nutrient limitation is not a general shield against heat, drought, flooding or disrupted markets. Its value is more concrete: an ability that researchers can investigate, farmers can assess and breeding programs may be able to use. Calling it a guarantee would make it less informative than the evidence deserves.

The climate studies and the Bere experiments belong together because they keep the future open in different ways. One set investigates how existing arrangements could become more difficult. The other shows why it matters to retain material and knowledge that a single measure of present-day performance might overlook. A system built around only the current winner has fewer answers when the test changes.

This is a modest kind of hope, but it has things to work with: surviving seed, measured differences, people with experience, and experiments that can be repeated or challenged. It asks for continued attention rather than a miraculous ingredient. The barley in Orkney did not solve the future of beer. It made a neglected possibility visible.

Following a beer closely leads outward through an astonishing number of lives and decisions. At the far end of this investigation there is still a plant, making grain in conditions that will not remain exactly the same. Someone will decide which seed to keep and where to sow it. A grain fits between two fingers. What can follow from it still needs a field.[9]

Afterword: An Empty Glass

When the drink is gone, something remains. A vessel needs washing. A producer has accounts to settle. A field will have to be planted again. In time, a container may reach a museum while the work that filled it becomes difficult to recover. What survives is seldom a fair sample of what mattered.

One pleasure of these investigations has been to follow the less conspicuous evidence. A qualification in a methods section can change the meaning of a result. A later source can explain how a story became familiar while failing to establish that it happened. A rule can reveal what an authority wanted without telling us how people behaved. Read closely, such details stop looking like obstacles between the reader and the story. They become events within it.

The work also changes the scale of attention. At one moment, the important distinction lies between two forms of a molecule. At another, it lies between the person named in an account and the people who did the work. Neither scale can do the other's job. A chemical explanation of an aroma cannot establish why a household keeps brewing. An account of a household cannot identify an unknown compound. The better questions let different kinds of evidence meet without making them interchangeable.

This is why a tidy verdict is sometimes less useful than a clearer description of the remaining problem. We can know which experiment would discriminate between two explanations. We can identify the document whose date matters, or the comparison that a small study did not make. We can learn that a proposed adaptation worked in one set of soils and still ask what happens elsewhere. An open case need not be an empty one.

Readers can continue the investigation without accepting an invitation to become suspicious of everything. A useful question is more specific than a general doubt. Which batch was measured? When was the account written? Does the label describe a procedure, an ingredient or a promise? Who supplied the name? Sometimes the answer will confirm the ordinary story. That is a result too. The purpose of checking is to find out, not to guarantee a reversal.

And there is still room simply to enjoy a drink, or to decline one. Knowing how an aroma can arise does not require turning every gathering into a demonstration. The understanding remains available when a question becomes

interesting. It may make a familiar smell newly noticeable, or lend weight to a name on a museum label that would otherwise have passed unread.

The empty glass is an inadequate archive. It says little about the field, the organism, the carrier or the brewer. Yet it has brought those histories close enough to ask about them. Somewhere beyond it, someone is keeping a culture alive, comparing a result, repairing equipment or deciding what to plant. The next clue may begin as a routine detail in that work.

Acknowledgments

This book depends on the authors, translators, researchers, collection staff and practitioners whose work is cited in its notes. Their publications made the investigations possible; citation does not imply their participation in, agreement with or endorsement of this book.

World Beer Wiki provided the research routes and an unusually useful record of its own evidence limitations. Open repositories, public collection catalogues and accessible scholarly versions made it possible to follow many of those routes beyond a summary. The project also benefits from the openly licensed work of the EB Garamond and Bricolage Grotesque font contributors.

Research Note: Following the Sources

The research began with World Beer Wiki at commit `3b985c19a9dcccef8d334be54b5405e42093e812`, edition 1.19. The repository's narratives, evidence metadata, cross-links and Atlas tooling helped identify connected cases. Its articles were treated as a map of claims to investigate, not as independent confirmation of those claims. The book is a separately maintained contribution under `books/the-beer-detective/`; it does not alter the underlying subject articles.

For each chapter, the research followed relevant links to original papers, historical texts, museum records, legal documents or situated accounts. The source register records what was actually accessible on 28 October 2026. "Full text" means the complete work was available and relevant passages were consulted; it does not mean every reference cited by that work was independently read. Abstracts, indexed excerpts, public previews and secondary descriptions are labeled separately. A publisher's citation of a source was never treated as proof that the source itself had been examined.

Evidence was checked at the scale at which it was obtained. A community sample was not expanded into a continent-wide finding. A laboratory mechanism was distinguished from its importance in every commercial beer. An artifact catalogue's identification was separated from a modern inference about the people who used the artifact. A model's conditional result was kept apart from a forecast of what must occur. Dates attached to fieldwork, photography, publication and later commemoration were compared where their differences could mislead a reader.

Conflicts remained visible. The proposed brewing interpretation at Raqefet is presented with objections as reported in the researchers' published response; the original commentary was not fully available. The two Jiahu investigations employed different evidence and do not establish that the same vessels were retested by identical means. Later accounts of Pilsen's condemned beer assign agency differently. The climate chapter does not substitute figures in a repository manuscript for the published article's final abstract. These distinctions change the stories that the evidence can support.

Chapter notes identify sources for the corresponding passages, including originating wiki routes. The separate *Sources and Chapters* document gives a more extensive map, and the machine-readable claim register retains access descriptions, evidence locators and unresolved questions. These records accompany the project package. Selected further reading offers a shorter route through the literature.

The original prose was drafted and revised with AI assistance. Internal passes addressed structure, factual accuracy, sentence-level style, cultural representation and skeptical-reader objections. Additional AI reviews by sessions that had not drafted the relevant chapters produced documented corrections and cuts. These reviews are internal quality work; they are not independent human review, peer review or community consultation.

The most consequential remaining needs are human developmental editing; specialist checks of the archaeological, historical and chemical interpretations; and community-informed readings of the chapters on living traditions. Published fieldwork can convey particular observations without supplying an entire community's judgment about this account. Source-language verification would improve passages reached through translations and English-language summaries. Detailed priorities and questions are listed in the editorial report.

All scenes are either descriptions supported by cited evidence or plainly signaled explanatory examples. No historical dialogue, private thought, interview or field visit has been invented. Short quotations are attributed in the relevant note. The cover's tablet is an original schematic illustration, not a reproduction of a specific object or a transcription of an inscription.

This edition closes its source record on 28 October 2026. It is complete as a book and open as a research project: corrections should identify the chapter, the disputed statement and the evidence that changes it.

Notes

Notes restart at 1 in each chapter. Each entry links to its source and back to the passage. Source access was checked on 28 October 2026 unless otherwise stated.

Introduction

1. The banana investigation and its original experimental sources are in chapter 7; measurements and expectation are investigated separately in chapters 8 and 22. For the core yeast result, see Kevin J. Verstrepen and colleagues, "Expression Levels of the Yeast Alcohol Acetyltransferase Genes ATF1, Lg-ATF1, and ATF2 Control the Formation of a Broad Range of Volatile Esters", *Applied and Environmental Microbiology* 69 (2003): 5228–5237. Full text and relevant methods and results were consulted. The opening glass is an illustrative situation, not a reported tasting session. Back to text.

2. These previews refer to the separately documented investigations of women in English brewing (chapter 13), Zoigl (chapter 14), and Korean nuruk starters (chapter 17). Their chapter notes distinguish primary documents, scholarly interpretations, practitioner accounts and laboratory studies; no additional field observation is implied here. Back to text.

1. Before the Field

1. Li Liu et al., "Fermented beverage and food storage in 13,000-y-old stone mortars at Raqefet Cave, Israel: Investigating Natufian ritual feasting," *Journal of Archaeological Science: Reports* 21 (2018), 783–793, doi:10.1016/j.jasrep.2018.08.008. Abstract and indexed methods/results excerpts consulted; the full publisher article and the university-hosted PDF could not be retrieved reliably. The brewing identification and feast setting are the authors' interpretations. Return 1 · Return 2.

2. Liu et al., "Fermented beverage and food storage," indexed experimental-methods discussion. The explanation here is a general account of malting, starch conversion and fermentation, not a proposed operating recipe for a Natufian brew. See also Peter Damerow, "Sumerian Beer: The Origins of Brewing Technology in Ancient Mesopotamia", *Cuneiform Digital Library Journal* 2012:2, §1.1. Back to text.

3. Li Liu et al., "Response to comments on archaeological reconstruction of 13,000-y old Natufian beer making at Raqefet Cave, Israel", *Journal of Archaeological Science: Reports* 28 (2019), 101914. Publisher abstract, introduction and indexed section excerpts consulted. Eitam's objections are reported here through this response; his complete original commentary was not examined. This is an access limitation, not a judgment that either side has settled the dispute. Back to text.

4. Dani Nadel et al., “Earliest floral grave lining from 13,700–11,700-y-old Natufian burials at Raqefet Cave, Mt. Carmel, Israel”, *PNAS* 110 (2013), 11774–11778. Abstract and indexed methods/results text consulted; the direct PMC page presented a browser challenge. The flower evidence is not independent proof of brewing. Back to text.

5. Ken-ichi Tanno and George Willcox, “How fast was wild wheat domesticated?”, *Science* 311 (2006), 1886, doi:10.1126/science.1124635. PubMed abstract consulted. The study supports distinguishing cultivation from the gradual appearance of domesticated forms; it does not identify a brewing motive. Back to text.

6. Amaia Arranz-Otaegui et al., “Archaeobotanical evidence reveals the origins of bread 14,400 years ago in northeastern Jordan”, *PNAS* 115 (2018), 7925–7930. Publisher indexed text and open full-text XML from Europe PMC consulted. “Bread-like” follows the authors’ description; the remains do not establish the date when bread was invented. Back to text.

7. Research route: World Beer Wiki, “Did brewing come before farming?” and “Raqefet Cave Brewing Evidence”. The chapter adds the published controversy and the distinction between cultivation and domestication. Sources accessed 28 October 2026. Back to text.

2. A Goddess in the Brewhouse

1. Electronic Text Corpus of Sumerian Literature, University of Oxford, “A hymn to Ninkasi (Ninkasi A)”, especially lines 1–16 and 41–48. Full English translation consulted; no independent translation from Sumerian is claimed. The wording and arrangement discussed here refer to this translation, including its marked gaps. Return 1 · Return 2.

2. Miguel Civil, “A Hymn to the Beer Goddess and a Drinking Song,” in R. D. Biggs and J. A. Brinkman, eds., *Studies Presented to A. Leo Oppenheim, June 7, 1964* (Chicago, 1964), 67–89, especially 67–69 and 75–76. The University of Chicago publication page supplies the open PDF. The chapter, its textual commentary and translated passages were consulted in the downloaded volume. The description of variants and interpretive uncertainty follows Civil’s account. Return 1 · Return 2 · Return 3.

3. Cuneiform Digital Library Initiative, “Beer: Hymn to Ninkasi”, public collection note; Oxford ETCSL, bibliography and cuneiform sources. Public catalog note and source list consulted. The broad dating applies to surviving textual witnesses. The CDLI note calls the work a recipe; this chapter examines the different meanings that label can bear rather than treating the label as settled technical evidence. Back to text.

4. Peter Damerow, “Sumerian Beer: The Origins of Brewing Technology in Ancient Mesopotamia”, *Cuneiform Digital Library Journal* 2012:2, §§6.4–6.9. Full open analytical article consulted. The discussion of circular reasoning develops the methodological problem, not a claim that all earlier scholarship was mistaken. Back to text.

5. Tate Paulette, “Bringing the Beers of Ancient Mesopotamia Back to Life”, *Peopling the Past*, 18 February 2022. Full firsthand account consulted, especially “Experimental brewing” and “Public engagement.” No personal participation in those experiments is claimed here. Back to text.

6. Research route: World Beer Wiki, “Is the Hymn to Ninkasi really a beer recipe?” and “Sumerian beer words are not modern style labels”. This chapter adds Civil’s full commentary and Paulette’s firsthand experimental account, and qualifies the wiki’s categorical “not a recipe” framing. Sources accessed 28 October 2026. Back to text.

3. The Accountant’s Beer

1. Metropolitan Museum of Art, “Cuneiform tablet: record of rations of beer, bread, oil, and onions for messengers”, object 1985.180.2. Full public catalog entry consulted, including dimensions, dating, credited gift and interpretation. The date “year 1 of Ibbi-Sin, month 2, day 20” is the museum’s reading; no independent reading of the tablet is claimed. Return 1 · Return 2.

2. Cuneiform Digital Library Initiative, “Beer: Means of Compensation”, public collection note identifying tablet P273413. Full public description consulted. Its statement about irregular allocations is retained; its suggestion that *dida* was a kind of pasteurized beer is not adopted as a demonstrated process. Back to text.

3. Maxwell Museum of Anthropology, “Object Monday: Cuneiform Tablet Messenger Text”, 24 April 2023, tablet MMA 67.134.3, with transliteration and translation supplied by Andrea Seri. Full public entry consulted. The chapter reports the published reading and does not independently translate Sumerian or turn the ancient quantities into modern drinking recommendations. Return 1 · Return 2.

4. Peter Damerow, “Sumerian Beer: The Origins of Brewing Technology in Ancient Mesopotamia”, *Cuneiform Digital Library Journal* 2012:2, §§5.17–5.22. Full open article consulted. The modern meal-and-loaves comparison is an explanatory analogy, not a claim about an ancient transaction. Back to text.

5. Ira Spar, ed., *Cuneiform Texts in The Metropolitan Museum of Art*, volume I (1988), catalog entry 45, p. 60, tablet MMA 57.16.6. The publisher’s catalog page and indexed PDF passage with transliteration and table-form translation were consulted; the full volume was not read. This is a different tablet from object 1985.180.2. Back to text.

6. British Museum, “The first writing: counting beer for the workers”, institutional collection text hosted on Google Arts & Culture, registration 1989,0130.4. Full public description consulted for the object’s date and signs. Its general assertion that beer was safer than water is not used here. Back to text.

7. Research route: World Beer Wiki, “The tablet was not the recipe” and “Ur III beer ration tablet”. Several separately identified museum records are added to prevent one object from carrying the whole account. Sources accessed 28 October 2026. Back to text.

4. What the Pottery Remembered

1. Patrick E. McGovern et al., “Fermented beverages of pre- and proto-historic China”, *PNAS* 101 (2004), 17593–17598. Publisher and PMC indexed abstract, methods, analytical results and discussion passages consulted. The direct PMC reader presented a browser challenge; a complete clean article download was not obtained. The paper’s own alternatives for fruit and wax sources and its lack of direct ethanol evidence are retained. Return 1 · Return 2 · Return 3.

2. Li Liu, Yahui He, Jingpu Li, Juzhong Zhang and Wuhong Luo, “Revisiting Jiahu’s 9000-Year-Old Fermented Beverage: New Microfossil Evidence for Early Qu-Based and Malt-Based Brewing Technologies in North China”, *Heritage* 9 (2026), 335, especially §§2.2–4.2 and 5.1. Substantial publisher-indexed sections, including methods, results and discussion, were consulted; direct HTML/XML opening was unsuccessful. Images and supplementary data were not independently evaluated. The study explicitly does not claim a parallel chemical reanalysis of the original samples. Return 1 · Return 2 · Return 3.

3. University of Pennsylvania, “Penn Museum Archaeochemist and International Scholars Confirm 9,000-Year History of Chinese Fermented Beverages”, 8 December 2004. Institutional announcement consulted as evidence of how the research was presented and of the distinction it makes between the Neolithic residues and later bronze-vessel liquids; it is not an additional independent experiment. Back to text.

4. Li Liu and colleagues, “Revealing the 2300-Year-Old Fermented Beverage in a Bronze Bottle from Shaanxi, China”, *Fermentation* 10 (2024), 365, introduction and discussion of *qu* preparation. Publisher-indexed full-text sections were checked during cross-review; direct access returned 429. This source supports the explanation of a starter and overlapping starch conversion and fermentation, not a claim that the later Qin beverage duplicated Jiahu’s drink. Back to text.

5. Jiajing Wang et al., “Revealing a 5,000-y-old beer recipe in China”, *PNAS* 113 (2016), 6444–6448. Publisher-indexed methods, results, discussion and supporting-methods excerpts consulted. This chapter uses the study to explain converging methods, without adopting its broader proposed motivation for barley’s arrival in China as established fact. Back to text.

6. Jiajing Wang, Leping Jiang and Hanlong Sun, “Early evidence for beer drinking in a 9000-year-old platform mound in southern China”, *PLOS ONE* 16 (2021), e0255833, doi:10.1371/journal.pone.0255833. PubMed abstract consulted. The account is confined to the reported setting, methods and interpretation; no new claim about an exact ritual or direct descent from Qiaotou to Jiahu is made. Back to text.

7. Research route: World Beer Wiki, “The Jiahu drink changed when scientists asked a new question”, “Jiahu archaeology: what changed between 2004 and 2026?”, and Jiahu pottery sampling. The chapter deliberately describes a shared site and research problem rather than asserting identical samples. Sources accessed 28 October 2026. Back to text.

5. The Last Service

1. Michael E. Moseley et al., “Burning down the brewery: Establishing and evacuating an ancient imperial colony at Cerro Baúl, Peru”, *PNAS* 102 (2005), 17264–17271. Abstract and indexed article passages concerning the colony, brewery and termination were consulted; direct PMC access was blocked by a browser challenge and a complete PDF was not successfully retrieved. Capacity is the authors’ estimate; ceremonial closure is their contextual interpretation. Return 1 · Return 2 · Return 3.

2. Nicole Pomerantz, “How a special beer played a part in ancient Peruvian politics”, Arizona State University, 3 January 2024. Institutional interview with excavation researcher Donna Nash, consulted for her explanation of labor, experimental work and the pin evidence. Her interpretation of the workers’ status is attributed rather than presented as direct identification of individuals. Back to text.

3. Donna J. Nash, Matthew E. Biwer and Emilee Witte, “Dreg Deposits and Domestic Beer Production: Assessing the Ubiquity of Chicha Production in the Wari Empire”, *Ethnoarchaeology* 16 (2024), 163–191. Institutional abstract and metadata consulted; full article not accessed. Its conclusions are restricted here to the stated Moquegua comparison, not generalized across the Andes. Return 1 · Return 2.

4. Patrick Ryan Williams, Donna J. Nash, Joshua M. Henkin and Ruth Ann Armitage, “Archaeometric Approaches to Defining Sustainable Governance: Wari Brewing Traditions and the Building of Political Relationships in Ancient Peru”, *Sustainability* 11 (2019), 2333. The direct publisher page was rate-limited, but a complete open PDF was subsequently retrieved and its site, methods, results and discussion sections inspected. Chemical findings are evidence consistent with molle, with compound-identification limitations discussed by the authors; the larger argument about political resilience remains an interpretation. Back to text.

5. Field Museum, “Field Museum Releases New Beer Inspired by Ancient Recipe”, institutional announcement, 2016. Public text consulted solely for the museum’s modern collaboration and recipe development. The modern product is not treated as independent evidence for ancient taste, exact ingredients or shelf life. Back to text.

6. Research route: World Beer Wiki, “The brewery they closed with fire”, Cerro Baúl brewery, and Wari hospitality. This synthesis has not received independent Andean archaeological or community review. Sources accessed 28 October 2026. Back to text.

6. The Inventor Arrives Late

1. Powerhouse, “Boulton and Watt rotative steam engine”, especially History and Production notes. The public collection record was read; it documents the 1784 order, 1785 installation, driven equipment, retention of the horse wheel, and later modifications. Accessed 28 October 2026. Back to text.

2. Martyn Cornell, “Myth 1: Ralph Harwood invented porter as a substitute for three-threads”, public article, read in full; James Sumner, “Status, scale and secret ingredients: the

retrospective invention of London porter," *History and Technology* 24, no. 3 (2008): 289–306, author-hosted full article, especially 297–99. Sumner identifies the February 1802 *Monthly Magazine* version as well as the guidebook tradition. The original 1802 magazine and guidebook pages were not directly inspected for this chapter; their wording and publication relationship are reported through these researchers. Accessed 28 October 2026. Back to text.

3. [William Ellis], *The London and Country Brewer*, second edition (1736), Project Gutenberg transcription: preface and chapters IV, VIII, and XVI. The full public text was available and these passages were inspected. The original appeared anonymously; the attribution follows Sumner. Historical medical explanations in the book are not adopted here. Accessed 28 October 2026. Back to text.

4. Sumner, "Status, scale and secret ingredients", 290–98, on existing scale, storage, product identity, and the transmission of the invention account. Full article accessed. The hypothetical monthly production calculation in this chapter is explanatory arithmetic, not a claim about a named brewery's accounts. Back to text.

5. Powerhouse, "Photograph of a drawing of a 1785 Boulton and Watt beam engine", object P1306, research note by Debbie Rudder, August 2007. Full public record read, including its correction of inherited horsepower figures. Read alongside the engine's own collection entry, which records the 1795 change to double action. Accessed 28 October 2026. Back to text.

6. The starting route was the wiki's "The porter invention story". This chapter extends the wiki's research by consulting Sumner's complete article, Ellis's primary brewing text, Cornell's public historical investigation, and the Powerhouse records. Cornell's CAMRA course, "Porter, the beer that came back from the dead", was checked only through its public introduction; the members' sections were not accessed and are not treated as evidence here. Back to text.

7. The Banana That Was Never Added

1. Kevin J. Verstrepen and colleagues, "Expression Levels of the Yeast Alcohol Acetyltransferase Genes ATF1, Lg-ATF1, and ATF2 Control the Formation of a Broad Range of Volatile Esters," *Applied and Environmental Microbiology* 69 (2003), 5228–5237, open article, especially the abstract and experimental/results sections. The full-text page and relevant methods and results were consulted; the genetic outcomes are restricted to the strains and conditions tested. Return 1 · Return 2.

2. Miki Harada, Yoshinori Ueda and Takashi Iwata, "Purification and Some Properties of Alcohol Acetyltransferase from Banana Fruit," *Plant and Cell Physiology* 26 (1985), 1067–1074, publisher record. The indexed publisher abstract was consulted; the full article was not accessible. Back to text.

3. Ambrose Atwine and colleagues, "Characterization of Physicochemical Properties, Volatile Compounds, and Sensory Attributes of Tonto, a Traditional Ugandan Banana-Based Alcoholic Beverage," *Food Science & Nutrition* 14 (2026), e72071, open article. Indexed full-text methods, sensory sections and conclusions were consulted. The observations

concern the sampled locations and productions, not all Ugandan makers. Return 1 · Return 2.

4. Stefan Coghe and colleagues, “Ferulic Acid Release and 4-Vinylguaiacol Formation during Brewing and Fermentation: Indications for Feruloyl Esterase Activity in Saccharomyces cerevisiae,” *Journal of Agricultural and Food Chemistry* 52 (2004), 602–608, PubMed abstract. Abstract-level access; no claim is made to have inspected the full experimental data. Back to text.

5. Joaquin F. Christiaens and colleagues, “The Fungal Aroma Gene ATF1 Promotes Dispersal of Yeast Cells through Insect Vectors,” *Cell Reports* 9 (2014), 425–432, PubMed abstract. The abstract supports the experimental link between the gene, fly attraction and dispersal; it does not settle the entire evolutionary function of ester production. Back to text.

6. Research route: World Beer Wiki, “The banana that was never added”, banana fruit versus yeast esters and clove aroma. These draft articles supplied leads; the papers above were checked separately. Back to text.

8. Two Beers, One Number

1. Toru Kishimoto, Satoko Teramoto, Akiko Fujita and Osamu Yamada, “Evaluation of Components Contributing to the International Bitterness Unit of Wort and Beer,” *Journal of the American Society of Brewing Chemists* 80 (2022), 53–61, publisher abstract. Published online in 2021. Only the publisher abstract was accessible; the reported 47.5/49.9 IBU and 39.9/2.2 ppm contrast comes from it. Return 1 · Return 2.

2. American Society of Brewing Chemists, “Beer 23: Bitterness Units (BU), Manual Isooctane Extraction”, public method summary consulted. The complete proprietary method was not inspected. This account explains the measurement rather than providing an extraction procedure. Back to text.

3. Victor Algazzali and Thomas Shellhammer, “Bitterness Intensity of Oxidized Hop Acids: Humulinones and Hulupones,” *Journal of the American Society of Brewing Chemists* 74 (2016), 36–43, publisher abstract. Abstract-level access; the platform also displays a later online migration date. The relative intensities are experiment-specific. Back to text.

4. Molly J. Higgins and John E. Hayes, “Discrimination of Isointense Bitter Stimuli in a Beer Model System,” *Nutrients* 12 (2020), 1560, open article. The full-text page, design and reported results were consulted. This model-beer experiment does not reproduce every commercial formulation. Back to text.

5. Olayide Oladokun and colleagues, “The Impact of Hop Bitter Acid and Polyphenol Profiles on the Perceived Bitterness of Beer,” *Food Chemistry* 205 (2016), 212–220, PubMed abstract. Abstract-level access. The analytical sample contained thirty-four commercial lagers; the sensory subset contained ten. Back to text.

6. Olayide Oladokun and colleagues, “Perceived Bitterness Character of Beer in Relation to Hop Variety and the Impact of Hop Aroma,” *Food Chemistry* 230 (2017), 215–224, publisher

abstract and indexed introduction. The full results tables were not inspected, so this chapter retains the abstract's qualitative finding. Back to text.

7. Research route: World Beer Wiki, "Two beers with the same IBU need not taste equally bitter", bitterness and perception and bitterness units and hop chemistry, consulted as research leads rather than independent verification. Back to text.

9. The Culprit Was Light

1. C. S. Burns, A. Heyerick, D. De Keukeleire and M. D. E. Forbes, "Mechanism for Formation of the Lightstruck Flavor in Beer Revealed by Time-Resolved Electron Paramagnetic Resonance," *Chemistry—A European Journal* 7 (2001), 4553–4561, PubMed abstract. Abstract-level access. The direct photolysis used model solvents and 308-nanometer light; it is not presented as an ordinary bottle-exposure trial. Return 1 · Return 2.

2. A. J. Irwin, L. Bordeleau and R. L. Barker, "Model Studies and Flavor Threshold Determination of 3-Methyl-2-Butene-1-Thiol in Beer," *Journal of the American Society of Brewing Chemists* 51 (1993), 1–3, publisher abstract. Indexed abstract consulted; the publisher's later online date is not the original publication year. The threshold is specific to the reported method and samples. Back to text.

3. Kevin Huvaere and colleagues, "Photooxidative Degradation of Beer Bittering Principles: A Key Step on the Route to Lightstruck Flavor Formation in Beer," *Journal of Agricultural and Food Chemistry* 53 (2005), 1489–1494, publisher abstract. The abstract was independently consulted in the publisher index and EPA HERO record; the full paper was not inspected. Back to text.

4. Kevin Huvaere and colleagues, "Flavin-Induced Photodecomposition of Sulfur-Containing Amino Acids Is Decisive in the Formation of Beer Lightstruck Flavor," *Photochemical & Photobiological Sciences* 5 (2006), 961–969, article PDF, DOI. The introduction, experimental methods and concluding mechanistic sections were consulted in full text. Model-system findings are distinguished from package testing. Return 1 · Return 2.

5. Richard Pozdrik, Felicity A. Roddick, Peter J. Rogers and Thang Nguyen, "Spectrophotometric Method for Exploring 3-Methyl-2-Butene-1-Thiol (MBT) Formation in Lager," *Journal of Agricultural and Food Chemistry* 54 (2006), 6123–6129, PubMed abstract. Abstract-level access; observations concern the tested lager and model mixtures. Back to text.

6. Barry Hucker and colleagues, "Vitamins in Brewing: Effects of Post-Fermentation Treatments and Exposure and Maturation on the Thiamine and Riboflavin Vitamer Content of Beer," *Journal of the Institute of Brewing* 122 (2016), 278–288, article. Full-text methods and the glass-transmission discussion and Figure 3 were consulted. This is evidence about the tested glasses, not all products of a nominal color. Back to text.

7. Research route: World Beer Wiki, lightstruck investigation, riboflavin and beer light and bottles and light barriers. The linked scientific sources were checked separately. Back to text.

10. The Bottle Kept Changing

1. A. J. Irwin, R. L. Barker and P. Pipasts, "The Role of Copper, Oxygen, and Polyphenols in Beer Flavor Instability," *Journal of the American Society of Brewing Chemists* 49 (1991), 140–149, publisher abstract. Abstract-level access; no process dosing or universal metal limit is inferred. Back to text.

2. European Brewery Convention, Analytica EBC method 11.5.1, "Total (Dissolved and Headspace) Oxygen in Packaged Beer". Public method scope consulted; the subscription procedure was not accessed. Back to text.

3. Alexandr Mikyška and Karel Štěrba, "Effect of Wort Boiling System and Hopping Regime on Wort and Beer Stale-Flavor Aldehydes," *Foods* 12 (2023), 3111, open article. Full-text methods, results and conclusions consulted. The proposed later stability difference was explicitly unconfirmed in this trial. Back to text.

4. P. Bustillo Trueba and colleagues, "Monitoring the Evolution of Free and Cysteinylated Aldehydes from Malt to Fresh and Forced Aged Beer," *Food Research International* 140 (2021), 110049, PubMed abstract. Abstract-level access. The distinction between the general bound-state concept and the study's negative finding for particular cysteinylated forms is retained. Back to text.

5. Carolina Maia and David Cook, "Binding of Staling Aldehydes to the Beer Matrix: Insights into the Equilibria Through Shelf-Life Which Drive Beer Flavor Instability," *Journal of the American Society of Brewing Chemists* 84 (2026), 96–107; online 12 December 2025, article. Indexed full-text abstract, experimental design and sample-preparation sections consulted. The concentrated reconstituted matrices differ from unaltered beer; an individual company funded the study and was not named in its funding statement. Back to text.

6. Arndt Nobis and colleagues, "A Comprehensive Evaluation of Flavor Instability of Beer (Part 2): The Influence of De Novo Formation of Aging Aldehydes," *Foods* 10 (2021), 2668, open article. Indexed full-text methods and results consulted. The six-beer experimental sequence supports simultaneous release and formation, not a fixed timetable for all products. Back to text.

7. Christian Schubert and colleagues, "The Influence of Storage Conditions on the Chemistry and Flavor of Hoppy Ales," *Food Chemistry* 395 (2022), 133616, PubMed abstract. Abstract-level access. Eleven commercially produced German hoppy ales and one lager were studied. Back to text.

8. Research route: World Beer Wiki, aging without a single culprit, bound aldehydes, and oxygen pathways. The original studies above were consulted independently. Back to text.

11. The Foam That Came from the Field

1. Anuradha U. Jayathissa and colleagues, "Impacts of Pathogen Strain and Barley Cultivar on Fusarium Head Blight in Barley and during Malting," *Plant Pathology* 73 (2024), 1874–1885, open article. Full-text methods, gushing results and discussion were consulted. The study used two cultivars, seven pathogen isolates and controls over three field seasons; the absence of a significant association is not proof of no possible biological relationship. Return 1 · Return 2.

2. Gérard Liger-Belair and Clara Cilindre, "How Many CO2 Bubbles in a Glass of Beer?," *ACS Omega* 6 (2021), 9672–9679, open article. Indexed full-text physical analysis and model limitations consulted; direct access intermittently returned a verification page. The model's simplifying assumptions preclude a universal bubble count. Back to text.

3. Tuija Sarlin and colleagues, "Identification and Characterization of Gushing-Active Hydrophobins from Fusarium graminearum and Related Species," *Journal of Basic Microbiology* 52 (2012), 184–194; online 2011, PubMed abstract. The abstract and the authors' institutional record were consulted. The purified-protein result supports a mechanism, not diagnosis by eye. Back to text.

4. Sylvie M. Deckers and colleagues, "Dynamic Light Scattering (DLS) as a Tool to Detect CO2-Hydrophobin Structures and Study the Primary Gushing Potential of Beer," *Journal of the American Society of Brewing Chemists* 69 (2011), 144–149, publisher abstract. Abstract-level access. The interpretation of the approximately 100-nanometer structures as stabilized nanobubbles is attributed to the investigators. Back to text.

5. Chen Xu and colleagues, "Interactions between Humulinone Derived from Aged Hops and Protein Z Enhance the Foamability and Foam Stability," *Food Chemistry* 434 (2024), 137449; online 2023, PubMed abstract and publisher preview. Abstract and indexed selected methods/conclusion excerpts consulted. This is a protein-solution study, not a whole-beer recipe comparison. Back to text.

6. Research route: World Beer Wiki, unexpected foam investigation, gushing and hydrophobins, grain contamination and foam defects and protein Z and humulinone. The original research was checked separately. No package-pressure experiment is proposed here. Back to text.

12. The Missing Parent

1. Sean A. Bergin and colleagues, "Identification of European Isolates of the Lager Yeast Parent Saccharomyces eubayanus," *FEMS Yeast Research* 22 (2022), foac053, open article. Full-text page and indexed results and conclusions consulted. Sampling was in September 2021; publication was in December 2022. The paper does not claim that the Irish isolates themselves founded lager yeast. Return 1 · Return 2.

2. Diego Libkind and colleagues, "Microbe Domestication and the Identification of the Wild Genetic Stock of Lager-Brewing Yeast," *Proceedings of the National Academy of Sciences* 108 (2011), 14539–14544, PubMed abstract. The abstract was consulted; the direct PMC article

returned a verification page. The identity figure describes the reported genome comparison, not a proven historical voyage. Back to text.

3. Jian Bing, Pei-Jie Han, Wan-Qiu Liu, Qi-Ming Wang and Feng-Yan Bai, "Evidence for a Far East Asian Origin of Lager Beer Yeast," *Current Biology* 24 (2014), R380–R381, publisher summary. The accessible publisher summary was consulted; its origin proposal is attributed rather than adopted as settled migration history. Back to text.

4. David Peris and colleagues, "Complex Ancestries of Lager-Brewing Hybrids Were Shaped by Standing Variation in the Wild Yeast Saccharomyces eubayanus," *PLOS Genetics* 12 (2016), e1006155, open article. Full-text abstract, author summary and relevant genomic interpretation consulted. The failure of one sampled wild strain to match all regions is central to the argument. Back to text.

5. EmilyClare Baker and colleagues, "The Genome Sequence of Saccharomyces eubayanus and the Domestication of Lager-Brewing Yeasts," *Molecular Biology and Evolution* 32 (2015), 2818–2831, open article; Alex N. Salazar and colleagues, "Chromosome Level Assembly and Comparative Genome Analysis Confirm Lager-Brewing Yeasts Originated from a Single Hybridization," *BMC Genomics* 20 (2019), 916, open article. Both full-text pages, abstracts and relevant comparative-background sections were consulted. The chapter attributes the changing conclusions and does not substitute a precise event count for an unobserved historical record. Back to text.

6. Mathias Hutzler and colleagues, "A New Hypothesis for the Origin of the Lager Yeast Saccharomyces pastorianus," *FEMS Yeast Research* 23 (2023), foad023, open article. Full-text abstract, historical discussion and concluding synthesis consulted. The underlying original archives were not independently examined for this book. The Munich date and place remain the authors' hypothesis. Back to text.

7. Research route: World Beer Wiki, lager yeast and hybrid ancestry and brewing yeast. This chapter expands beyond the wiki's inherited 2011 abstract with separately consulted genetic and historical research. A search through October 2026 did not supply documentary proof of a particular transoceanic founding voyage. Back to text.

13. The Women in the Fine Book

1. Judith M. Bennett, "Gender Rules: Women and the Regulation of Brewing," in *Ale, Beer, and Brewsters in England: Women's Work in a Changing World, 1300–1600* (Oxford University Press, 1996), publisher chapter abstract. Indexed publisher abstract accessed on 28 October 2026; the full chapter and original court rolls were not examined. The Wakefield and Exeter episodes are attributed to Bennett, without invented dialogue. Back to text.

2. *Judicium Pillorie*, reproduced in Owen Ruffhead's *Statutes at Large* (1763), digitized page and transcription; and *Assisa Panis et Cervisie*, public statutory preview accessed 28 October 2026. The traditional 1266 dating is not needed for the argument; the text records rules rather than their frequency of enforcement. Back to text.

3. Elizabeth Kuligowski, Museum of Oxford, "Alewives in Oxford: A History of Female Brewing", full museum article accessed 28 October 2026. Used narrowly for Maud London

and the discussion of documentary selection; broader origin stories and speculative motives in that article are not adopted. Back to text.

4. Bennett, book description and "Working Together: Wives and Husbands in the Brewers' Gild of London". Publisher descriptions and indexed chapter abstract only; not a full reading of the monograph. They support the attributed argument about a gradual and uneven change in household commercial production. Back to text.

5. *Goesaert v. Cleary*, 335 U.S. 464 (1948), United States Reports scan, Library of Congress, pp. 465–468. Full five-page opinion and dissent accessed as extracted PDF text. This is a historical account of the 1948 decision, not a statement of present law. Back to text.

6. Research route: World Beer Wiki, Oxford alewives and brewsters and change. These source-scoped drafts supplied the initial map, not independent verification. Back to text.

14. The Brewery with Many Front Doors

1. Schutz-Gemeinschaft Echter Zoigl vom Kommunbrauer, "An Introduction to the Tradition of Zoigl Beer", full practitioner account accessed 28 October 2026. Its more detailed process description locates fermentation at individual premises after wort production and cooling in the communal facility. This clarifies an abbreviated formulation in the heritage inventory. Back to text.

2. Stadt Windischeschenbach, "Freizeit & Tourismus", municipal text accessed 28 October 2026, section "Hauptstadt des Zoiglbiers." Used for the sign, calendar and local property-registration account; the book supplies no live opening schedule. Back to text.

3. Deutsche UNESCO-Kommission, "Oberpfälzer Zoiglkultur", full German-language national-inventory entry accessed 28 October 2026. Entry year 2018; institutional recognition is not a sensory specification or independent endorsement of this book. Back to text.

4. Schutz-Gemeinschaft, English introduction, paragraphs on ownership, Kesselgeld and property rights; corroborated locally by the municipal source in note 2. The chapter does not turn one town's access arrangements into a complete account of every exception elsewhere. Back to text.

5. Schutz-Gemeinschaft, "Der Zoigl", full practitioner text accessed 28 October 2026, particularly "Wer sind die 'Echten Kommunbrauer' der Oberpfalz?" and the stated logo conditions. Symbolic explanations are treated as the association's account, not independently established ancient origins. Dated 2017 counts on that page are not presented as current figures. Back to text.

6. Research route: World Beer Wiki, Zoigl communal brewing. The chapter adds direct consultation of practitioners' and municipal accounts; no brewer interviews or community review were conducted for this book. Back to text.

15. The Ingredient That Had to Be Carried

1. Joseph Alhassan, Divine Odame Appiah, Suale Iddrisu, Andrews Ofosu and Josephine Tetteh, "Incidence of water scarcity: understanding the triggers and implications for women's non-agricultural livelihoods in Northern Ghana", *Cogent Social Sciences* (2024), article 2433691, especially methods and section 4.5. Publisher text, including the Nagbo account, accessed through indexed full-article excerpts on 28 October 2026; direct HTML/PDF requests were blocked. Interview recordings and underlying survey data were not accessed. The participant's account is paraphrased, not presented as this book's interview. Back to text.

2. Aliata Issahaq Mumuni, Musah Dumah and Hawa Awuro Sam, "Modeling the Determinants in Pito Brewing of Small-Scale Enterprising in Ghana", *European Journal of Business and Management* 15, no. 16 (2023), DOI 10.7176/EJBM/15-16-06. Full nine-page PDF text accessed, including methods and tables. The methods specify Wa municipality, purposive/convenience contact and 362 returned questionnaires; the abstract's broader Upper West wording is not treated as evidence of regional representativeness. Reporting inconsistencies and unclear causal interpretation preclude policy estimates from its regressions. Back to text.

3. Aliata Issahaq Mumuni, Seidu Khalida and Kennedy Afenyo Biaku, "Assessing the Contribution of Pito Brewing to the Growth and Development of Small-Scale Enterprising in Wa Municipality" (2024). Publisher abstract accessed only. No profit amount, demographic percentage or national estimate is adopted, and the independence of its sample from the 2023 paper is not assumed. Back to text.

4. Isidore Lobnibe, "Drinking pito: conviviality, popular culture and changing agricultural production at the rural–urban interface in Brong Ahafo, Ghana", *African Geographical Review* 37, no. 3 (2018): 227–240; first published online in 2016. Indexed publisher abstract only; full article inaccessible. No reconstructed bar scene or unexamined claim about clientele is supplied. Back to text.

5. Hagrétou Sawadogo-Lingani and colleagues, "Sustainable Production of African Traditional Beers With Focus on Dolo, a West African Sorghum-Based Alcoholic Beverage", *Frontiers in Sustainable Food Systems* 5 (2021), 672410. Open review text consulted, including economic, resource and by-product sections. Used as a review, not presented as newly conducted fieldwork or a current census. Back to text.

6. Research routes: World Beer Wiki, pito enterprises in Wa, women brewers survey and dolo livelihoods. Further community-informed reporting remains outstanding. Back to text.

16. A Flag Outside the House

1. Frances M. Hayashida, "Ancient beer and modern brewers: Ethnoarchaeological observations of chicha production in two regions of the North Coast of Peru", *Journal of Anthropological Archaeology* 27 (2008): 161–174; and Felipe Vargas Faulbaum, "Chicha de jora: cauim de milho germinado nos Andes peruanos", *Equatorial* 6, no. 11 (2019): 1–25. Hayashida's publisher abstract and indexed methods/Piura excerpts were accessed; full article remained inaccessible. Vargas Faulbaum's PDF text was accessed, including the red markers on p. 5. Back to text.

2. Hayashida, "Ancient beer and modern brewers," publisher methods and Piura preview excerpts. Ten chicherías, named sales-day examples and purchases from mills belong to the documented north-coast study. They are not current opening information or a national survey. Back to text.

3. Vargas Faulbaum, "Chicha de jora," pp. 4–5 and 14–15, on Engracia Surco Escobedo, service signs and milling. Full extracted PDF text was consulted; this book claims no visit to the households and reproduces no field photographs. The ethnographer's older historical claims are not independently adopted. Back to text.

4. Vargas Faulbaum, "Chicha de jora," sections on *jora*, milling and cooking; and the Piura resolution in note 5, description of grain selection and germination. The explanation describes starch preparation in broad terms, not an operating recipe or evidence of identical techniques in every community. Back to text.

5. Ministerio de Cultura del Perú, Resolución Viceministerial N.º 159-2015-VMPCIC-MC, 4 November 2015, published in *El Peruano*, 19 November 2015. Full extracted official text accessed 28 October 2026; some accented characters are corrupted in the web transcription. Used for regional recognition, the distinction between institutions, and the documented white flag. Adjacent Cusco and Tumbes resolutions in the same extracted page are not conflated with the Piura declaration. Back to text.

6. Ministerio de Cultura, announcement of recognition for picanterías and chicherías, 23 November 2015. Indexed official announcement accessed; direct page retrieval was blocked. The named representatives are attributed to that dated ceremony. Back to text.

7. Research routes: World Beer Wiki, Piura household businesses, white flags and Cusco work. No independent review by the communities described has taken place. Back to text.

17. The Cake That Would Not Be a Strain

1. Sang Hoon Song and colleagues, "Analysis of Microflora Profile in Korean Traditional Nuruk," *Journal of Microbiology and Biotechnology* 23, no. 1 (2013): 40–46, DOI 10.4014/jmb.1210.10001; PubMed record. Accessed 28 October 2026: abstract and bibliographic metadata, not the full paper. The forty-two-sample count and recovery of *S. cerevisiae* from eighteen and *A. oryzae* from thirteen are abstract-level results; they are not population prevalence estimates for all nuruk. Back to text.

2. Changhoon Chai, Goo-Sang Lim, Yun-Ji Kim, and Se-Wook Oh, "Microbial Community Changes in Makgeolli during Brewing," *Journal of the Institute of Brewing* 121 (2015): 304–308, full text. Accessed 28 October 2026: publisher full HTML, including preparation methods, results, and discussion. The experiment used commercial nuruk **and commercial yeast starter**. Its community time course must not be represented as an uninoculated test of nuruk alone. The chapter's methodological comparison is original analysis of the two differently designed studies. Back to text.

3. Jin Ho Choo and colleagues, "Whole-Genome de novo Sequencing, Combined with RNA-Seq Analysis, Reveals Unique Genome and Physiological Features of the Amylolytic Yeast *Saccharomycopsis fibuligera* and Its Interspecies Hybrid," *Biotechnology for Biofuels* 9 (2016): 246, full text. Accessed 28 October 2026: full publisher HTML, particularly Background and "Isolation and growth characteristics" results. Two selected nuruk isolates support the functional example; they do not establish the composition of all starter cakes. Back to text.

4. Pooja Pradhan and Jyoti Prakash Tamang, "Phenotypic and Genotypic Identification of Bacteria Isolated From Traditionally Prepared Dry Starters of the Eastern Himalayas," *Frontiers in Microbiology* 10 (2019): 2526, full text. Accessed 28 October 2026: publisher full text, including collection locations, starter preparation, isolation methods, and sample counts. This is culture-dependent bacterial research. The chapter's discussion of what preservation could retain is explicitly a thought experiment, not a reported community conservation project. Back to text.

5. Cultural Heritage Administration of Korea, "Makgeolli Making and Sharing Designated as National Intangible Cultural Heritage" (2021), institutional text archived by ichLinks. Accessed 28 October 2026: complete public description. This concerns a Korean national designation, not inscription on UNESCO's international representative list. Historical origin claims in the institutional overview are not adopted here. Back to text.

6. Wiki research trail: A cake is a community, not a strain, Nuruk survey, Makgeolli time course, Himalayan starter diversity, and Starter transmission. The linked 2023 *FEMS Yeast Research* article on Korean yeast genomic features is a review; this chapter instead uses Choo and colleagues' original 2016 study for the amylolytic-yeast example. No maker interview or community review was conducted for this manuscript. Back to text.

18. The Beer Made in Two Directions

1. Ryosuke Kubo and Method Kilasara, "Brewing Technique of Mbege, a Banana Beer Produced in Northeastern Tanzania," *Beverages* 2, no. 3 (2016): 21, DOI and publisher record; open-access article PDF. Accessed 28 October 2026: publisher-indexed text, followed by the complete ten-page article as extracted PDF text through the Semantic Scholar copy. Methods and sections 3.1–3.3 were consulted; no claim of firsthand observation is made. The opening reports the documented operation without invented dialogue, weather, surroundings, or motives. The paper's inferred functional role for nyalu is not presented as a new microbial isolation result. Back to text.

2. Dorothy Machunda Gimbi and Naofumi Kitabatake, "Changes in Alpha- and Beta-Amylase Activities during Seed Germination of African Finger Millet," *International Journal of Food Sciences and Nutrition* 53, no. 6 (2002): 481–488, PubMed abstract, DOI. Accessed 28 October 2026: complete indexed PubMed and publisher abstracts; full article unavailable in this research pass. Thermal inactivation findings concern malt extracts at the specified experimental conditions, not a universal optimum for mso. Back to text.

3. Ryosuke Kubo, "The Reason for the Preferential Use of Finger Millet (*Eleusine coracana*) in Eastern African Brewing," *Journal of the Institute of Brewing* 122 (2016): 175–180, full text. Accessed 28 October 2026: publisher full HTML, methods, results, and discussion. The chapter limits conclusions to the compared cereals and Bupigu fieldwork. Reported health beliefs and speculative physiological explanations in the source are not adopted as medical facts. Back to text.

4. Ryosuke Kubo, "Production of Indigenous Alcoholic Beverages in a Rural Village of Tanzania," *Journal of the Institute of Brewing* 120 (2014): 142–148, full text. Accessed 28 October 2026: publisher full HTML, field methods and process descriptions. Nyambo and kikonde are names in that study's Bupigu setting; they are not substituted for nyalu and mso. Similar process organization does not establish a transmission route. Broad food-safety claims in the article are not generalized here. Back to text.

5. Kubo and Kilasara (2016), sections 2.3–2.4 and 3.3. The on-site comparison used field measurements of ethanol, Brix, and pH; the ethanol method employed a saliva alcohol kit with sample dilution. The manuscript therefore does not turn the reported values into a universal product specification or infer a microbial census from them. Proposed selective effects of bark constituents remained an explanation to investigate; the paper did not establish a generally applicable dose. This chapter is a process investigation, not a brewing or plant-use protocol. Back to text.

6. Andreas Hemp, "The Banana Forests of Kilimanjaro: Biodiversity and Conservation of the Chagga Homegardens," *Biodiversity and Conservation* 15 (2006): 1193–1217, publisher abstract. Accessed 28 October 2026: abstract and bibliographic record, not subscription full text. The regional landscape study is contextual evidence, not proof of the provenance of ingredients in the 2016 mbege observation. Back to text.

7. Knut Christian Myhre, "What the Beer Shows: Exploring Ritual and Ontology in Kilimanjaro," *American Ethnologist* 42, no. 1 (2015): 97–115, publisher abstract. Accessed 28 October 2026: abstract and reference list only. No detailed ritual sequence, speaker testimony, or full-text interpretation has been reconstructed. Back to text.

8. Wiki research trail: The beer with two porridges, Nyalu and mso, Mbege knowledge, Finger millet, and Rombo. The subsequent original-source work extended the wiki's abstract-bounded access to the complete 2016 process paper, and added the independent enzyme study and contextual comparisons. No local-language or community review was obtained. Back to text.

19. The Three-Ingredient Alibi

1. Jenny Gesley, "500 Year Anniversary of the Bavarian Beer Purity Law of 1516," Library of Congress, 21 April 2016, original-text section and translation. Public article and reproduced provision consulted. Its statement that yeast was not used before its later legal inclusion is not adopted here. Back to text.

2. Klaus Rupprecht, "Purity Law, 1516," Historisches Lexikon Bayerns, sections on earlier ordinances and the 1516 provision; full public entry consulted. The Bamberg oath is described and illustrated there; the original archival volume was not independently inspected. Back to text.

3. Rupprecht, "Purity Law, 1516," sections "Term" and "Further Development and Control." Dates and institutional changes are historical claims, not an account of current labeling law. Back to text.

4. Gesley, "500 Year Anniversary," section on the Reichsgericht, citing *Entscheidungen des Reichsgerichts in Strafsachen* 23, 409. The original reported decision was not accessed; no invented courtroom dialogue or claim about the brewer's intention is included. Back to text.

5. Court of Justice, *Commission v Federal Republic of Germany*, Case 178/84, judgment of 12 March 1987, ECLI:EU:C:1987:126, official record. Official search-indexed judgment excerpts, including paragraphs 24 and 37–41, consulted; direct page and PDF retrieval were blocked by a browser check. The chapter does not give present-day legal advice. Back to text.

6. World Beer Wiki research route: "Was the Reinheitsgebot a 500-year-old universal beer recipe?" and "Bavarian purity law and its afterlife". These draft syntheses supplied the starting questions; the sources above were checked separately. The chapter's comparisons and reasoning are this book's own. Back to text.

20. Thirty-Six Barrels and a Date

1. Compare Géraldine Lysser, "Cool beer — all year round," Swiss National Museum, 16 April 2026, opening account, with Evan Rail, "On the Founding of Pilsner Urquell," *Brewery History* 149 (2012), 20–29, especially 21. Both texts accessed. Their differing descriptions of the 1838 event are preserved rather than combined into an invented scene. Back to text.

2. Evan Rail, "On the Founding of Pilsner Urquell", full PDF available and relevant historical discussion and translated founding appeal consulted. Rail uses the 1892 jubilee volume and the appeal dated 2 January 1839. Those original Czech documents were not separately examined for this book. References to them follow his translation and analysis. Back to text.

3. Plzeňský Prazdroj, "History", public producer chronology, entries on the 1842 batch, transport and the 1898 trademark. The company's claim that modern beer has an unchanged historical flavor is not adopted. Back to text.

4. Plzeňský Prazdroj, "Pilsen bishop blesses Easter batch of Pilsner Urquell", 2011 company account, accessed through indexed text; distinguishes the 5 October brewing date from 11

November service. This is retrospective producer testimony, not a contemporaneous tasting record. Back to text.

5. Zuzana Lízcová, "No risk, no gain," Goethe-Institut, full public article consulted. Used for its account of the regional combination and later cultural framing; sensational assertions about older ingredients and unverified first-tasting quotations are not repeated. Back to text.

6. German Patent and Trade Mark Office, "Beer & IP", historical section, full public page consulted. Claims about the earliest brewing recipe, Groll's original product naming and precise universal lager temperatures are not adopted. Back to text.

7. World Beer Wiki route: "What actually became new in Pilsen in 1842?", historical evidence, and burgher brewing rights. The original wiki had only an excerpt of Rail; access was improved for this edition. Back to text.

21. The Water in the Bowl

1. Alejandra Bussalleu and colleagues, "Cultural Values and the Coliform Bacterial Load of 'Masato,' an Amazon Indigenous Beverage," *EcoHealth* 17 (2020), 370–380, DOI 10.1007/s10393-020-01498-5. Open full text consulted, especially methods and Tables 1–2. The twenty-six waters include nineteen household samples and seven outdoor sources; they are not twenty-six independent communal sources. The paper is CC BY 4.0; this chapter provides an original, selective paraphrase and interpretation, without reproducing its photographs. Back to text.

2. Mario Çobo, Ann Charles-Vegdahl, Kaylyn Kirkpatrick and Randy Worobo, "Survival of Foodborne Pathogens in Low and Nonalcoholic Craft Beer," *Journal of Food Protection* 86 (2023), 100183. Publisher and PubMed abstract and highlights consulted; full methods were not independently inspected for this chapter. No operating limit or home safety test is inferred from the abstract. Back to text.

3. US Centers for Disease Control and Prevention, "About Water Disinfection with Chlorine and Chloramine", 14 February 2024, public page consulted 28 October 2026. Used to explain treatment and distribution as a sequence, not to recommend a treatment protocol for masato. Back to text.

4. Bussalleu and colleagues, methods on participatory photography and community dissemination, and discussion of further research. The microbiological collection took place in 2015, the later photographic study in 2017, and community validation workshops in January 2018. The full text also records unknown fermentation durations for the sampled drinks and dilution of fermented mash before consumption. The study's community participation does not constitute community review or endorsement of this book. Specific preparation and cultural descriptions remain limited to its participants. Back to text.

5. World Beer Wiki routes: "Why a fermented drink is not a water-safety certificate", "Shawi masato and water", and "Nonalcoholic beer pathogen challenge". The commentary on what different study designs establish is this book's reasoning, not a new experiment. Back to text.

22. The Beer the Eyes Ordered

1. Felipe Reinoso Carvalho, Pieter Moors, Johan Wagemans and Charles Spence, "The Influence of Color on the Consumer's Experience of Beer," *Frontiers in Psychology* 8 (2017), 2205, DOI 10.3389/fpsyg.2017.02205. Open full text consulted, including methods, results and discussion. The preliminary triangle test involved twenty-five participants; the main study involved 136. Failure to detect a difference is not a proof of identity. This chapter selectively paraphrases the study and adds its own interpretation; no figures are reproduced. Back to text.

2. Helena Blackmore, Claire Hidrio and Martin R. Yeomans, "A taste of things to come: The effect of extrinsic and intrinsic cues on perceived properties of beer mediated by expectations," *Food Quality and Preference* 94 (2021), 104326. Publisher-indexed abstract and highlights consulted; direct full-text retrieval returned an access error. No unexamined effect size or subgroup result is supplied. Back to text.

3. Blackmore, Hidrio and Yeomans, "How sensory and hedonic expectations shape perceived properties of regular and non-alcoholic beer," *Food Quality and Preference* 99 (2022), 104562. Publisher abstract, introduction, design excerpt and discussion excerpt consulted; full participant-level methods and data were not independently examined. The article discloses industry funding of the PhD project and an industry-affiliated coauthor; it also states the coauthor did not conduct the analysis or interpretation. Back to text.

4. Edward W. Wolfe, "Beer Evaluation and the Judging Process," Beer Judge Certification Program, public educational guide consulted. The page explicitly says much of the material has been superseded. It is used as evidence of practical concern about context, not as a current exhaustive standard or controlled psychological study. Back to text.

5. World Beer Wiki routes: "Colour and label expectations can influence beer perception" and "Difference tests versus quality scores". The 2017 null result is independently added to prevent a one-directional account of color's effects. Back to text.

23. The Borrowed Winter

1. University of Chicago Library, "Refrigeration," *Something's Brewing*. Full public exhibition text consulted. This institutional account provides the ice-house, Linde and refrigerated-railway context; the underlying nineteenth-century patent was not independently examined. Return 1 · Return 2.

2. Géraldine Lysser, "Cool Beer—All Year Round," Swiss National Museum, 16 April 2026. Full public article consulted. The chapter uses its harvesting, transport and Hürlimann chronology, without adopting its ranking of early European installations or reproducing collection images. Back to text.

3. Ingebjørg Eidhammer, "Export of Beer from the 1860s—Enabling Technologies," Norwegian Museum of Science and Technology. Full public research overview consulted. The URL's reference to oil is misleading; the actual page concerns beer. The Ringnes archival holdings

are described by their researcher, not presented as archives independently examined for this book. Back to text.

4. International Energy Agency, "How a Heat Pump Works," *The Future of Heat Pumps*, 2022. Full public explanatory section consulted. Its account supports the transfer mechanism; household performance figures are not assigned to an industrial brewery. Return 1 · Return 2.

5. Roger Padullés and colleagues, "Generalized Electrification Solutions for Breweries: A Pinch Analysis-Based Approach to Heat Pump Integration and Process Decarbonization," *Energy Conversion and Management* 357 (2026), 121292, DTU research record and abstract, DOI. The full repository entry and journal abstract were consulted; the linked PDF returned 403. Model results are not reported as measured savings at operating breweries. Back to text.

6. Greenhouse Gas Protocol, "Calculation Tools FAQ", particularly the public explanation of direct and indirect emissions and Scopes 1, 2 and 3. Relevant guidance consulted on 28 October 2026. No numerical historical-versus-modern lifecycle comparison is available from the sources used here. Back to text.

7. Research route: World Beer Wiki, winter ice to heat pumps, ice houses and mechanical cooling and industrial heat-pump model boundaries. The institutional pages and original engineering abstract were checked separately. Back to text.

24. The Field Ahead

1. Sidsel Birkelund Schmidt and colleagues, "Ancient Barley Landraces Adapted to Marginal Soils Demonstrate Exceptional Tolerance to Manganese Limitation," *Annals of Botany* 123 (2019), 831–843; online December 2018, open text. Methods, field results and discussion consulted, including the 2017 Burray/Dundee comparison and the Bere West Orkney exception. These are specific genotype-by-environment results. Return 1 · Return 2.

2. Martin Mozny and colleagues, "Climate-Induced Decline in the Quality and Quantity of European Hops Calls for Immediate Adaptation Measures," *Nature Communications* 14 (2023), 6028, open article. Full-text results and methods consulted. Historical regional comparisons are separated from projections using three climate models under RCP4.5; alpha-acid concentration is not treated as a complete account of aroma or future beer flavor. Back to text.

3. Prossie Nakawuka, Troy R. Peters, Stephen Kenny and Doug Walsh, "Effect of Deficit Irrigation on Yield Quantity and Quality, Water Productivity and Economic Returns of Four Cultivars of Hops in the Yakima Valley, Washington State," *Industrial Crops and Products* 98 (2017), 82–92, article PDF. Abstract, experimental-site and treatment methods, and reported conclusions consulted. The economic finding concerns the study's local prices and costs. Back to text.

4. Wei Xie and colleagues, "Decreases in Global Beer Supply Due to Extreme Drought and Heat," *Nature Plants* 4 (2018), 964–973, UCL research record and abstract, DOI. The final publication abstract was checked; publisher full text was unavailable. A linked manuscript

differs from that abstract, so its numerical ranges are not substituted for published results. Back to text.

5. Peter Martin and colleagues, "Back to the Future: Using Ancient Bere Barley Landraces for a Sustainable Future," *Plants, People, Planet* 7 (2025), 546–561; online 7 September 2023, open review. Historical, genetic and conservation sections consulted. This synthesis supplied routes to the original experiments and social research, which were checked separately; no exact ancient introduction date is asserted. Back to text.

6. J. E. Cope and colleagues, "Assessing the Variation in Manganese Use Efficiency Traits in Scottish Barley Landrace Bere (Hordeum vulgare L.)," *Annals of Botany* 126 (2020), 289–300, open text. Indexed full-text abstract, methods summary and discussion consulted; direct page access returned verification screens. Associated genomic regions are candidates requiring validation, not completed new varieties. Back to text.

7. Niamh Mahon, Shawn McGuire and Md. Mofakkarul Islam, "Why Bother with Bere? An Investigation into the Drivers behind the Cultivation of a Landrace Barley," *Journal of Rural Studies* 45 (2016), 54–65, authors' university record and abstract. The complete institutional abstract was consulted. The interviews belong to the original researchers; no interview or direct quotation was created for this book. Back to text.

8. Peter Martin and colleagues, "Trends, Challenges and Opportunities in the In Situ Conservation of Cereal Landraces in Scottish Islands," *Genetic Resources* 4 (2023), 32–45, UHI research record and abstract, DOI. Institutional abstract consulted; it reports historical trends rather than certifying current farm numbers or guaranteed future survival. For the named milling example, see Birsay Heritage Trust, "About the Trust", and Barony Mill, "History and Millers", both full public practitioner accounts read on 28 October 2026. These accounts support the 1998 restoration purpose, named transfer of milling skills, and production of beremeal; no interview or endorsement was obtained for this book. Their broader claims about Bere's antiquity or worldwide uniqueness are not adopted. Return 1 · Return 2.

9. Research route: World Beer Wiki, climate and the future pint, Bere crop resilience and Orkney knowledge and revival. The closing reflections are the book's interpretation of the evidence, not a forecast or a community's endorsement. Back to text.

Selected Further Reading

These starting points offer a shorter route into the investigations. The chapter notes and accompanying source map contain the fuller bibliography and precise access limits. Inclusion does not imply endorsement of this book.

What Survives

• **Chapter 1:** Liu et al. (2018), Fermented beverage and food storage in 13,000-y-old stone mortars at Raqefet Cave. Abstract plus substantial publisher-indexed introduction, methods and results excerpts; direct full-text/PDF retrieval unsuccessful.

• **Chapter 2:** Oxford ETCSL, A hymn to Ninkasi (Ninkasi A). Full English translation; no independent Sumerian translation performed.

• **Chapter 3:** Metropolitan Museum of Art, Cuneiform tablet: record of rations of beer, bread, oil, and onions for messengers. Full public collection record.

• **Chapter 4:** Revealing the 2300-Year-Old Fermented Beverage in a Bronze Bottle from Shaanxi, China. Publisher-indexed full-text introduction and relevant process discussion consulted during cross-part review; direct page returned 429. Not a claim to inspect all figures or supplementary data.

• **Chapter 5:** Field Museum (2016), Field Museum Releases New Beer Inspired by Ancient Recipe. Full public institutional press release.

• **Chapter 6:** Powerhouse, Boulton and Watt rotative steam engine. Full public object catalog.

Inside the Glass

• **Chapter 7:** Expression Levels of the Yeast Alcohol Acetyltransferase Genes ATF1, Lg-ATF1, and ATF2 Control the Formation of a Broad Range of Volatile Esters. Full-text page; relevant abstract, strain methods and results consulted

• **Chapter 8:** Evaluation of Components Contributing to the International Bitterness Unit of Wort and Beer. Indexed publisher abstract only; full article not accessible

• **Chapter 9:** Vitamins in Brewing: Effects of Post-Fermentation Treatments and Exposure and Maturation on the Thiamine and Riboflavin Vitamer Content of Beer. Full-text methods and glass-transmission discussion consulted

• **Chapter 10:** Effect of Wort Boiling System and Hopping Regime on Wort and Beer Stale-Flavor Aldehydes. Full-text methods, results and conclusions consulted

• **Chapter 11:** Impacts of Pathogen Strain and Barley Cultivar on Fusarium Head Blight in Barley and during Malting. Indexed full-text methods, gushing results and discussion

• **Chapter 12:** Microbe Domestication and the Identification of the Wild Genetic Stock of Lager-Brewing Yeast. PubMed abstract; direct PMC full text returned verification page

A Living Trade

• **Chapter 13:** Elizabeth Kuligowski, Alewives in Oxford: A History of Female Brewing, Museum of Oxford. Full public museum article. Used narrowly; its speculative witch imagery and broad causal story were excluded.

• **Chapter 14:** Echter Zoigl vom Kommunbrauer, An Introduction to the Tradition of Zoigl Beer. Full practitioners' association HTML. Historical-date discrepancies and dated establishment counts were not adopted.

• **Chapter 15:** Aliata Issahaq Mumuni, Musah Dumah and Hawa Awuro Sam, Modeling the Determinants in Pito Brewing of Small-Scale Enterprising in Ghana (2023). Full nine-page PDF text, including methods and tables.

• **Chapter 16:** Frances M. Hayashida, Ancient beer and modern brewers: Ethnoarchaeological observations of chicha production in two regions of the North Coast of Peru (2008). Publisher abstract plus detailed indexed methods and Piura excerpts; full paper inaccessible. Indigenous CHIRAPAQ catalogue consulted for metadata, not a community endorsement.

• **Chapter 17:** Changhoon Chai, Goo-Sang Lim, Yun-Ji Kim and Se-Wook Oh, Microbial Community Changes in Makgeolli during Brewing (2015). Full publisher HTML, especially preparation methods and time-course results.

• **Chapter 18:** Dorothy Machunda Gimbi and Naofumi Kitabatake, Changes in alpha-and beta-amylase activities during seed germination of African finger millet (2002). Complete indexed PubMed and publisher abstracts; full article unavailable.

The Unfinished Case

• **Chapter 19:** Klaus Rupprecht, Purity Law, 1516, Historisches Lexikon Bayerns. Drafter recorded full public entry. Independently reopened in the fresh AI review; relevant historical passages and illustrated-document descriptions consulted. Underlying archival volumes not separately examined.

• **Chapter 20:** Zuzana Lízcová, No risk, no gain, Goethe-Institut. Drafter's recorded access: full public article. Not independently reopened by the fresh reviewer.

• **Chapter 21:** Alejandra Bussalleu et al., Cultural Values and the Coliform Bacterial Load of Masato, an Amazon Indigenous Beverage, EcoHealth 17 (2020), 370–380. Drafter recorded full open text. Independently opened and checked during the fresh AI review, including methods, Tables 1–2, photographic account, discussion and limitations. No source photographs reproduced.

• **Chapter 22:** Felipe Reinoso Carvalho, Pieter Moors, Johan Wagemans and Charles Spence, The Influence of Color on the Consumer's Experience of Beer, Frontiers in Psychology 8 (2017), 2205. Drafter recorded full open text. Independently reopened during the fresh AI review; stimulus preparation, triangle test, main design and results checked. No figures reproduced and participant-level data not reanalyzed.

• **Chapter 23:** Refrigeration — Something's Brewing, University of Chicago Library. Full public institutional exhibition text consulted; the historical patent and cited scholarly book were not independently examined

• **Chapter 24:** Ancient Barley Landraces Adapted to Marginal Soils Demonstrate Exceptional Tolerance to Manganese Limitation. Open full text and indexed full-text results consulted; relevant introduction, experimental methods, field results and discussion read